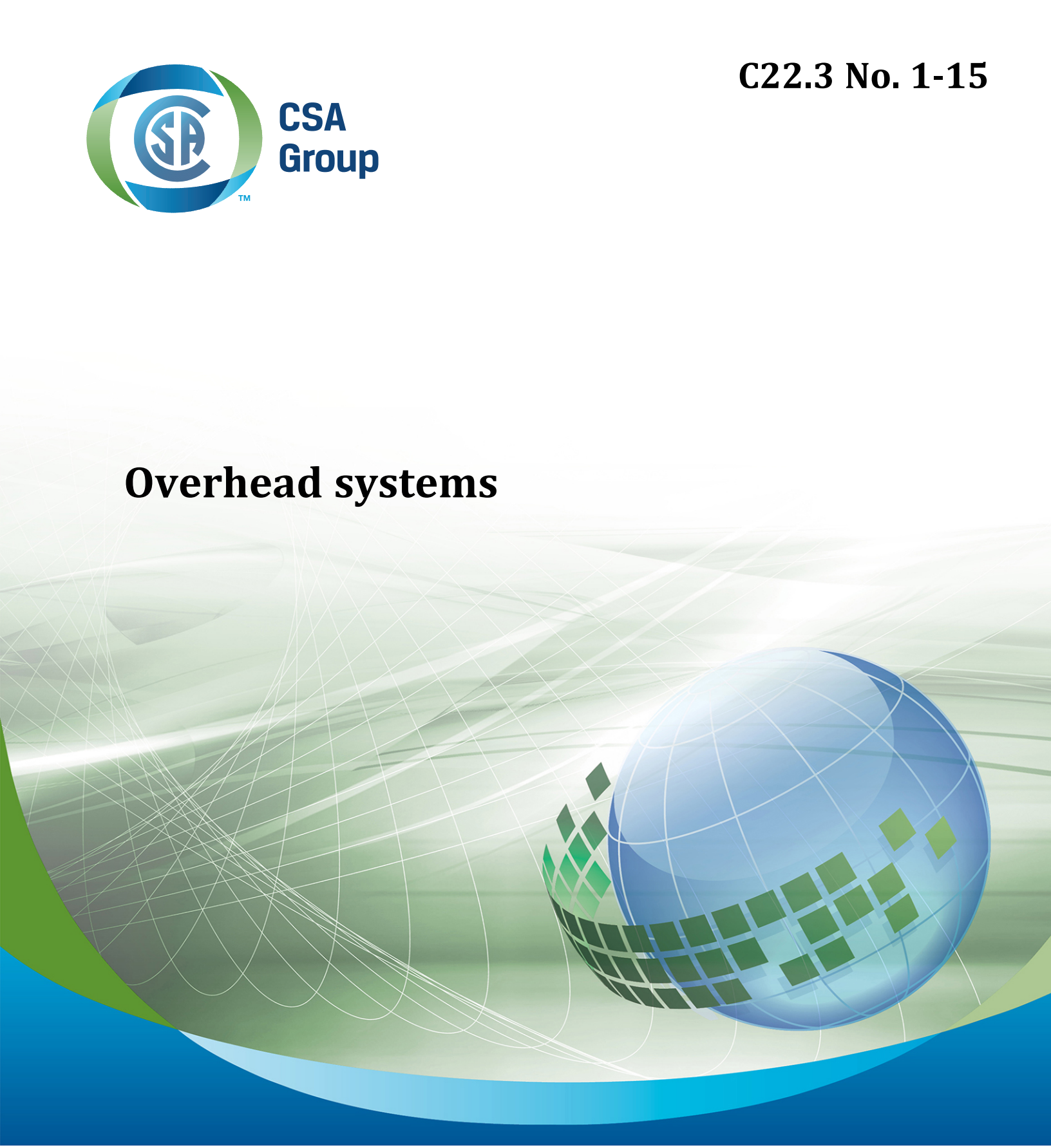
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C22.3 No. 1-15

Overhead systems



REVISED APRIL 2016

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Preface

This is the tenth edition of CSA C22.3 No. 1, *Overhead systems*, one of a series of Standards issued under the *Canadian Electrical Code, Part III*. It supersedes the previous editions, published in 2010, 2006, 2001, 1987, 1985, 1979, 1976, and 1970, and the original edition, which was published as a series of five Standards in 1959, 1953, 1947, and 1940.

Significant changes in this edition include

a) The linear design method has been removed from the Standard completing an updating process that was initiated in 2003.
b) Definitions and use of terminology have been updated including
   i) the following new definitions: “pole brace”, “rated tensile strength”, and “vegetation control cycle”; and
   ii) revised definitions for “vegetation management” and “communications drop”.
c) A new clause and guidance in Annex A regarding clearance between supply lines and spray irrigation systems has been included.
d) New clauses have been included addressing
   i) poles with internal supply riser cables; and
   ii) point of attachment of service conductors at customer premises.
e) Clearances between wires crossing each other and supported by different structures (Table 13) have been rationalized and updated. Expanded examples and guidance has been provided in Annex A.
f) Coefficients of variation (COV) in the headings of Table 31 specifying load factors for non-linear design of structures have been coordinated with CAN/CSA-O15.
g) Clauses dealing with communication circuits used exclusively in the operation of supply lines have been expanded to include communication equipment used exclusively in the operation of supply circuits.
h) A clause dealing with assumed transverse load due to wind pressure has been revised to permit reduction in loading for sheltered distribution circuits.
i) Clauses dealing with vegetation management, protection against mechanical damage, and assumed longitudinal load have been re-worded to improve clarity.
j) Clearances over aerial tramways have been updated.
k) Revisions have been made to clauses dealing with obstruction of climbing and working space to improve clarity
l) Load factors for guy assembles have been revised to reflect the transition to the non-linear method of analysis.
m) Criteria regarding tower head clearances on transmission line structures have been included.
n) The Standard has undergone general updating.

The reliability-based design of transmission lines is covered by CAN/CSA-C22.3 No. 60826.

The purpose of the annexes is to provide the background information and reasoning necessary for the application of this Standard. It is not the intent of the annexes to modify the requirements of any of the clauses in this Standard.

This Standard was prepared by the Technical Committee on Overhead Systems, under the jurisdiction of

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the Strategic Steering Committee on Power Engineering and Electromagnetic Compatibility, and has been formally approved by the Technical Committee.

Notes:

1) *Use of the singular does not exclude the plural (and vice versa) when the sense allows.*

2) *Although the intended primary application of this Standard is stated in its Scope, it is important to note that it remains the responsibility of the users of the Standard to judge its suitability for their particular purpose.*

3) *This Standard was developed by consensus, which is defined by CSA Policy governing standardization — Code of good practice for standardization as "substantial agreement. Consensus implies much more than a simple majority, but not necessarily unanimity". It is consistent with this definition that a member may be included in the Technical Committee list and yet not be in full agreement with all clauses of this Standard.*

4) *To submit a request for interpretation of this Standard, please send the following information to inquiries@csagroup.org and include "Request for interpretation" in the subject line:*

a) *define the problem, making reference to the specific clause, and, where appropriate, include an illustrative sketch;*

b) *provide an explanation of circumstances surrounding the actual field condition; and*

c) *where possible, phrase the request in such a way that a specific "yes" or "no" answer will address the issue.*

Committee interpretations are processed in accordance with the CSA Directives and guidelines governing standardization *and are available on the Current Standards Activities page at* ***standardsactivities.csa.ca****.*

5) *This Standard is subject to review five years from the date of publication. Suggestions for its improvement will be referred to the appropriate committee. To submit a proposal for change, please send the following information to inquiries@csagroup.org and include "Proposal for change" in the subject line:*

a) *Standard designation (number);*

b) *relevant clause, table, and/or figure number;*

c) *wording of the proposed change; and*

d) *rationale for the change.*

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C22.3 No. 1-15
Overhead systems

1 Scope

1.1
This Standard applies to electric supply and communication lines and equipment located entirely outside of buildings and fenced supply stations.

1.2
Existing installations (including maintenance replacements and maintenance alterations) meeting the original designs that currently comply with prior editions of this Standard, need not be modified to comply with this edition of the Standard, except as might be required for safety reasons by the authority having jurisdiction.

1.3
This Standard, which forms part of the *Canadian Electrical Code, Part III*, provides requirements for the construction of overhead systems. It covers electric supply and communication circuits that

a) are installed alone;
b) are in joint use;
c) are in proximity to each other or other facilities;
d) cross each other or other facilities; and
e) cross railways, highways, navigable waterways, or land that is likely to be traversed by vehicles or pedestrians.

1.4
This Standard presents a choice between deterministic and reliability-based design methods. Reliability-based design methods are covered by CSA C22.3 No. 60826.

1.5
The requirements contained in this Standard do not constitute complete design and construction specifications, but rather prescribe the minimum design requirements that are most important to the

a) safety of persons;
b) continuity of service; and
c) protection of property.

1.6
Conditions not covered by this Standard are governed by equivalent Standards in common use or by the authority having jurisdiction.

1.7
In some cases in this Standard, specific types of construction are envisaged. This does not preclude the use of other types of construction, provided that the engineering representatives involved can demonstrate the safety and suitability of these alternatives.

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1.8
The use of terms such as “where practicable” is not intended to provide an opportunity for not meeting the requirements of this Standard, but indicates the preferred clearance or method. Where an alternative is not specified, the engineering solution that most closely adheres to the preferred method should be used.

1.9
Where the requirements of more than one clause apply, all should be satisfied.

1.10
In this Standard, “shall” is used to express a requirement, i.e., a provision that the user is obliged to satisfy in order to comply with the standard; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express an option or that which is permissible within the limits of the Standard.

Notes accompanying clauses do not include requirements or alternative requirements; the purpose of a note accompanying a clause is to separate from the text explanatory or informative material.

Notes to tables and figures are considered part of the table or figure and may be written as requirements.

Annexes are designated normative (mandatory) or informative (non-mandatory) to define their application.

2 Reference publications
This Standard refers to the following publications, and where such reference is made, it shall be to the edition listed below, including all amendments published thereto.

CSA Group

CAN/CSA-A14-07 (R2012)
Concrete poles

B149.2-10
Propane Storage and Handling Code

C22.1-15
Canadian Electrical Code, Part I

C22.2
Canadian Electrical Code, Part II

C22.2 No. 41-13
Grounding and bonding equipment

C22.3
Canadian Electrical Code, Part III

CAN/CSA-C22.3 No. 3-98 (R2013)
Electrical coordination

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C22.3 No. 5.1-93 (R2013)
Recommended practices for electrical protection — Electric contact between overhead supply and communication lines

C22.3 No. 60826-10*
Design criteria for overhead transmission lines
**A new edition is currently under development and will supersede the 2010 edition.*

C57-98 (R2011)
Electric power connectors for use in overhead line conductors

C83-96 (R2011)
Communication and power line hardware

CAN/CSA-C411.1-10 (R2014)
AC suspension insulators

CAN/CSA-C411.4-10 (R2015)
Composite suspension insulators for overhead lines ≥ 75 kV

CAN/CSA-C411.5-10 (R2014)
Dead-end/suspension composite insulators for overhead lines ≤75kV

CAN/CSA-C411.6-11
Line post composite insulators for overhead distribution lines

CAN/CSA-C411.7-11
Composite insulator for guy wires

CAN/CSA-C61089-11
Round wire concentric lay overhead electrical stranded conductors

G12-14
Zinc-coated steel wire strand

O15-15
Wood utility poles and reinforcing stubs

Z98-07 (R2012)
Passenger ropeways and passenger conveyors

ANSI/NEMA (American National Standards Institute/National Electric Manufacturers Association)

C29.1-1988 (R2002)
American National Standard — Test Methods for Electrical Power Insulators

C29.11-2012
American National Standard for Composite Suspension Insulators for Overhead Transmission Lines — Tests

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C29.12-2013
American National Standard for Insulators — Composite — Suspension Type

C29.13-2012
American National Standard for Insulators — Composite — Distribution Deadend Type

ASCE (American Society of Civil Engineers)

Manual 104-2003
Recommended Practice for Fiber-Reinforced Polymer Products for Overhead Utility Line Structures

ASTM International (American Society for Testing and Materials)

B524/B524M-99
Standard Specification for Concentric-Lay-Stranded Aluminum Conductors, Aluminum-Alloy Reinforced (ACAR, 1350/6201)

B230/230M-07
Standard Specification for Aluminum 1350-H19 Wire for Electrical Purposes

B232/B232M-11
Standard Specification for Concentric-Lay-Stranded Aluminum Conductors, Coated-Steel Reinforced (ACSR)

B398/B398M-14
Standard Specification for Aluminum-Alloy 6201-T81 and 6201-T83 Wire for Electrical Purposes

B500-12
Standard Specification for Metallic Coated or Aluminum Clad Stranded Steel Core for use in Overhead Electrical Conductors

D1036-99 (2012)
Standard Test Methods of Static Tests of Wood Poles

CIGRE (International Council on Large Electric Systems)

Technical Brochure 273 (2005)
Overhead conductor safe design tension with respect to Aeolian vibrations

Health Canada

Safety Code 6 (2009)
Limits of Human Exposure to Radiofrequency Electromagnetic Fields in the Frequency Range from 3 kHz to 300 GHz

IEC (International Electrotechnical Commission)

IEC/TS 60815-1 (2008)
Selection and dimensioning of high-voltage insulators intended for use in polluted conditions — Part 1: Definitions, information and general principles

IEC/TS 60815-2 (2008)
Selection and dimensioning of high-voltage insulators intended for use in polluted conditions — Part 2: Ceramic and glass insulators for a.c. systems

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IEC/TS 60815-3 (2008)
Selection and dimensioning of high-voltage insulators intended for use in polluted conditions — Part 3: Polymer insulators for a.c. systems

IEEE (Institute of Electrical and Electronics Engineers)

80-2000
IEEE Guide for Safety in AC Substation Grounding

487-2007
IEEE Recommended Practice for the Protection of Wire-Line Communication Facilities Serving Electric Supply Locations

1542-2007
Guide for Installation, Maintenance, and Operation of Irrigation Equipment Located Near or Under Power Lines

ANSI/IEEE 738-2006
IEEE Standard for Calculating the Current-Temperature of Bare Overhead Conductors

ANSI/IEEE C2-2012
National Electrical Safety Code

C95.6-2002
IEEE Standard for Safety Levels with Respect to Human Exposure to Electromagnetic Fields, 0 to 3 kHz

National Research Council Canada

National Building Code of Canada, 2010

NERC (North American Electric Reliability Corporation)

Standard FAC-003-1 (2006)
Transmission Vegetation Management Program

Transport Canada

Canadian Aviation Regulations 2009-1, Part VI — General Operating and Flight Rules,
Standard 621.19 — Standards Obstruction Markings

TP 1247
Land Use in the Vicinity of Airports — Part I — Obstacle Limitation Surfaces

Navigation Protection Act, R.S.C. 1985, c. N-22

Minor Works And Waters (Navigation Protection Act) Order, April 22, 2009 (Canada Gazette, Vol. 143, No. 19 — May 9, 2009)

Other publication

Deno, D.W. and L.E. Zaffanella. 1982. Field Effects of Overhead Transmission Lines and Stations. In *Transmission Line Reference Book,* 3rd ed., Palo Alto: Electric Power Research Institute

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3 Definitions and abbreviations

3.1 Definitions

The following definitions shall apply in this Standard:

Across or cross — crossing over or under.

Active electrical equipment — equipment used in primary metering, transformer, regulator, capacitor, and recloser installations.

All-dielectric self-supporting (ADSS) fibre optic cable — a communication cable consisting of coated glass optical fibres contained in a protective dielectric fibre optic unit that is surrounded by, or attached to, suitable dielectric strength members and jackets.

Note: *The dielectric properties of the outermost jacket can be of a class that is*

a) *suitable only for application in electric fields where the level of electrical stress on the jacket does not exceed 12 kV space potential; or*
b) *suitable for application in electric fields where the level of electrical stress on the jacket can exceed 12 kV space potential.*

Caution*: Some cables are manufactured with semiconducting jackets appropriate for either Items (a) or (b).*

Alongside or along — within the boundaries of the area under consideration or within a specified distance from it.

Ballast — crushed stone or gravel placed between and below the ties of railway tracks.

Bonding — the electrical interconnecting of metallic parts or conductors in order to maintain them at the same potential and achieve a desired distribution of currents within a grounding system.

Cable — an assembly of one or more insulated electrical or optical conductors, or a combination thereof, in a compact form, enclosed in a covering consisting of a combination of metal, plastic, or other materials used to provide mechanical and electrical protection.

Catenary parameter (C = H/w) — the horizontal tension (H) in a conductor (expressed in N) divided by the conductor unit weight (w) (expressed in N/m).

Circuit — a conductor or system of conductors through which electric current or optical energy is intended to flow.

Civil infrastructure — structures, trenches, and components used to support, enclose, and/or protect lines, cables, and equipment.

Clearance — the distance, under specified design conditions, between the nearest points of two objects at points where at least one object is movable.

Note: *See also* ***Separation*** *and* ***Spacing****.*

Climbing space — the space reserved to climb on a structure that permits line workers to have access to equipment and conductors located on the structure and that allows equipment to be hoisted to its desired position.

Coefficient of variation (COV) — the standard deviation of a sample divided by its mean.

Common-structure crossing — a crossing of lines wherein a structure acts as a common support to, and forms an integral part of, each line at the point of crossing.

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Communication cable — a cable used for a signal or communication service.

Communication circuit — a circuit used for a signal or communication service.

Communication conductor — a conductor in a communication circuit or cable.

Communication drop — the portion of a communication circuit that runs from the line to the service location.

Note: *These are typically paired conductors, coaxial wires, or 1 or more optical fibers; designed to span distance similar to electric service conductors.*

Communication line — see **Line**.

Conductor — a material used for the transmission of electrical, electromagnetic, or optical energy.

Coordinated electrical protection (as applied to supply and communication lines in close physical proximity) — the protective measures applied to the plant of both utilities so that

a) the supply and communication lines are designed, constructed, operated, and maintained in such a way that the supply voltage is promptly removed from the communication plant (including lines) by de-energization or other means, both initially and following subsequent breaker operations, in the event of contact between the supply line and the communication plant (including lines); and
b) in the event that the communication plant comes into contact with the supply conductors, the voltage and current impressed on the communication plant (including lines) do not exceed the safe operating limit of the communication protective devices.

Creep — the permanent elongation of suspended wire due to the effects of tension over a period of time.

Cross — see **Across**.

Crossarm assembly — an assembly that consists of crossarms and the devices used to attach them to a structure.

Dead-end assembly — the component parts, such as clamps or devices, that secure the conductors to the structure in dead-end construction.

Dead-end construction — the attachment to structures of wires or cables so as to transmit their tensions directly to the structures in one direction only.

Deterministic design — a line design in which the load, strength, and load factors are specified and not necessarily related to statistical data.

Distribution line — a supply line that is used primarily for supplying electrical energy to local area circuits and services.

Effectively grounded — permanently connected to earth through a ground connection of sufficiently low impedance and having sufficient current-carrying capacity to prevent the buildup of voltages that can result in undue hazard to connected equipment or to persons.

Fastening — a tie, clamp, or other device used to secure wire or cable attachments to a structure.

Final unloaded tension — the tension in the conductor when in the condition of final unloaded sag.

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Full-tension compression splice — a device that is applied to a conductor in order to join together two separate ends of conductor that are of the same size and characteristics.

Notes:

1) *A full-tension compression splice is installed through the application of pressure; compressive energy may be supplied through the use of*
   a) *mechanical press and suitable dies;*
   b) *implosive devices; or*
   c) *other means as might be developed in the future.*

2) *A properly applied full-tension compression splice will have a tensile strength equal to or greater than 95% of the rated tensile strength of the conductor. See Mechanical Duty Class 1 in CSA C57.*

Grounded — electrically connected to earth through a grounding electrode or through an extended conducting body.

Grounded conductor — a line conductor that is grounded at one or more points.

Grounding — the provision of a permanent and continuous conductive path to the earth that
a) has sufficient ampacity to carry any fault current liable to be imposed on it;
b) has a sufficiently low impedance to limit the voltage rise above ground potential; and
c) facilitates the operation of the protective devices in the circuit.

Grounding conductor — a conductor used to connect a grounding electrode directly to the device, component, or system that is being grounded.

Grounding conductor carried in aerial spans — a conductor carried in one or more spans that is used to interconnect equipment needing a common ground to an adequate grounding electrode.

Note: *Grounding conductors carried in aerial spans do not include supply-line neutral conductors or lightning protection wires.*

Grounding electrode — a metallic conductor, rod, or other object, or an assembly of objects, that is embedded in or suitably connected to the earth.

Note: *Grounding electrodes are used for maintaining ground potential on associated grounding conductors and dissipating current into the earth.*

Grounding system — an interconnected system of grounding electrodes, grounding conductors, and bonding conductors that is designed to provide an acceptable connection between an electrical system or component and the earth.

Group of similar conductors — conductors that
a) occupy the same crossarm;
b) form part of a single circuit; or
c) are otherwise considered to function as a unit having common structural behaviour.

Guarded — protected by mechanical means of strength adequate to inhibit contact by the public or damage by the public or workers to a structure, attachment, wire, or conductor.

Guy —

Span guy — a guy that spans the distance between structures.

Working guy — a guy installed for the purpose of supporting a structure under normal working loads.

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Guy assembly — the strand, insulators, clamps, anchor rods, anchors, and devices used to secure the guy to the structure.

Guy strand — a wire or strand whose main function is to provide mechanical support to a structure.

Highway — a road to and from which access is permitted only at designated entrances and exits.

Ice drop-off — a condition that can occur with ice-loaded conductors in mild weather, where the ice coating drops off suddenly and causes a conductor to rise to a level significantly higher than the normal stationary level.

Inaccessible — see **Isolated**.

Initial unloaded tension — the tension in a conductor when the conductor is in the condition of initial unloaded sag.

In-span crossing (as applied to two lines) — a crossing without use of a common structure.

Insulated — separated from other conducting surfaces by a dielectric material or air space that offers a high resistance to the passage of current and to disruptive discharge through the material or air space.

Inverted level arrangement — the relative vertical positions of circuits or lines that cross or are attached to the same structure such that

a) communication facilities are above one or more of the supply lines or circuits; or
b) lower-voltage circuits are above higher-voltage circuits.

Note: *See Normal level arrangement.*

Isolated — not readily accessible to persons unless special means for access are used.

Jacket — a protective non-metallic covering that is applied to cables with or without sheaths.

Joint use — the supporting or enclosing of supply and communication lines within or on a common civil infrastructure.

Lateral conductor — a conductor that extends in a generally horizontal direction at an angle to the general direction of the line conductors.

Lightning protection wire — a grounded overhead wire used for lightning protection.

Note: *Lightning protection wires are usually located at or near the top of the supporting structure.*

Limited- or controlled-access highway — a fully controlled highway to which access is controlled by a governmental authority for purposes of improving traffic flow and safety.

Note: *Limited-access highways have no grade crossings and have carefully designed access connections.*

Line — conductors and cables, including their associated equipment and supporting structures, that are located entirely outside of buildings.

Communication line — a line used for a signal or communication service.

Supply line — a line used for transmitting a supply of electrical energy.

Linear analysis — a structural analysis method that does not consider P-delta (secondary moment) effects of structure displacements.

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Line conductor — a wire that is used to carry electric current and is supported by structures for one or more spans.

Live — electrically connected to a source of potential difference or electrically charged to have a potential that is significantly different from that of the surrounding earth.
Note: *In this Standard, the term "live" is sometimes used in place of "current-carrying" or "energized".*

Load —

Assumed load —
a) for supports, taps, and splices — the force resulting from the assumed conductor load and other applicable forces, multiplied by the applicable load factor specified in this Standard; and
b) for wires and cable attachments — the force in the wire or cable attachment resulting from the applicable weather load.

Thermal load — the heating effect produced in a wire or conductor by which its temperature is raised above the ambient due to
a) the absorption of heat from the sun; and
b) the heat produced by the current flowing in the conductor.

Weather loads — loads that result from low temperature, ice, and/or wind acting on conductors and structures.

Load effects — loads acting jointly or separately.
Note: *Load effects can be axial forces, shear forces, bending moments, or a combination of the three.*

Load factor — the factor by which a load is multiplied to create an assumed load.
Note: *A load factor takes into account uncertainties in loading conditions and, in the case of deterministic designs, the strength of materials.*

Loading — see Load.

Luminaire — a complete lighting unit consisting of a light source and its direct appurtenances, such as globes, reflectors, housings, and those supports integral to housings.
Note: *Luminaires do not include brackets that are separate units.*

Main line — the supply or communication line involved when strength requirements are being considered for a specific crossing, proximity, or joint-use section that calls for the strongest grade of construction.

Maximum planned summer operating current — the maximum rms current that an individual conductor of a circuit is designed to carry under specified summer maximum ambient conditions.
Note: *The maximum planned summer operating current does not include the current under fault conditions or short-duration emergency conditions.*

Mean (μ) — the arithmetic sum of a sample of data divided by the number of units (observations) in the sample.

Messenger — a wire or strand for supply or communication lines that supports, in addition to its own weight, the weight of one or more conductors or cables.
Notes:
1) *The messenger does not form part of the electrical circuit.*
2) *See also **Supporting conductor**.*

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Minimum (as applied to a value in this Standard) — all deviations are positive.
Note: *For example, a minimum value of 1 means 1.0000 ..... or greater.*

Multi-grounded neutral — a supply-line neutral conductor that is grounded at multiple locations so that the neutral is effectively grounded.

Near — in such proximity as to increase the possibility of interference.

Neutral conductor — a metallic conductor that forms a continuous return path from source to load for a supply line.
Note: *Neutral conductors are generally grounded.*

Non-energized supply plant — supply plant equipment or parts that are not intended to be connected directly to a source of voltage but that could become energized accidentally.
Note: *Non-energized supply plant includes through-bolts, braces, and transformer and other supply equipment cases.*

Non-linear analysis — a structural analysis method that considers geometric non-linearities and P-delta (secondary moment) effects of structure displacements.

Normal level arrangement — the relative vertical positions of lines that cross or are attached to the same structure, as follows (reading from the uppermost position downwards):
a) supply (higher voltages);
b) supply (lower voltages);
c) communication; and
d) trolley contact conductors and associated feeders.

Normally inaccessible — not readily accessible to persons unless special means for access are used.

Open (as applied to conductors) — bare or uninsulated.

Optical groundwire (OPGW) — a grounded overhead wire consisting of optical fibres integrally constructed into stranded metallic wires.

Ordinary high-water mark (OHWM) — the limit or edge of the bed of a body of water.
Note: *In the survey industry, the line of vegetation along a body of water is used to identify the OHWM. In the case of non-tidal waters, the OHWM is sometimes known as the "bank" or "limit of the bank".*

Overcurrent protection — the devices applied to a circuit and arranged to disconnect the protected circuit or equipment when the current exceeds a predetermined value for a predetermined length of time.

Overhang — the longitudinal suspension of one or more of the conductors on a line over a route normally traversed by vehicles.

P-delta effect — the additional bending moment applied to a vertical pole by a vertical load, P, multiplied by the horizontal deflection, delta (δ), at its point of application, as measured from the vertical axis of the pole.

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Pipeline — a line of piping, generally of circular cross-section, for the transmission of gases, liquids, or solids outside of buildings.

Notes:

1) *Pipelines include all associated branches, extensions, tanks, reservoirs, pumps, racks, compressors, loading facilities, and other outside works.*

2) *A pipeline does not include pipe-type electric power cables.*

3) *Pipelines can be above-ground or underground.*

Pole brace — a rigid structural component that provides lateral support to a pole by normally acting in compression. A pole brace includes the entire assembly necessary for attachment to the pole on one end and to either another pole or a foundation at the other end, to support the expected loads.

Proximity — the relative location of lines such that the failure of a conductor or any part of the structure (such as overturning at the ground line) interferes with the normal use, operation, or maintenance of public or private property, plant, or other facilities, through contact or encroachment on minimum clearance requirements by the conductor or structure.

Qualified person — a person who is familiar with the construction and operation of an apparatus and with the hazards involved, and who is authorized by the applicable utility to perform certain functions in connection with the plant under consideration.

Railway — the track(s) and control facilities of any transportation system operating on a fixed rail or track.

Rated ampacity — the current that results in a temperature rise in a bare conductor of 60 °C above an ambient of 40 °C and, for a covered conductor, a rise of 40 °C above an ambient of 40 °C, with the following parameters:

a) a wind velocity of 0.61 m/s;
b) an emissivity of 0.5;
c) full sunshine;
d) a solar absorption coefficient of 0.5; and
e) an elevation at sea level.

Rated tensile strength (RTS) — a number, calculated according to a specific standard, intended to provide a minimum theoretical tensile strength value of a conductor.

Readily accessible — able to be reached, entered, or influenced without difficulty, obstruction, or the use of special means.

Reliability-based design — a line design in accordance with CSA C22.3 No. 60826.

Right-of-way — a strip of land reserved for public roads, railway tracks, pipelines, supply or communication lines, or other services.

Riser — a cable, grounding, or neutral conductor that rises or descends to interconnect an overhead system and an underground system.

rms — the root mean square value of an alternating current or voltage.

Sag (as applied to a conductor at any point in the span) — the vertical distance from a particular point on a conductor to a straight line between its two points of support.

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Final unloaded sag — the sag of a conductor after
a) it has been subjected to a load equivalent to that prescribed for the loading district in which it is situated and subjected to the effects of creep; and
b) the load has been removed.

Initial unloaded sag — the sag of a conductor prior to the first application of an external load or prior to the effects of creep.

Maximum sag (as applied to any point of a wire or conductor) — the larger of the following:
a) the sag under the thermal loading conditions specified in this Standard; or
b) the sag under a vertical load that is equivalent in magnitude to the total resultant loading calculated from the ice, wind, and temperature conditions specified in this Standard for the loading district in which the line is located.

Unloaded sag (as applied to a conductor) — the sag of a conductor in the absence of an external load and at a specified conductor temperature.

Self-supporting cable — a cable that has a messenger integrated into its construction that is suitable for supporting the cable.

Separation — the distance between the nearest points of two objects, where both objects are fixed.
Note: *See also* ***Clearance*** *and* ***Spacing****.*

Service conductor — a conductor that forms part of a circuit used for delivering electrical energy at 750 V or less from the secondary distribution line or distribution feeder, or from the transformer, to the wiring systems of a premises.

Spacing — the distance between the centres of two objects where both objects are fixed.
Note: *See also Clearance and Separation.*

Span guy — see **Guy**.

Span wire — a wire or strand between two structures whose main function is to supply mechanical support to trolley conductors, luminaires, and similar equipment.

Standard deviation (σ) — the positive square root of the variance of a sample of data.

Strand — one or more uninsulated wires whose main function is to provide mechanical support.

Structure — the main supporting unit for a supply line and/or communication line.
Note: *A structure can be*
a) *a pole made of wood, concrete, fibre-reinforced composite, or metal;*
b) *a lattice steel tower; or*
c) *a cable suspension system.*

Supply cable — a cable in a supply line.

Supply circuit — a circuit used for transmitting an electrical energy supply.

Supply conductor — a conductor in a supply line.

Supply line — see **Line**.

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Support — a pole, tower, foundation, guy, crossarm, pin, insulator, fastening, or other component used to support wire or cable attachments.

Supporting conductor — a conductor whose purpose is to support other conductors.

Supporting strand — see **Messenger**.

Supporting structure — a structure and its supports.

Suspension strand — see **Messenger**.

Swing — the horizontal displacement (resulting from any loads specified in this Standard) of any point on a conductor from its position at rest.
Note: *Swing includes the deflection of insulators that are free to swing.*

Tension change — when wire or cable attachments have different tensions on each side of a structure.
Note: *Tension change is usually accomplished by means of dead-end construction.*

Termination — the termination of wire or cable attachments on a structure.
Note: *Termination is usually accomplished by means of dead-end construction.*

Transmission line — a supply line used for transmitting bulk electrical energy between power stations, switching stations, or substations.

Variance — the sum of the squares of the deviations of the data points from the mean value of the sample, n, divided by (n – 1).

Vegetation control cycle — the period of time, as determined by the utility, to perform vegetation management activities.

Vertical conductor — a conductor extending in an approximately vertical direction on a pole or structure.

Voltage — rms voltage-to-ground, unless otherwise specified.
Notes:
1) *The voltage of a grounded circuit is the highest effective voltage between any conductor of a circuit and the point or conductor of the circuit that is grounded, unless otherwise specified.*
2) *The voltage of an ungrounded circuit is the highest effective voltage between any two conductors of the circuit, unless otherwise specified.*

Wet flashover voltage — the voltage at which the air surrounding a clean wet insulator or shell completely breaks down, determined in accordance with ANSI C29.1 and the applicable insulator standard.

Wire and cable attachments — conductors, messengers, cables, guys, and any wires and hardware attached to line supports.

Working guy — see **Guy**.

Working space — the space reserved at the side of a structure that
a) permits line workers to have access to equipment and conductors located on the structure; and
b) allows for the installation of equipment.

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Zone of influence — the area around a power station (or any ground electrode carrying any return portion of a ground fault current) that is raised in potential with respect to remote ground.

3.2 Abbreviations

The following abbreviations shall apply in this Standard:

AASC — aluminum alloy stranded conductor

ACSR — aluminum conductor, steel reinforced

ADSS — all-dielectric self-supporting

ASC — aluminum stranded conductor

COV — coefficient of variation

OHWM — ordinary high-water mark

OPGW — optical groundwire

RTS — rated tensile strength

UTS — ultimate tensile strength

4 General requirements

4.1 General design and maintenance

4.1.1 Multiple requirements

Where two or more requirements apply to a situation, the requirement specifying the greater clearance, separation, spacing, or strength shall take precedence.

4.1.2 Service and operating conditions

All electric supply and communication lines and equipment shall be of suitable design and construction for the service and conditions under which they are to be operated.

4.1.3 Accessibility

4.1.3.1 Normal access

Parts of lines or equipment that need to be examined, adjusted, or maintained during operation shall be arranged so as to be readily accessible to qualified persons by the provision of adequate climbing and working spaces, as well as by the various other clearances and separations specified in this Standard. All structures shall be kept free of unauthorized attachments that could be hazardous and obstruct or otherwise hamper the operations of the utilities using the structures.

4.1.3.2 Access by hydraulic lift

At the time that a structure is designed, the means of access for maintenance shall be determined, and necessary space and facilities shall be provided for access by means of a hydraulic boom truck, lift, or equivalent.

4.1.3.3 Access by helicopter

On higher-voltage lines, the need for accessing structures and conductors by helicopter shall be considered. Where necessary, special landing surfaces shall be provided, taking into account all safe

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clearances to surrounding vegetation and other potential hazards. Where necessary, provisions shall be made for access to the centre phase in a flat-configuration structure and to conductors immediately adjacent to ground wires.

4.1.4 Climbing space

4.1.4.1

Where climbing space is required in accordance with Clause 4.1.3.1, it shall be free from obstructions and shall provide sufficient space, adjacent to the structure, to de-energized conductors or to adequately protected and/or insulated conductors, in order to allow safe access to circuits, equipment, and other attachments on the upper part of the structure. Except as specified in Clauses 4.1.4.2 and 4.1.4.3, the climbing space shall be not less than 0.75 m × 0.75 m adjacent to any conductors, cables, crossarms, or other attachments of the party using any lower part of the structure, and shall extend at least 1 m above and 1 m below the limiting attachments. The climbing space at the various levels of the structure shall be so arranged that transformers, crossarms, and other equipment can be hoisted to the upper position of the structure.

4.1.4.2

Where climbing space is needed adjacent to energized circuits that are not protected or insulated, sufficient additional space shall be provided to allow for climbing safety.

Note: *Though this requirement also applies to normal level arrangements, it is particularly important in inverted level arrangements, where the circuits on the lower part of the structure generally need greater allowances of space than if the circuits were located in the "upper" or "normal" position.*

4.1.4.3

Where a pole carries only communication circuits, or where only supply service conductors are attached to the top of a common crossing pole, the climbing space through a communication line shall be not less than 0.40 m × 0.75 m.

4.1.5 Working space

Working space shall be provided on the climbing face of the structure and shall have the same horizontal dimensions as the climbing space. The height of the working space shall be not less than that required by this Standard for the vertical separation of line conductors carried at different levels on the same structure, except that along crossarms the working space shall extend to the outermost pin position of the crossarms. On joint use poles the working space shall extend below the lowest communications line plant by 1.3 m.

Note: *In cases where workers are expected to make use of ladders for climbing the structure, the working space will be near the top of the climbing space.*

4.1.6 Obstruction in climbing space or working space

Attachments on the structure shall not encroach on the climbing space or working space, with the exception of

a) pole steps at the side of the climbing space or working space on non-wooden poles, or crossarms along the side of the working space or climbing space;
b) longitudinal communication runs along the side of the working space or climbing space, provided that their vertical arrangement does not obstruct climbing of the structure;

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c) longitudinal runs of supply conductors of 0 to 750 V on racks, supply cables of 0 to 750 V on messengers, or supply cables of 0 to 750 V having effectively grounded continuous metal sheaths, along the side of the climbing space;
d) risers or vertical runs in the working space or climbing space that are suitably protected and do not interfere with the use of climbing equipment; and
e) other equipment or items that the parties involved may agree upon that do not interfere with the use of climbing equipment.

4.1.7 Vegetation management

Where vegetation is located near supply-line conductors, the vegetation shall be managed (for example by pruning or cutting), where practicable, accommodating the period of time represented by the vegetation control cycles to maintain the flashover distance specified in Table 35. This distance shall be applied in accordance with the conditions specified in Clause 5.2. Other field conditions can warrant consideration of an additional buffer. Where vegetation management is impracticable, the conductors shall be protected as necessary to prevent damage and electrical hazards.

Notes:

1) *For supply lines operating at 200 kV or higher, refer also to NERC Standard FAC-003-1.*

2) *Where vegetation is located near conductors or messengers on joint-use lines, the vegetation should be managed where practicable, such that under normal operating conditions, there is no load-bearing contact between the conductors or messengers and the vegetation.*

4.2 Structures and attachments

4.2.1 Protection against fires

Structures should be located, guarded, and maintained such that exposure to brush, grass, rubbish, or building fires is minimized.

4.2.2 Protection against mechanical damage

When structures are located in areas where there is an increased risk of impact damage where a single incident could substantially affect the structural integrity of the structure, the provision of guards to minimize the possible damage should be considered.

Note: *This Clause is not intended to address loss of control of vehicles travelling on roadways.*

4.2.3 Insulation of energized conductors attached to structures

Bare and covered energized conductors attached to steel, wood, concrete, or fibre-reinforced composite structures shall be insulated in accordance with Clause 8.18. Insulators are not required for multi-grounded neutral conductors or grounding conductors.

4.2.4 Risers

4.2.4.1 Separation between risers of communication and supply systems

If it is necessary to install risers for communication systems and risers for supply systems on the same pole, they shall be placed on different quadrants of the pole and grouped such that they do not interfere with the use of climbing equipment. On structures other than poles, the separation between risers of communication and supply systems shall be great enough to permit maintenance of each plant

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without interference. When located on streets or roadways, risers shall be placed such that the risk of damage by traffic is minimized.

Note: *Placing risers for communication systems and supply systems on the same pole should be avoided where practicable.*

4.2.4.2 Mechanical protection of supply cables

Riser cables of supply systems shall be protected by a covering that provides suitable mechanical protection for the full length of the run, starting at least 0.3 m below the surface of the earth.

Supply riser cables that have a grounded metal sheath or concentric neutral need not have further protective covering above the 2.5 m level for non-joint-use poles. On poles jointly used with communication systems, the protective covering shall extend at least 1 m above the communication plant.

4.2.4.3 Mechanical protection of communication cables and conductors

Although communication riser cables and conductors need not have mechanical protection for considerations of safety, adequate protection shall be provided to prevent damage or service interruptions.

4.2.4.4 Marking of poles with internal supply riser cables

A pole, other than a dedicated street light pole, that uses its interior as a raceway for supply riser cables shall have visible signage posted on the pole at a height of 2.5 m from the base of the pole at finished grade to warn of the presence of hidden supply conductors. Signage shall be affixed to the pole prior to the installation of any internal supply riser cable.

4.2.5 Protection against climbing

4.2.5.1 Playgrounds and schoolyards

Structures located on or adjacent to playgrounds or schoolyards and carrying open conductors of supply or communication lines shall be constructed, located, or guarded so as to prevent climbing by anyone other than qualified persons.

4.2.5.2 Other locations

Structures that can be climbed without the use of special means and that carry supply-line conductors that exceed 750 V shall be protected by fences or other means against climbing, or shall carry signs in the predominant languages of the locality that warn against trespassing and call attention to the hazards. Bridge fixtures supporting such conductors shall also be guarded and marked. Unguarded poles carrying conductors that exceed 750 V shall not have permanent steps that are located less than 3 m above the ground or surface and that provide access to the pole.

4.2.6 Protection against corrosion

4.2.6.1

All metal supporting structures and hardware subject to corrosion, including bolts, nuts, washers, guys, guy rods, and similar items, shall be protected by galvanized coating, paint, or other treatments that effectively retard corrosion, in accordance with the appropriate CSA Standard for such items. Alternatively, these items shall be manufactured from materials that are corrosion resistant or shall be

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reinforced with additional material such that minimum strength requirements specified in Clause 8 or in CSA C22.3 No. 60826 are met until the item is replaced.

Metallic components shall be
a) compatible with one another in order to minimize galvanic corrosion; or
b) provided with appropriate cathodic protection.

4.2.6.2

Where gas or oil pipelines are located in the vicinity of supply or communication lines, they shall be assessed to determine whether the cathodic protection on the pipelines will cause corrosion to towers, guys, and other buried equipment and, where necessary, mitigating measures shall be implemented. For wood pole lines where insulators are installed in the guys, this assessment might not be necessary.

4.2.7 Grounding and insulating of guys

4.2.7.1 General

Guys located such that the failure of the guys or nearby supply conductors could result in contact with supply conductors shall be effectively grounded or insulated. Where a guy is connected to an effectively grounded metallic structure or ground connection on a wood or fibre-reinforced composite pole, an insulator is not required. Where gas or oil pipelines are involved, an assessment should be performed in accordance with Clause 4.2.6.2.

Where insulators are used or are necessary to prevent electrochemical corrosion of anchor rods or hazards from grounded guys in the supply space, the requirements of Clauses 4.2.7.2 and 4.2.7.3 shall be met.

4.2.7.2 Use of guy insulators

Where a guy could fail and come in contact with the supply conductors of the structure to which it is attached, only one insulator location shall be required.

Where the guy could come in contact with the supply conductors of a separate supply line, two insulator locations shall be placed so as to include the exposed section of the guy between them.

Where the guy could fail and come in contact with a supply conductor above a communication circuit, an insulator shall be placed above the communication circuit.

The guy insulator(s) shall be located so that, in the event of a guy failure, the location of the insulator(s) will not be less than 2.5 m above the ground or accessible surface.

4.2.7.3 Electrical strength of guy insulators

Guy insulators shall have a dry flashover voltage rating of at least double the voltage-to-ground of the highest-voltage supply circuit with which the guy could come in contact and a wet flashover voltage rating equal to the highest voltage between any two conductors. Two or more insulators may be used in series to comply with this requirement.

4.2.8 Marking of guys

Except for guys in areas where cross-country, vehicular, or pedestrian traffic is not expected, all guys shall be marked in a substantial and conspicuous manner. If there is more than one guy at the same anchor, a guy marker shall be applied on both the innermost and outermost down-guy.

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4.2.9 Luminaires, luminaire span, and supply wires

4.2.9.1
Luminaire span or supply wires shall be provided with insulation suitable for the applicable voltage and type of service.

4.2.9.2
Current-carrying metal parts of a luminaire (except lamp leads) shall be adequately insulated from non-current-carrying metal parts.

4.2.9.3
Strain insulators having a dry flashover voltage rating of at least double the voltage-to-ground of the circuit and a wet flashover voltage rating equal to the highest voltage between any two conductors shall be inserted in the span wires supporting luminaires. Such insulators shall be located not less than 1.8 m from the structure.

4.2.9.4
Where luminaires are wired internally, the conductors used shall be provided with insulation for the applicable voltage, temperature, and type of service.

4.2.10 Traffic lights
Traffic lights and associated brackets shall satisfy the requirements for luminaires specified in this Standard.

4.2.11 Trolley span wires and brackets
A strain insulator shall be inserted in each trolley span wire and, in the case of trolley brackets, an insulator shall be placed on each side of the trolley contact conductor support. These insulators shall have a wet flashover voltage rating of at least twice the voltage of the trolley circuit. Such insulators shall be placed not less than 1.8 m from the structure.

Where trolley feeders are to be attached to a structure below communication attachments, the trolley feeders shall be placed on a crossarm.

4.3 Overhead conductors

4.3.1 Identification
Conductors, cables, and equivalent longitudinal strands of supply and communication lines shall be installed in such a manner as to facilitate identification by qualified persons. This may be achieved by one or more means, including

a) occupation of definite relative levels throughout the line;
b) attachment of distinctive insulators or crossarms;
c) consistent standards of construction; and
d) marking or numbering.

4.3.2 Common neutral
Primary and secondary supply-line neutrals may use a single conductor as a common neutral, provided that such a conductor meets the requirements of a multi-grounded neutral.

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4.3.3 All-dielectric self-supporting (ADSS) fibre optic cables

An all-dielectric self-supporting fibre optic cable is normally considered a communication cable with respect to separations and clearances. Reduced separations and clearances may be considered where the issues of ADSS cable jacket dielectric strength, worker qualifications, and public safety have been addressed in accordance with Clause 1.7.

4.3.4 Optical groundwire (OPGW)

An OPGW, where installed as a supply transmission-line lightning protection wire, shall be considered a part of the supply system with respect to clearance, separation, loading, and strength requirements. Special electrical protection measures can be warranted when the OPGW is routed from the supply system space to a communication space or other space.

4.4 Coordinated electrical protection

Where supply and communication lines are located in sufficiently close proximity that electrical contacts are possible, electrical protection shall be applied in accordance with CSA C22.3 No. 5.1. Where electrical protection is not possible, stronger grades of construction shall apply (see Clause 6).

4.5 Inductive coordination

4.5.1 Supply and communication circuits

Supply and communication circuits and their connected apparatus shall be designed, constructed, operated, and maintained with due regard to avoiding or minimizing interference to the service provided by the communication circuits and hazards to persons using, operating, or maintaining the communication circuits. Where excessive inductive interference or induced voltages are anticipated or experienced, the methods of coordination specified in CAN/CSA-C22.3 No. 3 shall be applied.

4.5.2 Other wire facilities

If other wire facilities exist where supply parallels are proposed (e.g., those associated with ski tows or aerial tramways), an assessment shall be performed to determine and address the possible hazards of electrostatic or electromagnetic induction.

4.6 Special rules for communication lines

4.6.1 Communication lines (including supply circuits used exclusively for the operation of communication circuits)

4.6.1.1

Lines shall have grades of construction and clearances in accordance with the requirements for communication lines as specified in this Standard, provided that they meet the voltage and power requirements specified in Clause 4.6.1.2 or the conditions specified in Clause 4.6.1.3. Otherwise, they shall meet the requirements specified for supply lines of the applicable voltage.

In the case of interconnection of communication lines that are owned by separate utilities, all parties shall conform to the requirements of this Standard concerning voltage and power limitations.

Note: *Communication lines include those for telephone, telegraph, railway signal, messenger call, clock, fire, police alarm, community antenna, and other systems.*

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4.6.1.2

Where open wire or non-metallic sheathed cable is used, the communication circuits or supply circuits used exclusively for the operation of communication circuits shall operate at voltages not exceeding 150 V line-to-ground or 300 V between any two points of the circuit, and the transmitted power shall not exceed 150 W under normal operating conditions or fault conditions.

4.6.1.3

Where metallic sheathed cable is used, the voltage and power limitations specified in Clause 4.6.1.2 shall not apply, provided that

a) the metallic sheath is continuous and effectively grounded;
b) the metallic sheath is capable of carrying, without damage, the maximum current that could develop from a fault within the cable;
c) supply circuits in such cables are promptly de-energized in the case of a cable fault;
d) supply circuits in such cables are terminated at points accessible only to qualified persons;
e) communication circuits brought out of such cables and not terminating at points only accessible to qualified persons are protected and arranged such that, in the event of failure within the cable, the voltage on these circuits does not exceed 300 V line-to-ground and is promptly removed; and
f) all circuits in such cables or sections of cable are operated by one party and are accessible only to qualified persons.

4.6.1.4

Communication cable messengers shall be effectively grounded to protect customers in the case of accidental contact between supply conductors and communication facilities. The metal sheaths or shields of cables shall be bonded to the messenger to avoid voltage differences due to lightning surges or supply contacts.

The communication messenger shall be electrically continuous throughout the length of the facility. Where there are several communication messengers on the same pole line, they shall be bonded together.

Lateral messengers shall be bonded to the main cable route messenger. When a communication drop facility has a messenger or supporting conductor, it shall be bonded to the messenger on the main communication facility.

The thermal rating of the messenger and bonding shall be sufficient to withstand induced circulating currents and the induced currents under short-circuit conditions.

4.6.2 Communication circuits and communication equipment used exclusively in the operation of supply lines

4.6.2.1 General

Communication circuits and communications equipment used exclusively in the operation of supply lines or operation of an electrical power distribution system may be run as ordinary communication circuits or supply circuits under the conditions specified in Clauses 4.6.2.3 and 4.6.2.4. Any section of the communication system that is isolated or separated by transformers shall use a consistent type of construction throughout the isolated or separated section.

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4.6.2.2 Guarding

Communication circuits used in the operation of supply lines shall be isolated by elevation or otherwise guarded at all points accessible to the public.

4.6.2.3 Communication line construction

Communication circuits used in the operation of supply lines may be run as ordinary communication conductors, provided that

a) such circuits comply with Clause 4.6.1.2 or 4.6.1.3;
b) such circuits occupy the position indicated for normal level arrangement;
c) such circuits meet the requirements for clearance from supply circuits, as specified in this Standard, at crossings, proximities, or on jointly used structures; and
d) adequate protective measures are applied to prevent the communication circuit from exceeding 300 V line-to-ground under normal low-frequency induction conditions.

4.6.2.4 Supply line construction

Communication circuits and communication equipment used exclusively in the operation of supply lines or operation of an electrical power distribution system shall be classified as supply circuits where they do not comply with Clause 4.6.2.3 and, in the case of an electrical power distribution system, are attached directly to any supply conductor or to the structure between the primary and secondary/ neutral supply conductors. These circuits shall comply with all requirements for the supply lines with which they are used.

4.7 Changes in the vicinity of a line

Where a line exists (or is under construction), no changes should be made in its vicinity such that any clearance and separation with respect to that line will be reduced below the minimum clearance and separation required by this Standard or pertinent regulations. Reduction below required minimum clearances and separations can be avoided by using one or both of the following precautions:

a) any desired changes are planned or carried out such that the line's clearances and separations will not be affected; and
b) any necessary alterations to the line are arranged in advance by agreement with the owner of the line.

Notes:

1) *Changes that can affect the clearances and separations of a line include the following:*
   a) *construction or alteration of any structure or other line in the vicinity;*
   b) *changes in the elevation or surface condition of the land under or near the line; and*
   c) *changes in the use or accessibility, or both, of the land under and near the line.*

2) *When a line is being designed or constructed in a location where there are definite and specific plans for other construction or changes in the topography, and these changes are made known to the utility, sufficient additional clearance or separation should be provided, by mutual consent of the parties involved, to accommodate such construction or changes, where practicable.*

5 Clearances, separations, and spacings

5.1 Scope

Clause 5 specifies clearances, separations, and spacings involving overhead line components, and their relationship to each other, to the ground, and to other plant. The clearances, separations, and spacings specified in Clause 5 are the basic values required for public safety and are not intended to address the

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minimum approach distance to electrical installations as specified in occupational health and safety regulations.

5.2 General application

5.2.1 Construction and day-to-day clearances

The clearances specified in Clause 5 for wires and conductors are minimum values related to maximum specified loads and service conditions and represent design limits rather than clearances for construction or day-to-day operation. Clearances under day-to-day conditions are greater than the minimum clearances specified in Clause 5 when loads and service conditions are less severe than specified maximum conditions. Clearances provided at the time of construction shall by design be sufficiently greater than the minimum clearances specified in Clause 5 to ensure that the actual clearances under maximum specified loads and service conditions meet minimum clearance requirements.

Note: *Clearances specified in the* Canadian Electrical Code, Part I*, apply at the time of installation rather than under specified maximum conditions and are therefore larger than those specified in the* Canadian Electrical Code, Part III*, for the reasons given in this Clause.*

5.2.2 Vertical design clearances

Unless otherwise specified, vertical clearances shall apply under conditions of maximum sag of the wires, conductors, or cables, whether thermally loaded (see Clause 5.2.5) or physically loaded under wind and/or ice (see Clause 7). The calculation of final maximum thermal sag for a wire, conductor, or cable shall be based on its final unloaded ambient air sag condition. The calculation of sags and tensions shall consider the non-linear stress-strain characteristics of the wire, conductor, or cable. (See Clause A.7.3.6 and Annex B).

5.2.3 Horizontal design clearances

Horizontal clearances shall apply under the swing conditions specified in Clause 5.2.7.

5.2.4 Conductor classification for clearances

Supply cables insulated for the applicable voltage and having a continuous metallic sheath that is effectively grounded and capable of carrying the full fault current available at the applicable voltage may be classified as bare 0 to 750 V line-to-ground conductors. All other supply cables or assemblies shall have the same clearances as open conductors operating at the applicable voltage.

5.2.5 Conductor temperature for thermal loading conditions

5.2.5.1

For wires or conductors carrying currents less than 1/3 of their rated ampacity, the design conductor temperature for the thermal loading condition shall be 50 °C.

Note: *See the definition of rated ampacity in Clause 3.*

5.2.5.2

For wires or conductors carrying currents greater than 1/3 of, but not exceeding, their rated ampacity, the design conductor temperature for the thermal loading condition shall be

a) calculated in accordance with ANSI/IEEE 738 for the anticipated worst-case conditions; or
b) 100 °C for bare conductors and 80 °C for covered conductors.

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5.2.5.3

For wires or conductors carrying currents greater than their rated ampacity, the design conductor temperature for the thermal loading condition shall be calculated in accordance with ANSI/IEEE 738 for the anticipated worst-case conditions.

5.2.6 Neutral conductors and lightning protection wires

Neutral conductors and lightning protection wires associated with supply circuits shall have the same clearances as the phase wires of the circuit with which they are associated. However, neutral conductors that are effectively grounded throughout their length and associated with circuits whose voltage does not exceed 26 kV, and lightning protection wires that are effectively grounded, may have the same clearances as 0 to 750 V conductors, except where lesser clearances are permitted by this Standard.

5.2.7 Wire or conductor swing for horizontal design clearances

Where specified horizontal clearances are required by this Standard to include an allowance for the swing of conductors, the deviation of the conductor shall be calculated as follows:

a) In exposed and open terrain, the horizontal deviation of the wire or conductor from its position at rest shall be as specified in Table 1.
b) In locations where the span of wire or conductor is partly sheltered from the full effect of transverse winds by buildings or other construction, as in built-up commercial or residential areas, the horizontal deviation of the wire or conductor from its position at rest shall be as specified in Table 1.
c) Where wires or conductors are fully sheltered from transverse wind loads that could cause them to swing, such as along narrow urban streets or lanes and between buildings that reach at least the height of the conductor or wire on both sides, no allowance is required for horizontal deviation.
d) For the purpose of determining wire or conductor swing, the sag at the point under consideration shall be taken as the final unloaded sag at a conductor temperature of 40 °C or, alternatively, the 2.5% July design temperature (as specified in the *National Building Code of Canada*) applicable to the area under consideration. The 2.5% July design temperature is the temperature at or above which 2.5% of the July hourly ambient temperature occurs.
e) Where suspension insulators that are not restrained from swinging transversely are used, the length of the insulator string shall be added to the conductor sag for the purpose of determining the sag used in Table 1.

5.2.8 Insulator string swing for clearance calculations — Transmission line structures

Where suspension insulators that are not restrained from swinging transversely are used, sufficient clearances between the live end of an insulator string in a swing-out position and the supporting structure shall be maintained. The following conditions shall be considered to determine the required structure head geometry to accommodate swinging insulator string:

a) high wind with power frequency voltage — operational requirement;
b) moderate wind with switching surge overvoltage — operational requirement; and
c) low wind with switching surge overvoltage — live-line maintenance requirement (where applicable).

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5.3 Vertical design clearances and separations

5.3.1 Vertical design clearances of wires and conductors above ground or rails

5.3.1.1 Basic clearances

The minimum vertical clearances of wires and conductors above ground or rails shall be as specified in Clause 5.2 and Tables 2 and 4, except that

a) the clearances over roadways or other areas where vehicles are expected to be used are based on a combined vehicle and load height of 4.15 m. For provinces and territories that permit the combined vehicle and load height to exceed 4.15 m, the applicable clearance specified in Tables 2 and 4 shall be increased by the amount by which the allowable combined vehicle and load height exceeds 4.15 m;

b) for altitudes exceeding 1000 m and where voltages exceed 50 kV, the clearances specified in Table 2 shall be increased by 1% for each 100 m increase in excess of 1000 m above mean sea level;

c) because the rail level of a railway where ballast is used is not fixed, when any line that crosses a railway is constructed or altered, an additional 0.3 m of vertical clearance above rails shall be provided, unless a lesser amount is mutually agreed upon, to permit normal subsequent ballast adjustments without encroaching on the specified minimum clearance;

d) when a line that crosses or will cross any public thoroughfare likely to be travelled by road vehicles is constructed or altered, an additional 0.225 m of vertical clearance shall be provided to permit the road surface to be raised by this amount during subsequent road work operations without encroaching on the specified minimum clearance; and

e) where snow is not cleared away, additional clearance equal to the mean annual maximum snow depth shall be provided, making due allowances for maximum conductor sag at prevailing temperatures and compression of snow due to vehicle travel.

Note: *See Table D.1 for information on mean annual maximum snow depth.*

5.3.1.2 Special equipment and operations

Specified vertical clearances above ground or rails do not normally provide for the operation of special equipment such as cranes or other aerial devices used for construction or maintenance underneath the conductors, nor for operations that are carried out in special locations such as railway yards, materials handling and storage yards, truck loading or unloading areas, farm feed lots, and similar premises. At locations where special operations are frequent, suitable additional clearance beyond the minimum specified in this Standard might have to be provided as mutually agreed upon between the parties concerned.

5.3.2 Vertical separations (heights) of supply equipment from ground

5.3.2.1

Minimum vertical separations from ground for supply equipment such as cable terminals, arresters and line switches, transformers, regulators, and capacitors shall be as specified in Table 5.

5.3.2.2

There is no vertical separation requirement for communication terminals, supply switch handles, and control, police, and fire-alarm boxes, provided that they are effectively grounded and do not obstruct pedestrian traffic.

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5.3.3 Clearances over waterways

5.3.3.1 General

Clause 5.3.3 specifies requirements for vertical clearances of wires and conductors above navigable waterways, including commercial, recreational, and secondary waterways.

Notes:

1) *Navigable waters are governed by the* Navigation Protection Act *under the authority of Transport Canada. Crossings over navigable waterways are subject to the provisions of the* Navigation Protection Act *and might therefore require approval by Transport Canada through regional offices.*

2) *Obstruction markings of wires, conductors, and structures are not covered by this Standard. The Civil Aviation Branch of Transport Canada should be consulted to determine aerial marking requirements. Transport Canada Standard 621.19 provides guidance in this area.*

5.3.3.2 Basic vertical clearances

The minimum vertical clearances of wires and conductors above waterways shall be as specified in Table 3, except that

a) for areas where vessel heights are known to exceed the reference vessel height at the time of design, the clearance specified in Table 3 shall be increased by the difference between the taller vessel and the reference vessel height specified in Table 3; and

b) for elevations above 1000 m and where voltages exceed 50 kV, the clearances specified in Table 3 shall be increased by 1% for each 100 m or part thereof in excess of 1000 m above mean sea level.

5.3.3.3 Determining minimum vertical clearances

When determining minimum vertical clearance, the sag of the wire or conductor shall be calculated in accordance with Clause 5.2.2, except in the case of uncharted waterways that are used exclusively for recreational navigation, for which the vertical clearance shall be based on the maximum sag of the thermally loaded conductor above the reference water level. The reference water level used shall be the ordinary high-water level (OHWL).

OHWL elevations shall be determined by means of establishing survey control points tied into the existing Geodetic Survey of Canada or Canadian Hydrographic Service benchmarks.

Note: *The OHWL elevation and the data used to determine it should be recorded and included in the application for approval.*

5.3.4 Maximum height of wires and equipment above ground in the vicinity of airports

The heights of wires, supporting structures, and equipment on land adjacent to, surrounding, or near an airport in Canada are subject to the regulations of Transport Canada and are not covered by this Standard. Transport Canada TP 1247 provides guidance in this area.

5.4 Horizontal design clearances of wires and conductors from railway tracks

5.4.1

Where wires and conductors are along a railway track or tangential to a curved track and where, under maximum sag conditions, the wires and conductors provide less than the minimum vertical clearance above rails required by Clause 5.3.1, minimum horizontal clearances shall be provided in accordance with Clauses 5.4.2 and 5.4.3. This minimum horizontal clearance shall be the distance between the vertical projection of the inside edge of the top of the nearest rail and the nearest position of the wire under the conditions of swing specified in Clause 5.2.7.

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5.4.2
Where there is no curvature of railway tracks, the horizontal clearances shall be as specified in Table 6.

5.4.3
At a point of curvature of a railway track, the horizontal clearances shall be as specified in Table 6, with an increment of 25 mm added for each degree of curvature. In addition, where the wire is closer to the low side of tracks that are at different elevations, a further increment of 2.5 mm shall be added for each millimetre of superelevation. (The total of these two increments shall not exceed 0.75 m, and the value of 0.75 m may be used in place of calculations.)

5.5 Horizontal separations of supporting structures from railway tracks
Note: *Side separations as specified in Clause 5.5 are not customary construction separations but minimum separations for circumstances where greater separations cannot be achieved. To allow working space for normal maintenance operations, railways do not ordinarily permit poles, guys, etc., to be placed on the railway right-of-way less than 9 m from the track centreline (where space permits). On narrower rights-of-way, structures and guys should be placed as far as possible from the centreline of the nearest track.*

5.5.1
Any part of a supporting structure not complying with the vertical clearance requirements of Clause 5.3.1 shall be provided with minimum horizontal separations from railway tracks in accordance with Clauses 5.5.2 to 5.5.5. The minimum horizontal separation shall be the distance between the vertical projection of the inside edge of the top of the nearest rail and the nearest position of the structure.

5.5.2
Where there is no curvature of the railway tracks, the minimum horizontal separation between the tracks and supporting structures shall be as specified in Table 7, but shall be increased, where applicable, to comply with Clauses 5.5.4 and 5.5.5.

5.5.3
At a point of curvature of the railway tracks, the minimum horizontal separation shall be that specified in Table 7, plus the increment required by Clause 5.4.3. This minimum horizontal separation shall be increased, where applicable, to comply with Clauses 5.5.4 and 5.5.5.

5.5.4
At loading sidings, sufficient space shall be left for a driveway, in accordance with the stated needs of the railway.

5.5.5
In all cases, horizontal separations shall be great enough to permit an unobstructed view of signals, signs, and similar equipment.

5.6 Horizontal separations of supporting structures from fire hydrants, street corners, curbs, and other buried services

5.6.1 Horizontal separation from fire hydrants
Supporting structures, including guys or attachments, that are less than 2 m above the top of a fire hydrant shall have a minimum horizontal separation from the hydrant of 1 m.

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5.6.2 Horizontal separation from street corners

Supporting structures, including guys, shall be located as far as practicable from the beginning of curvature of the roadway.

5.6.3 Horizontal separation from curbs

5.6.3.1

Structures shall be located at least 150 mm from the edge of a curb, measured away from the travelled portion of the roadway.

5.6.3.2

Where crossarms and other attachments such as guys provide less than the minimum vertical separation above roads required for guys in Clause 5.3.1, such attachments shall have the minimum horizontal separation specified in Clause 5.6.3.1.

5.6.4 Horizontal separation from other buried services

A minimum separation of 0.6 m shall be maintained between the edge of a pole and the nearest face of other buried services in order to permit pole replacement without damage to other services.

5.7 Clearances of wires, conductors, and equipment from buildings, signs, bridges, swimming pools, and similar plant

5.7.1 General

Wires and cables shall not interfere with the normal use of balconies, doors, fire escapes, windows, permanent ladders, catwalks, etc.

5.7.2 Clearances of supply conductors permanently attached to buildings

5.7.2.1

Where the permanent attachment of supply conductors to buildings is necessary (e.g., for a service conductor), the minimum clearance of conductors from the surface of buildings as specified in Table 8 shall apply under the following conditions:

a) vertical clearances shall apply under conditions of maximum sag of the conductor; and
b) horizontal clearances shall apply where the conductor is in the position of swing, as determined in accordance with Clause 5.2.7.

5.7.2.2

The point of attachment for supply conductors shall be on the same side of the building as the consumer’s service head and shall be located such that

a) the point of emergence of consumer’s service conductors from the consumer's service head or the equivalent is a minimum of 150 mm and a maximum of 300 mm above the support for attachment of the overhead service conductors; and
b) a minimum of 150 mm and a maximum of 300 mm horizontally from the consumer's service head.

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5.7.3 Clearances of wires and conductors passing by or over buildings, signs, billboards, lamps, traffic signs, standards, and antennas (not attached)

5.7.3.1 Basic clearances

The minimum horizontal and vertical clearances from wires and conductors to buildings, signs, and similar plant specified in Table 9 shall apply under the following conditions:

a) clearances shall apply to any part of a building, including balconies, fire escapes, antennas and their supporting structures, and other permanent fixtures;
b) vertical clearances shall apply under conditions of maximum sag of the wire or conductor;
c) horizontal clearances shall apply when the wire or conductor is in the position of swing, as determined in accordance with Clause 5.2.7; and
d) guy wires, communication cables, and drop wires shall not be allowed to rub against buildings or other plant.

5.7.3.2 Application of clearances

Note: *In this Clause, the word "building" is used to refer to any of the plant specified in Clause 5.7.3.1.*

The minimum required vertical clearance at any given point on a building shall apply when the wire or conductor is above that point and closer horizontally than the specified minimum horizontal clearance.

The minimum required horizontal clearance shall apply when the wire or conductor is closer vertically to the point in question than the specified minimum vertical clearance.

The vertical clearance from a wire passing under a building projection to the nearest point of that projection shall be not less than the horizontal clearance requirement.

Furthermore, the minimum clearances specified in this Clause shall apply to all points on the wire or conductor and all points on the building.

Where buildings contain flammable materials or explosives, the need for additional clearances shall be investigated.

When clearances as specified in Table 9 are applied, allowance shall be made for the swing of the sign or similar plant.

5.7.3.3 Conductors energized above 200 kV

When buildings are located adjacent to a transmission corridor with lines operating above 200 kV, a study shall be conducted to determine suitable clearances between conductors and buildings greater than those specified in Table 9, taking into account electrostatic induction. In these situations, additional measures might be necessary to mitigate the effects of electrostatic induction on the building, its occupants, or construction and maintenance personnel. Measures that may be taken include

a) change in conductor height;
b) increased right-of-way width;
c) installation of shielding between the transmission line and buildings constructed at the edge of the right-of-way;
d) modification of the building configuration; or
e) grounding of metallic objects.

Note: *The list in this Clause is not comprehensive.*

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5.7.4 Clearances from supply wires and conductors to bridges

5.7.4.1 General

Clause 5.7.4 specifies clearances for supply wires and conductors that are attached to or pass under, over, or near a bridge. Clause 5.7.4 does not cover clearances over a bridge used solely for a railway or over portions of a bridge normally traversed by pedestrians, road vehicles, or trains. Clearances over or adjacent to portions of the bridge that are normally traversed by pedestrians, road vehicles, or trains are specified in Tables 2 and 4.

5.7.4.2 Basic clearances for supply conductors

The minimum clearance from supply conductors to bridges as specified in Table 10 shall apply under the following conditions:

a) vertical clearances shall apply under conditions of maximum sag for a conductor above the portion of the bridge involved, and shall apply to the initial sag at the lowest anticipated temperature for the area for a conductor under the portion of the bridge involved; and
b) horizontal clearances shall apply when a wire is in the position of swing, as determined in accordance with Clause 5.2.7.

5.7.4.3 Operation of bridges

Wires or cables shall not interfere with the use or operation of a bridge.

5.7.4.4 Guarding of trolley contact conductors

Where the trolley contact conductor is located such that a trolley pole leaving the conductor could make simultaneous contact between the conductor and the bridge structure, suitable guards shall be provided. Guards shall be substantial inverted troughs of non-conducting material located above the contact conductor or other suitable means of preventing contact between the trolley pole and the bridge structure.

5.7.5 Minimum separation of equipment and clearances of conductors over and adjacent to swimming pools

Conductors and equipment shall not be located over or adjacent to a swimming pool unless it is not practical to locate them elsewhere. If equipment and conductors need to pass over a swimming pool or the area surrounding a pool, the minimum clearance of conductors and separation of equipment from swimming pools shall be as specified in Table 11 and as shown in Figure 1.

Conductor sags shall be determined in accordance with Clauses 5.2.5 and 5.2.7 for vertical and swing conditions.

5.7.6 Horizontal separations from supply equipment to buildings

The minimum horizontal separations from supply equipment to buildings shall be as specified in Table 12. The supply equipment shall not inhibit the normal use of balconies, doors, windows, fire escapes, permanent ladders, catwalks, and other such structures.

5.7.7 Clearances from propane tanks

In order to conform to CSA B149.2, clearances from propane tanks shall be as follows:

a) supply lines shall not be installed over propane tanks of an aggregate capacity of 7600 L or greater;
b) overhead supply conductors less than or equal to 22 kV shall have a horizontal clearance at rest of 7.6 m from propane tanks that have an aggregate capacity of 7600 L or greater; and

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c) overhead supply conductors greater than 22 kV shall have a horizontal clearance at rest of 15 m from propane tanks that have an aggregate capacity of 7600 L or greater.

5.7.8 Clearances from other flammable hazards

All potentially hazardous areas, including gasoline filling stations, storage tanks, fuel pumping and regulator stations, or above-ground gas pipelines, shall be reviewed and suitable safety measures shall be agreed upon by the parties involved. Construction of lines over such installations shall be avoided where practicable. Where overhead lines need to be used for such installations, Grade 1 construction shall be used.

Note: *Devices that can emit sparks or glowing embers, such as fuses and arresters, should not be located on poles adjacent to fuelling stations.*

5.7.9 Clearances for other wire facilities

The minimum clearance between an aerial tramway, ski tow, T-bar, chairlift, or similar equipment and an overhead power line greater than 750 V, measured horizontally between the vertical planes passing through the closest points of the equipment and the overhead power line, shall be 15 m.

Δ

5.7.10 Clearance from irrigation systems — Lines operating at >66kV

Clearances shall be maintained between overhead supply conductors and irrigation systems to address the risk of flashover between the energized supply conductors and the irrigation system. The required distance will depend on the voltage of the conductors, the conductivity of the water spray being produced by the irrigation system, and whether or not water from the irrigation system is allowed to contact energized conductors. Clause A.5.7.10 provides information from several sources on the required clearance from irrigation systems to energized conductors.

In some jurisdictions, the clearance requirements and interaction of irrigation systems and supply systems is governed by provincial or other regulatory authorities.

For supply lines operating at >66 kV phase-to-phase, located parallel to and within 30 m of long irrigation systems such as wheel move or pivot systems, induction from the power line can create a hazard for persons touching the pipe of the irrigation system. In these situations, the owner of the transmission line shall conduct a study to determine the current and voltage induced on the irrigation system, and shall work with the owner of the irrigation system to install suitable grounding or other mitigation.

5.8 Clearances between wires and conductors of one line and wires, conductors, and structures of another line

5.8.1 Vertical clearances at crossings of line wires and conductors supported by different structures

5.8.1.1 Basic clearances

The minimum vertical clearance between wires or conductors crossing each other and on different structures shall be as specified in Table 13. Table 14 covers the clearances for crossings over aerial tramways, including gondolas, T-bars, and chairlifts.

The clearances specified in Tables 13 and 14 shall apply under the following conditions:

a) the upper wire or conductor is in its maximum sag position; and

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b) the lower wire or conductor is assumed to form a straight line between its points of support.

Note: *This straight line is called the "line of sight" of the lower line.*

5.8.1.2 Reduced clearances

Where the ice- and wind-loaded sag of the lower wire or conductor exceeds 6 m at the point where it crosses under the upper wire or conductor, the clearance in Table 13 may be reduced by one-half the difference between the ice- and wind-loaded sag and 6 m. Particular attention shall be paid to the possibility of sudden ice drop-off, which could cause the lower conductor to jump upwards.

5.8.1.3 In-span vertical clearances between supply and communication conductors

Where supply conductors between 0.75 and 22 kV cross communication conductors, a common structure at the point of crossing shall be provided, where practicable, to which the supply and communication conductors shall be attached. Where coordinated electrical protection cannot be achieved, a higher grade of construction shall be used.

5.8.2 Clearances between conductors supported by different structures but not crossing each other

Where conductors are supported by different structures but do not cross each other, they shall have

a) a minimum vertical clearance in accordance with Clause 5.8.1; or
b) minimum horizontal clearances with the conductors in the at-rest position that are equal to the sum of the calculated conductor displacement due to a transverse conductor swing plus the appropriate increment from Table 15. This conductor displacement shall be calculated in accordance with the following:
   i) where the conductor of one line is free to swing so that it approaches the conductor of a second line at a point where this conductor is not free to move (e.g., a structure of one line near the midspan position of the second line), the conductor displacement at midspan shall be calculated in accordance with Clause 5.2.7; and
   ii) where lines run alongside each other and the adjacent conductors are free to approach one another, the conductor of the line having the greater sag shall be considered to be displaced one-half the distance calculated in accordance with Clause 5.2.7, and the conductor of the second line shall be considered at rest.

5.8.3 Clearances in any direction between conductors of one line and supporting structures of another line

5.8.3.1 Basic clearances

The minimum clearances in any direction between conductors of one line and supporting structures of another line, where conductors of the first line are not attached to the supporting structure of the second line, shall be as specified in Table 16. The clearances shall apply for the more applicable of the following conditions:

a) the conductor is at its maximum sag condition; and
b) the conductor is in a position of swing, as specified in Clause 5.2.7.

5.8.3.2 Climbing space and working space

The climbing space (see Clause 4.1.4) and working space (see Clause 4.1.5) on a structure of one line shall not be reduced by a conductor of another line.

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5.8.3.3 Joint-use common structure

Where communication circuits are in joint use with supply conductors rated up to 50 kV, midspan structures that support only the communication conductors shall be avoided. To ensure adequate clearances are maintained, the supply conductors and communication facilities shall be attached to the common structure.

5.8.4 Conductors energized at 230 kV phase-to-phase or greater

When overhead or underground low-voltage or communication wiring is to be located under a transmission line or adjacent to a transmission corridor, with the transmission lines operating at 230 kV (phase-to-phase) or greater, an electromagnetic induction study shall be conducted to determine suitable clearances between transmission conductors and low-voltage wiring, less than or equal to 22 kV (line-to-ground). In specific situations, measures might be necessary to mitigate the effects of electromagnetic induction on low-voltage and communication wiring for the safety of workers and/or building occupants and the supply of undisturbed regulated low-voltage power supply. Measures that may be taken include

a) avoiding installation of low-voltage or communication wires parallel to high-voltage lines;
b) installing low-voltage or communication wires perpendicular to high-voltage lines;
c) changing low-voltage or communication feeds to multiple sources;
d) increasing plant's supply voltage from nominal 110 V; or
e) increasing separation between low-voltage or communication wires and high-voltage lines.

5.9 Separations of supply-line conductors and conductor supports on the same supporting structure

5.9.1 Horizontal separations or separations within 45° of the horizontal of supply-line conductors on the same supporting structure

5.9.1.1

The minimum separations between supply-line conductors of the same circuit or different circuits that are at the same level or within 45° of the horizontal on the same supporting structure shall be as specified in Table 17.

5.9.1.2

When applying the separations specified in Table 17, voltages shall be

a) for dc, the voltage differences between the conductors involved;
b) for ac conductors in the same circuit, normal operating phase-to-phase voltages; and
c) for ac conductors in two different circuits, the sum of the normal operating phase-to-ground voltages of both circuits.

5.9.1.3

Where suspension insulators are used and are not restrained from swinging transversely, the sags specified in Table 17 shall include the length of the insulator string.

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5.9.2 Vertical separations and clearances of supply-line conductors attached to the same supporting structure

5.9.2.1 Minimum vertical separations between supply-line conductors attached to the same supporting structure

Where supply-line conductors of the same or different voltage classifications are supported by the same structure, but at different levels, the minimum vertical separations shall be as specified in Table 18. Where galloping or ice drop-off are likely to occur, increased separation should be considered to ensure adequate in-span clearance.

5.9.2.2 Vertical spacings between supply-line conductors of 0 to 750 V on vertical brackets

The average vertical spacings between supply-line conductors of 0 to 750 V that are supported on vertical brackets shall be not less than those specified in Table 19 and shall apply under the following conditions:

a) conductors shall be erected and maintained in a manner that prevents in-span contact under normal operating conditions; and
b) the minimum vertical spacing between the top conductors supported on the vertical bracket and the centre of the nearest crossarm above shall be 600 mm.

5.9.2.3 In-span vertical clearances between supply-line conductors of the same circuit

The minimum in-span vertical clearance between supply-line conductors of the same circuit that are attached to the same supporting structure shall be as specified in Table 20, and shall apply

a) at maximum sag for conductors of the same voltage class;
b) at final unloaded sag at a 40 °C design conductor temperature for neutrals or conductors of 0 to 750 V, which are considered to be of the same circuit of a voltage class other than 0 to 750 V, and are attached below the other circuit conductors; and
c) where the maximum sag of the circuit conductors (other than 0 to 750 V) associated with the neutral (i.e., the circuit phase conductors) is determined by the ice-loading conditions, the conditions for the neutral or conductors of 0 to 750 V shall be the final unloaded sag at a 0 °C design conductor temperature.

5.9.2.4 In-span vertical clearances between supply-line conductors of different circuits

The minimum in-span vertical clearance between supply-line conductors of different circuits of the same or different voltage classes and attached to the same supporting structure shall be as specified in Table 21, and shall apply under the following conditions:

a) the upper conductor shall be assumed to be at maximum sag; and
b) the lower conductor shall be assumed to be at initial unloaded sag at the mean annual temperature.

5.9.3 Separations in any direction other than horizontal or vertical between supply-line conductors

The separations in any direction other than horizontal or vertical between supply-line conductors supported by the same supporting structure shall be not less than the greater of

a) the value for vertical separations between conductors specified in Clause 5.9.2; or
b) the value for horizontal separations between conductors specified in Clause 5.9.1.

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5.9.4 Clearances in any direction from communication-line conductors to guys, span wires, or grounding conductors carried in aerial spans and attached to the same supporting structure

The minimum clearance in any direction in the vicinity of a structure from communication-line conductors to guys, span wires, or grounding conductors carried in aerial spans shall be 75 mm.

5.9.5 Separations or clearances in any direction from supply conductors to other supply plant attached to the same supporting structure

The minimum separations or clearances in any direction between supply conductors and supply plant (e.g., vertical or lateral conductors, guys, and lightning protection wires) shall be as specified in Table 22. These separations or clearances shall apply under the following conditions:

a) The voltage specified in Table 22 for ac conductors shall be taken as
   i) voltage-to-ground for clearance between conductors and crossarms, structures, or span or guy wires;
   ii) phase-to-phase voltage for clearance between conductors of the same circuit; and
   iii) the sum of the voltages-to-ground for any two conductors of different circuits.
b) Voltage for dc conductors shall be taken as
   i) voltage-to-ground for clearance between a conductor and a supply, span, or guy wire; and
   ii) the sum of the voltages-to-ground for clearance between any two conductors.
c) Where suspension insulators that are not restrained from swinging transversely are used, the clearances shall apply with one conductor at rest and the other conductor in the position of swing as determined in accordance with Clause 5.2.7.

5.10 Joint-use clearances and separations — Supply and communication plant

5.10.1 Vertical separations at the structure — Normal level arrangement

5.10.1.1

The minimum vertical separations at the structure and within the working space shall be in accordance with Table 23. Greater separation might be necessary in order to meet the clearances in the span specified by Clause 5.10.3.

5.10.1.2

Luminaires and associated brackets shall be effectively grounded unless they are located more than 1 m above the communication plant. Supply cables and wires associated with the luminaire shall be insulated and protected by a covering that provides suitable mechanical protection, unless they are located more than 1 m above the communication plant.

5.10.1.3

Switch handles shall be separated by at least 100 mm from the communication plant. The metal frames and operating rods in the vicinity of the joint-use space shall be effectively grounded or shall be insulated above the communication working space.

5.10.1.4

Supply service to the communication power supply shall be provided by an effectively grounded, continuous metal-sheathed cable or enclosed in a conduit.

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5.10.1.5
Common structure crossings shall be classified as joint use.

5.10.1.6
Option A of Table 23 shall be used except where the conditions of Clause 5.10.1.7 are met.

5.10.1.7
Option B of Table 23 may be used, provided that
a) work on the communication plant is carried out without the use of a platform and/or a tent; and
b) communication personnel are restricted to positions where only the head and shoulders project above the communication plant.

5.10.2 Vertical separations at the structure — Inverted level arrangement

5.10.2.1 General
The minimum vertical separations at the structure and within the working space shall be in accordance with Clause 5.10.2.2.

5.10.2.2 Inverted level arrangement
For inverted level arrangement, the minimum vertical separation between the uppermost attachment of supply plant of 0 to 750 V and the lowest attachment of communication plant shall be at least 1200 mm, and preferably 1500 mm. Inverted level arrangement with supply plant of 0.75 to 22 kV should be avoided; however, where it needs to be used, the minimum separation shall be 1500 mm. Inverted level arrangement with supply plant of greater than 22 kV should not be used.

5.10.2.3 Wireless communication antenna
For a wireless communication antenna installed on a joint-use structure above supply plant, the minimum separation between the antenna and the supply plant shall be
a) 1200 mm for supply plant of 0 to 750 V (only secondary on pole);
b) 1500 mm for supply plant greater than 750 V and less than or equal to 22 kV; and
c) 2400 mm for supply plant greater than 22 kV and less than or equal to 50 kV.

Wireless communication attachments that do not comply with Clause 4.6.2.4 shall not be attached between primary and secondary/neutral supply conductors.

Installation of communication antennae on structures supporting supply plant greater than 50 kV shall be subject to agreement between the parties involved and the criteria of the owner of the supply plant.

5.10.3 In-span vertical clearances

5.10.3.1
The requirements of Clause 5.10.3 shall apply outside of the working space of the supporting structures.

Clause 5.10.3 shall not apply to installations covered by Clause 5.10.6.

5.10.3.2
The minimum in-span vertical clearances between supply conductors and communication wire or cable under the condition of maximum sag in the supply conductors shall be as specified in Table 24.

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5.10.3.3

For effectively grounded neutral conductors of multi-grounded neutral distribution circuits that do not exceed 26 kV and are located 300 mm or more below phase conductors,

a) on spans less than or equal to 75 m, minimum separations at the structure shall be adjusted such that the neutral does not fall below the line of sight of the highest communication conductor; and
b) on spans greater than 75 m, the common grounded neutral may sag below the line of sight of the communication attachments, provided that it is a minimum of 300 mm above the in-span communication conductors under
    i) maximum sag due to maximum temperature or to maximum winter loads in the supply circuit; and
    ii) sag at 45 °C for the communication circuit.

5.10.4 Vertical runs attached to surface of structure

5.10.4.1

The vertical run shall be separated from span or guy wires by at least 50 mm. Vertical runs enclosed in a moulding of adequate electrical insulating and mechanical properties do not require separation in addition to that provided by the moulding; however, the guy or span wire shall not abrade the moulding.

5.10.4.2

Vertical runs of communication and supply systems shall meet the requirements for risers specified in Clause 4.2.4.

5.10.5 Vertical runs not attached to surface of structure

Lamp leads or trolley feeder taps that run directly from a supply crossarm through or within 1 m vertically from the communication line plant shall be held taut at least 1 m from the surface of the structure, unless insulated. Where passing through the communication plant, these conductors shall have a minimum clearance of

a) 0.3 m beyond the end of communication crossarms;
b) 0.15 m from the communication drop wire; and
c) 0.5 m from the communication cable, where practicable.

Where the clearances of Items (a) to (c) cannot be achieved, conductors shall be provided with insulation for the applicable voltage and type of service.

5.10.6 Clearances and separations between drops and service conductors on buildings

5.10.6.1

Clause 5.10.6 applies where both drops and service conductors are attached to the exterior surface of a building or to an exterior support attached to a building (e.g., a service mast).

5.10.6.2

The minimum clearance or separation between drops and service conductors attached to buildings shall comply with the requirements of the *Canadian Electrical Code, Part I*, Rule 60-510. For the purpose of Clause 5.10.6, bare neutrals associated with insulated service wires may be constructed according to the clearance or separation specified for insulated conductors in this Standard.

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The service conductors and drops shall not come into contact under normal operating conditions. Supply service conductors and communication service drops shall have a minimum separation of 300 mm at any point in the span, including the point of attachment to the building, at the time of installation. The conductors shall be insulated and the clearance specified in Table 23 shall be maintained between the two services at the pole.

5.11 Guys and guy attachments

5.11.1 Guy attachments — Joint-use structures

5.11.1.1 Supply guy attachments

Supply guy attachments on joint-use structures shall be attached not less than 0.75 m above or below communication attachments. Where this separation is not possible, the largest practical separation shall be used.

5.11.1.2 Communication guy attachments

Communication guy attachments shall be located such that they do not restrict the working space and climbing space around the communication attachments.

5.11.2 Guys attached to joint-use structures

5.11.2.1 Clearances between guys and plant of another system

The clearance between a guy that is attached to a joint-use structure and plant of another system (not supported by the same joint-use structure) shall be as specified in Table 25. Guys shall not abrade one another.

5.11.2.2 Clearances or separations between guys and other plant attached to a joint-use structure

At any point, with the exception of attachments, the clearance or separation between a guy and other plant attached to a joint-use structure shall be as specified in Table 26. For vertical separations between attachments at joint-use structures, see Clause 5.11.1.

5.11.3 Guys attached to remote structures

Guys that extend from supply structures (not involved in joint use) to joint-use structures shall be constructed such that they will not create a hazard to workers on either structure. This may be accomplished by effectively grounding the guy or by placing one or more suitable strain insulators in the guy.

5.11.4 Guys attached above current-carrying supply plant

Where the guy attachment to a joint-use structure is located above any item of live or current-carrying supply plant, and where the guy passes, or would pass under failed guy conditions, within 1 m of the communication line plant or communication equipment (excluding communication guys), the guy shall be effectively grounded or insulated.

Where the guy cannot be grounded,

a) the portion of the guy having inadequate clearance from the communication line plant shall be electrically isolated with guy insulators; or
b) equivalent insulation shall be applied.

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6 Minimum grades of construction

6.1 General

For the purposes of strength requirements for communication and supply lines less than or equal to 69 kV nominal voltage (see Clause 8), lines shall be classified under the grades specified in this Clause on the basis of relative hazards. Grades of construction apply to structures, guys, crossarms, conductors (including cables and suspension strands), messengers, insulators, pins, and fastenings.

Where two or more conditions affect the grade of construction, the grade used shall be the strongest required under any of the conditions.

For supply lines at nominal voltages greater than or equal to 70 kV with no communication circuits in their proximity, the strength requirements of Clause 8 or of CSA C22.3 No. 60826 may be applied.

6.2 Order of grades

Grades of construction for supply and communication lines range from 1 through 3, with Grade 1 being the strongest.

6.3 Minimum grades of construction

6.3.1 General

Tables 27, 28, and 29 specify the minimum grades of construction for crossings, lines in proximity, and joint use, respectively.

6.3.2 Crossings

6.3.2.1

Table 27 specifies the minimum grades of construction for in-span crossings of supply and communication lines across highways and roads, private and public property, pipelines, waterways, railways, aerial tramways, and other communication and supply lines.

6.3.2.2

Common-structure crossings are classified as joint use. Span guys that form a crossing shall have the same grade and strength requirements as would the wire or cable attachments they support, if such attachments were to form the crossing themselves.

6.3.3 Proximities

Table 28 specifies the minimum grades of construction where structures or conductors of communication and supply lines are in proximity to any of the following: railways, highways and roads, pipelines, waterways, aerial tramways, other communication or supply circuits or lines, and other private or public property.

6.3.4 Joint use

Table 29 specifies the minimum grades of construction for joint use.

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6.4 Insulated cabled supply conductors

Insulated supply conductors supported on, and cabled together with, an effectively grounded supporting conductor shall have the same grade of construction as open supply conductors of the same voltage.

6.5 Multiple crossings

Where a line crosses in one span over two other lines, the grade of construction of the uppermost line shall be not less than the strongest grade that would be required if one of the lower lines crossed the other lower line.

7 Weather loads and assumed loads according to deterministic design methods

7.1 General

Supply and communication lines shall be designed using either deterministic design methods or reliability-based design methods. Clause 7 specifies the deterministic method for calculating the loads to be applied to the design. Clause 8 specifies the deterministic method for calculating the strength of the structures required to support these loads safely.

The reliability-based methods outlined in CSA C22.3 No. 60826 should be used for supply lines greater than 70 kV (phase-to-phase).

Clause 7 specifies the assumed weather loads to be applied to the design of wire and cable attachments and to supports, including poles, towers, crossarms, pins, posts, insulators, and fastenings. For thermal loads, see Clause 5.2.5.

7.2 Weather loads

This Standard recognizes the following four deterministic weather load conditions (see Table 30):

a) severe;
b) heavy;
c) medium loading A; and
d) medium loading B.

Loading classification shall be based on local experience and weather records. Annex C provides maps for guidance; the loads for the areas shown are considered the minimum, and local experience and information might permit the adjustment of these loads.

7.3 Assumed loads for wire and cable attachments

7.3.1

The assumed loads for wire and cable attachments shall be as follows:

a) the assumed vertical load shall be the load, in N/m, of ice-covered wire and cable attachments, using the radial thickness of ice specified in Table 30 and assuming the density of ice to be 900 kg/m^3; and
b) the assumed transverse load shall be the load, in N/m, created by the wind (in N/m^2, as specified in Table 30) acting horizontally on ice-covered wires and cable attachments.

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7.3.2
The assumed maximum tensions of wire and cable attachments shall be the tension determined by the vector resultant of the assumed vertical and transverse loads specified in Items (a) and (b), respectively, of Clause 7.3.1.

7.3.3
The assumed load for conductor splices and dead-end fittings shall be 1.6 times the assumed maximum tension of the conductor containing the splice or dead-end fitting. The tap shall be of a type that does not reduce the capability of the conductor to withstand 1.6 times the assumed maximum tension.

7.3.4
When calculating the loads on conductors, the coating of ice shall be considered to be a hollow cylinder around the outer circumference. For cables or cable bundles with or without suspension strand (excluding cable assemblies in which the conductors are spaced more than 50 mm apart), the maximum diameter of the assembly in any direction shall be used for calculating the wind and ice loads.

7.3.5
The values for ice, wind, and temperature specified in Table 30 shall be used to determine the assumed loads and maximum tensions.

7.3.6
When calculating the tension of a wire or cable, its non-linear characteristics of stress and strain shall be taken into account.

7.4 General requirements — Loads on supports

7.4.1
The loads specified in Clauses 7.5 to 7.9 shall be used for calculating the strength requirements for supports, including poles, towers, foundations, guys, anchors, crossarms, pins, posts, insulators, and fastenings. The loads are divided into five categories, as follows:

a) vertical loads: see Clause 7.5;
b) transverse loads: see Clause 7.6;
c) loads at angles: see Clause 7.7;
d) longitudinal loads: see Clause 7.8; and
e) loads for additional lines at angles to the main line: see Clause 7.9.

When calculating the loads to be applied to the structure, the assumed loads shall be combined vectorially, except for that portion of the assumed longitudinal load specified in Clause 7.8.2, which shall be considered separately.

7.4.2
The assumed loads for wood, concrete, and fibre-reinforced composite poles, for metal structures, and for guys shall be calculated by multiplying the loads of each category in Clause 7.4.1 by the appropriate load factors in Tables 31 and 32 for the type of load being considered.

7.4.3
For construction using pin-type insulators, the minimum assumed load in any direction on the pins, insulators, ties, or other conductor fastenings shall be 2.2 kN for each pin or conductor fastening.

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7.5 Assumed vertical load on supports

7.5.1 General

The vertical load on poles, towers, foundations, crossarms, pins, posts, insulators, and fastenings shall be the vertical force produced by their own mass plus the mass of all attachments, including the ice-coated wire and cable attachments in the parts of adjacent spans carried by the support. The radial thickness of ice, as specified in Table 30, shall be applied only to wire and cable attachments. Structures and foundations shall have additional vertical loads applied due to the vertical component of associated guys or due to the difference in the elevation of supports.

7.5.2 Assumed vertical load on wood, concrete, and fibre-reinforced composite poles, metal structures, and towers

The assumed vertical load on wood, concrete, and fibre-reinforced composite poles, metal structures, and towers shall be calculated by multiplying the vertical load specified in Clause 7.5.1 by the load factors for vertical loads in Table 31.

7.5.3 Assumed vertical load on crossarm assemblies, pins, posts, insulators, and fastenings

7.5.3.1

The assumed vertical load on crossarm assemblies shall be the vertical load specified in Clause 7.5.1 multiplied by a load factor of 2.0. Crossarm assemblies shall be capable of supporting a vertical load of 1.0 kN at either extremity, in addition to supporting the mass of conductors without ice covering.

7.5.3.2

A built-up metal crossarm of a metal structure shall be designed to withstand the specified loads with load factors at least as large as those required for the tower or structure itself.

7.5.3.3

The assumed vertical load on pins, posts, insulators, and fastenings shall be the vertical load specified in Clause 7.5.1.

7.6 Assumed transverse load due to wind pressure

7.6.1 Assumed transverse load due to wind pressure on wire and cable attachments

The assumed transverse load on supports, due to wind pressure on the wire and cable attachments, shall be the wire loads specified in Clause 7.3.1(b), for the loading area concerned and using the transverse wire load due to 1/2 of the sum of adjacent spans.

If communication wire and cable attachments or supply conductors up to 50 kV phase-to-ground are partially sheltered from the full effects of wind, the associated wind pressure on these items may be reduced to 40% of these values.

7.6.2 Assumed transverse load on a structure

7.6.2.1

The assumed transverse load on a structure shall be the wire load specified in Clause 7.6.1 plus the load created by wind pressure (as specified in Table 30) on the surfaces of the structure without an ice

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covering. Where applicable, the wind load shall be calculated in accordance with Clauses 7.6.2.2 to 7.6.2.4.

7.6.2.2

For flat surfaces, latticed or otherwise, the assumed unit wind pressure shall be increased by 100%. Where flat-surfaced lattice structures are involved, the actual exposed area of one lateral face shall be increased by 50% to allow for the pressure on the front of the opposite face, and the wind pressure increase of 100% shall then be applied. The pressure calculated in this manner need not exceed the pressure that would be incurred on a solid structure of the same outside dimensions. The results obtained by more precise calculations may be substituted for the values obtained in accordance with the requirements specified in this Clause.

7.6.2.3

For structures carrying more than 10 wires, not including cables supported by strand, where the pin spacing does not exceed 380 mm, the transverse loads shall be calculated on 2/3 of the number of wires, with a minimum of 10 wires.

7.6.2.4

Transverse loads due to snow creep shall be taken into account in the design of the structure.

7.6.3 Assumed transverse load on pins, posts, insulators, and fastenings

The assumed transverse load on pins, posts, insulators, and fastenings shall be as specified in Clause 7.6.1.

7.6.4 Assumed transverse load on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guys

The assumed transverse load for wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guys shall be calculated by multiplying the transverse loads specified in Clause 7.6.2 by the appropriate transverse load factors in Tables 31 and 32.

For guyed wood or concrete poles, the pole at the point of guy attachment shall be designed to withstand the total assumed loads.

7.7 Assumed loads on supports at angles

7.7.1 General

7.7.1.1

Where a change in direction of wire or cable attachments occurs, the load shall be taken as the resultant load equal to the vector sum of the following:

a) the transverse load as specified in Clauses 7.6.1 and 7.6.2;
b) the load due to the change in direction of the line, i.e., the resultant of the longitudinal tensions in each direction away from the corner structure under the appropriate loading conditions; and
c) where Grade 1 construction is called for, the longitudinal load specified in Clause 7.8.2.

7.7.1.2

The wind pressure shall be taken in such a direction that the loads specified in Clause 7.7.1.1 produce the maximum stress in the supports. As the wind does not usually act at right angles to each adjacent

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half-span simultaneously, proper reduction may be made to the transverse load to account for the application of the wind to the wires at the appropriate angle, where the wind is assumed to be acting along or near the bisector of the angle.

7.7.2 Assumed loads at angles on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guy assemblies

The assumed loads on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guy assemblies at angles in the line shall be calculated by multiplying the loads specified in Clause 7.7.1 by the appropriate load factors in Tables 31 and 32.

7.7.3 Assumed loads at angles on crossarms, pins, posts, insulators, and fastenings

The assumed loads at angles on crossarms, pins, posts, insulators, and fastenings shall be the load specified in Clause 7.7.1 for the wire and cable attachments that they support.

7.8 Assumed longitudinal loads on supports

7.8.1 Assumed longitudinal load — Terminations or tension changes: Grades 1, 2, and 3

7.8.1.1 General

For terminations or tension changes, the longitudinal load on supports at crossing structures or within sections where Grade 1, 2, or 3 construction is called for in Table 27, 28, or 29 shall be understood to be as follows:

a) for terminations at angles, or where the overhead wires end on a structure (e.g., where dead-end construction is used), the longitudinal load shall be taken as an unbalanced pull equal to the tensions of all conductors and supporting strands (including overhead ground wires) so terminated, under the appropriate loading conditions; and
b) for terminations with spans in each direction, such as those employed in tension changes or at switch locations, the longitudinal load shall be understood to be the vector sum of the tensions under the appropriate loading conditions.

7.8.1.2 Assumed longitudinal load for terminations or tension changes on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guy assemblies

The assumed longitudinal load for terminations or tension changes on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guy assemblies shall be the load specified in Clause 7.8.1.1 multiplied by the appropriate load factors in Tables 31 and 32.

Note: *For angles affected by terminating, Clause 7.7.2 may be applied in place of Clause 7.8.1.2.*

7.8.1.3 Assumed longitudinal load for terminations or tension changes on crossarms, pins, posts, insulators, and fastenings

The assumed longitudinal load for terminations or tension changes on crossarms, pins, posts, insulators, and fastenings shall be the load specified in Clause 7.8.1.1 for the wire and cable attachments that they support.

The minimum assumed longitudinal load for terminations or tension changes on dead-end assemblies where Grade 1 or 2 construction is called for shall be 1.6 times the maximum tension of the conductor or conductors, as applicable.

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The maximum allowable longitudinal load on crossarm assemblies using pin-type or post-type insulators shall be 3 kN, applied at the outer pins.

7.8.2 Assumed longitudinal load — Without terminations or tension changes

Note: *The degree of longitudinal loading is determined by examining the strength of the structures adjacent to the sections required to be of Grade 1 construction.*

7.8.2.1 General

Clause 7.8.2 applies to those wires and cable attachments that are straight and continuous across the crossing or end structures (i.e., where normally balanced tensions are involved).

7.8.2.2 Lines of Grade 1 construction

Where adjacent structures are built to Grade 1 construction on both sides of the section requiring Grade 1 construction, longitudinal loads need not be considered.

7.8.2.3 Lines of Grade 2 construction or weaker

Where supply lines or communication lines built to Grade 2 construction or weaker are adjacent to a section requiring Grade 1 construction, the longitudinal load for wires, self-supporting cables, and messengers shall be taken as an unbalanced pull in the direction of the section requiring the stronger grade, in accordance with 33-1/3 percent of the sum of all the tensions, under the appropriate loading conditions.

The assumed longitudinal load shall have a means of support as specified in Clause 8.4.4.

7.8.2.4 Assumed longitudinal load — Wire and cable attachments not involved in terminations or tension changes where Grade 2 or 3 construction is called for by Table 27, 28, or 29

Longitudinal load (as specified in Clause 7.8.2) need not be considered where Grade 2 or 3 construction is called for by Table 27, 28, or 29.

7.8.2.5 Assumed longitudinal load on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guy assemblies

The assumed longitudinal load on wood, concrete, and fibre-reinforced composite poles, metal structures, towers, and guy assemblies shall be calculated by multiplying the loads specified in Clauses 7.8.2.2 and 7.8.2.3 by the appropriate load factors in Tables 31 and 32.

7.8.2.6 Assumed longitudinal load on crossarms, pins, posts, insulators, and fastenings

The assumed longitudinal load on crossarms, pins, posts, insulators, and fastenings shall be as specified in Clauses 7.8.2.2 and 7.8.2.3. For conductors supported by pin-type insulators, the assumed longitudinal load at each point of support shall be determined by dividing the wire load specified in Clauses 7.8.2.2 and 7.8.2.3 by the number of points of support, with each group of similar conductors being treated separately.

The minimum longitudinal load on crossarm assemblies using pin-type or post-type insulators shall be 2.2 kN, applied at the outer pin or post.

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7.9 Assumed loads on a main line at a common structure

7.9.1
The line whose circuit is attached to the common structure requiring the strongest grade shall be called the main line. Where a main-line structure also supports wire and cable attachments of one or more other lines (either by crossing or as drops or service conductors), the assumed loads for each line to be applied to the common structure shall be determined using the strongest grade called for in Table 27, 28, or 29. The longitudinal loads of the additional line(s) shall be assumed to act in each direction across the main line.

7.9.2
The load on the common structure shall be calculated by taking the vector sum of the forces due to the loads on each line calculated in accordance with Clauses 7.4 to 7.8, using the load factors for the type of load being considered.

7.9.3
In the additional line(s), where conductors, messengers, cables, or guys have balanced tensions at the common structure, the assumed longitudinal load for such conductors, messengers, cables, or guys may be reduced to 50% of the load specified in Clause 7.8.2.

7.9.4
The wind shall be assumed to act in directions producing the maximum stress in each line for each type of load considered.

7.9.5
The loads on one line shall not be reduced as a result of loads on another line.

8 Strength of supporting systems according to deterministic design methods

8.1 General

8.1.1
Clause 8 specifies the strength requirements for supply and communication lines designed to withstand the loads specified in Clause 7. The strength requirements in Clause 8 shall apply to all grades of construction unless otherwise specified.

8.1.2
The longitudinal strength requirements specified in Clause 8 generally apply only to the structure immediately adjacent to a crossing, or at the ends of a proximity or joint-use section. The requirements specified in Clause 8 for transverse strength and strength at angles generally apply to each of the structures within the crossing, proximity, or joint-use section.

8.1.3
The method of analysis for structures shall be the non-linear method including stability (buckling) check.

Note: *See Clause A.8.1.3 for information regarding non-linear design.*

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Deformation, deflection, or displacement of parts of the structure can sometimes change the effects of the assumed loads. Where changes in the effects of assumed loads can be estimated due to deformation, deflection, or displacement of parts of the structure, they may be included in the non-linear analysis.

Similar movements of other structures connected by span attachments shall not be considered in these changes to assumed loads, unless the methods used have been mutually agreed upon by the parties involved.

8.2 Materials

8.2.1

Materials, including those used in foundations and settings, shall be able to withstand and maintain safely the loads specified in Clause 7.

8.2.2

Tie wires and fastenings shall have no sharp edges at points of contact with conductors and shall be applied such that they do not damage line conductors. Tie wires shall be made of a material that does not cause corrosion. The side pull of the conductor due to a change in the direction of the conductors shall be against the insulator rather than the tie wire.

8.2.3

Messenger clamps and messenger fastenings shall be of sufficient strength to withstand the assumed loads for messengers specified in Clauses 7.4 to 7.9.

8.3 Strength of supports

8.3.1 Structures — All grades

8.3.1.1

Poles, structures, and towers shall be capable of withstanding the assumed loads specified in Clauses 7.4 to 7.9 and any unbalance due to uneven spans or uplift under the wire loads specified in Clause 7.3.1 for new structures and those where additional attachments are made.

8.3.1.2

The strength requirements for poles and towers may be met by the structures alone or with the aid of guys or braces.

8.3.1.3

When the strength of a wood pole structure has deteriorated to 60% of the required design capacity, the structure shall be reinforced or replaced.

8.3.2 Vertical pull due to changes in elevation

The change in elevation of the conductors of a line approaching a crossing shall be such that there will be no uplift on the pins, posts, insulators, or ties on the crossing structures or on the immediately adjacent structures, unless special precautions are taken to ensure that the conductors, insulators, pins, or posts will not become detached from the supports and that vertical displacement of the structures themselves will not occur.

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8.3.3 Spliced wood poles

Spliced wood poles may be used, provided that recognized methods of splicing are employed and that the splice and the whole assembly meet the strength requirements for wood poles.

8.3.4 Stub-reinforced poles

Stub-reinforced poles may be used, provided that the above-ground section of the pole is sound and free from defects that would materially affect its strength, and provided that the reinforcement method used ensures that the whole assembly meets the strength requirements for wood poles.

8.3.5 Pole top supports and pole top pins

Pole top supports and pole top pins and posts may be used, provided that they do not impair the strength of the pole. They shall be considered an integral part of the structure and shall be subject to the same load factors as the pole itself.

8.3.6 Pole mounts

Pole mounts and their anchorages shall be of sufficient strength to withstand the load transferred by the pole under the loads specified in Clause 7.

8.3.7 Special wood structures

Wood structures such as H-frame structures shall have a transverse or longitudinal strength corresponding to the number of poles used. Where fully braced, poles shall be in either tension or compression and, where suitable foundations are provided to prevent their overturning, such structures may be considered unit-trussed structures, and the additional strength provided may be used in meeting transverse strength requirements.

8.3.8 Wood poles

8.3.8.1

Wood poles shall comply with CAN/CSA-O15. Poles not covered by CSA Standards but having adequate strength and durability may be used when agreed to by the parties involved. Damage and failure limits of wood poles are specified in Table A.4.

Note: *Where wood poles are used, they should be treated with a preservative in accordance with applicable CSA Standards.*

8.3.8.2

Where Grade 1 construction is required, no wood pole shall be weaker than CSA Class 6, except that CSA Class 7 may be used to support a single supply conductor or communication drop wire, provided that the requirements of Grade 1 construction are met.

8.3.9 Concrete poles

Concrete poles shall be in accordance with applicable CSA Standards and shall not be weaker than the equivalent of a CSA Class 6 wood pole where Grade 1 construction is required. Damage and failure limits of concrete poles are specified in Table A.4.

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8.3.10 Metal poles, metal towers, and metal supports

8.3.10.1

In the design of metal poles, metal towers, and metal supports, the “load factor” specified in Table 31 shall be interpreted such that the completed structure will support, with only negligible deformation of component parts (see Table A.4), the assumed loads to which it will be subjected, as specified in Clauses 7.3 to 7.9.

8.3.10.2

The absence of a permanent set in the structure indicates that no part has been stressed beyond the yield point. Allowance should be made for bolt slip and anchor consolidation. Metal structures fully supported by guys shall meet the requirements for self-supporting structures, and such guys shall be in accordance with the minimum load factors specified in Table 32.

8.3.11 Foundations and settings

Foundations and settings shall be of sufficient strength to withstand the load transferred by the pole or structure under the assumed loads specified in Clause 7. Where necessary, protection against washouts or debris slides shall be provided.

8.3.12 Fibre-reinforced composite poles

8.3.12.1

Fibre-reinforced composite poles shall comply with the applicable requirements of ASCE Manual 104.

8.3.12.2

In the design of fibre-reinforced composite poles, the “minimum load factor“ of Table 31 shall be interpreted such that the completed structure will support the assumed loads to which it will be subjected, as specified in Clauses 7.3 to 7.9. Fibre-reinforced composite poles may use the minimum load factors for steel poles, provided that the manufacturer proves through testing that the fibre-reinforced composite poles meet or exceed the minimum 5th percentile strength (i.e., the 5% lower exclusion limit). In the event that the manufacturer cannot meet the 5th percentile strength requirements, the fibre-reinforced composite poles shall use as default values the minimum load factors for wood poles.

8.3.12.3

Fibre-reinforced composite structures fully supported by guys shall meet the requirements for self-supporting structures. Such guys shall meet the load factors specified in Table 32. Damage and failure limits of fibre-reinforced composite poles are specified in Table A.4.

8.3.12.4

To verify the adequacy of composite poles to meet the required loads (i.e., working loads multiplied by the appropriate load factors), the manufacturer shall have and make available actual test data from full-scale bending tests of similar composite poles, conducted in accordance with ASTM D1036. Additional bending or other verification tests may be requested by the purchaser.

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8.4 Guys, guy assemblies, and braces

8.4.1 Application of guys

8.4.1.1
Guys may be used with structures when needed to provide necessary strength. Where poles or other flexible supports are used, the required transverse and longitudinal strength shall be achieved, where practicable, by the use of side and head guys. Guys and their fastenings shall meet applicable CSA Standards. Those guys and fastenings that are not covered by CSA Standards but that have adequate strength and durability may be used when agreed to by the parties involved.

8.4.1.2
The guy shall be attached as close as practicable to the centre of the load that the guy is designed to sustain.

8.4.1.3
Pole braces should not be used as substitutes for guys and anchors. However, where they offer advantages, they shall be designed to support the required compressive forces and to not result in uplift of the pole they are bracing.

8.4.1.4
To avoid abrasion, guys shall not be placed or maintained in contact with other guys, stubs, or anchors, nor maintained in contact with metallic covered cables, suspension wires, or overhead ground wires, except where the guys have been attached to the same point and protected at the point of contact.

Δ **8.4.1.5 —*Deleted***

8.4.2 Installation of anchorages

8.4.2.1
Anchor rods shall be installed such that they are in line with the pull of the attached guy when under load, except that in rock or concrete the rod may be at right angles to such pull.

8.4.2.2
Guy strand shall be attached to the eye of the anchor rod above ground.

8.4.2.3
Where direct attachment to anchor rods is not practicable, guys may be attached to stubs or poles that are themselves properly guyed, or to other secure structures. Guys shall not be attached to trees.

8.4.3 Side-guyed structures — Grades 1, 2, and 3
Where conditions make it impracticable to side-guy a crossing or other transversely weak structure in order to meet transverse strength requirements, these requirements may be met by the use of special structures or by side-guying the line at each side of, and as near as practicable to, the crossing or transversely weak structure, provided that

a) the distance between the side-guyed structures does not exceed 250 m or two spans (where the distance of two spans exceeds 250 m);

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b) the side-guyed structures are capable of withstanding the calculated transverse load due to wind on the supporting structures and ice-covered conductors in the entire section between the side-guyed structures;
c) the line between the side-guyed structures is substantially straight; and
d) the entire section between the transversely strong structures complies with the required grade of construction, except for the transverse strength of intermediate poles or towers.

8.4.4 Head-guyed structures

8.4.4.1 General

Longitudinal strength requirements shall be met by placing structures of the required strength at the crossing or at the ends of proximity or joint-use sections. Where this is impracticable, the alternatives specified in Clauses 8.4.4.2 and 8.4.4.3 may be adopted, using the loads specified in Clauses 7.4 to 7.9.

8.4.4.2 Grade 1

8.4.4.2.1

For conductors, messengers, cables, ground wires, and other such items with a rated tensile strength greater than 12.4 kN (except at terminations), structures of the required longitudinal strength may be located one or more span lengths away from the crossing structure or end of proximity or joint-use sections.

The line between the longitudinally strong structures shall be approximately straight or suitably guyed and shall meet the requirements for transverse strength. This requirement shall not apply to terminations.

8.4.4.2.2

For conductors, messengers, cables, and ground wires with a rated tensile strength less than 12.4 kN, and for terminations where it is impracticable to guy the crossing structure or the structure at the end of the proximity or joint-use section, strength requirements may be met by head-guying the line as near as practicable to, but at a distance not exceeding 150 m from, the nearest crossing pole or structure at the end of the proximity or joint-use section, provided that the line is approximately straight and that a steel wire strand of strength equal to that of the head guy is run between the guyed poles. The strand shall be attached to the guyed poles at the point at which the head guys are attached and shall be secured to every pole between the guyed poles.

8.4.4.3 Grades 2 and 3

The requirements of Clause 8.4.4.2 shall apply to Grades 2 and 3 at terminations or at angles using dead-end construction.

8.4.5 Longitudinal guying requirements for combinations of messenger cable and wire at crossings

For lines carrying both open wire and cable, the cable messenger may be used to provide the necessary longitudinal strength, provided that
a) the messenger is dead-ended and guyed in accordance with the crossing requirements;
b) messenger fastenings at crossing poles are capable of transmitting the assumed loads from the pole to the messenger;

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c) the assumed load due to the open wire plus the maximum tension due to the cable, as specified in Clause 7, does not exceed 60% of the rated tensile strength of the strand, and the messenger has a rated tensile strength not less than 26 kN; and
d) the pole is capable of withstanding the assumed loads imposed at the point of messenger attachment, using the appropriate load factor.

8.5 Crossarms, pins, posts, and fastenings — Grades 1, 2, and 3

8.5.1 Strength

Crossarms, crossarm assemblies, pins, posts, and fastenings shall have sufficient strength to withstand the assumed loads specified in Clauses 7.4 to 7.9.

8.5.2 Strength of crossarms

Crossarms shall meet the requirements of the applicable CSA Standard. Crossarms shall be attached to wood, concrete, fibre-reinforced composite, or metal structures by devices of adequate strength to support the specified loads. Where through-bolts are used, they shall be not less than 15 mm in diameter. Lag screws shall not be used.

8.6 Strength of insulators

8.6.1 Pin-type and post-type insulators

Pin-type and post-type insulators shall withstand the stress incurred when the conductor is subjected to the loads specified in Clauses 7.4 to 7.9.

8.6.2 Suspension-type insulators

8.6.2.1

Suspension-type insulators that are normally in the vertical position shall withstand, without exceeding 1/2 of their rated ultimate strength, the resultant vertical and transverse loads.

8.6.2.2

Suspension-type insulators in the strain position shall withstand, without exceeding 1/2 of their rated strength, the tension that occurs when the conductor is subjected to the loads specified.

8.7 Supply conductors, supporting conductors, and messengers

8.7.1 Materials

Conductor or messenger materials shall meet the requirements of applicable CSA Standards.

8.7.2 Minimum rated strength — Grade 1

For Grade 1 construction supply conductors, supporting conductors and messengers shall have a minimum rated strength not less than 12.4 kN.

8.7.3 Wire tension limits

8.7.3.1 Reliability-based design cases where limit loads are used

Note: *Limit loads are loads inclusive of all load factors.*

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8.7.3.1.1

The initial tension limit at average temperatures during the coldest month (January) should not exceed a catenary parameter of 2000 m for single conductor spans properly equipped with vibration dampers. In the case of bundled conductors, the catenary parameter may be increased to 2200 m.

Note: *This limit does not apply to special conductors such as self-damping conductors where different limits may be used in accordance with past experience and appropriate studies.*

8.7.3.1.2

For economical reasons, consideration may be given to reducing the limit of 2000 m when spans are less than 400 m, because such reduction will not greatly affect the conductor sag, but will provide additional safety for fatigue damage to conductors due to aeolian vibration and will reduce loads on angle structures. Reductions of the catenary parameter of 2000 m that should be used are specified in Table 33.

8.7.3.1.3

The final tension limit after creep or permanent stretch due to ice and wind loads should not exceed 70 to 80% RTS.

Note: *A value of 75% RTS may be used and has been applied to many Canadian lines.*

8.7.3.2 Deterministic design

8.7.3.2.1

Conductor tensions under deterministic ice and wind loads shall not exceed 60% RTS of the conductor.

8.7.3.2.2

The initial conductor tension limit shall not exceed the values specified for reliability-based design. For spans of 75 m or less in length, which are typical for distribution lines, consideration may be given to further reducing the initial tension limits because it lowers the risk of vibration problems and reduces loads on angle structures, while having insignificant impact on conductor sag.

8.8 Lightning protection wires

8.8.1 General

Lightning protection wires that run parallel to line conductors shall comply with Clause 8.7.2 where Grade 1 construction is required, except that galvanized steel wire strand shall comply with Clause 8.8.2.

8.8.2 Galvanized steel wire strand

Galvanized steel wire strand shall

a) comply with CAN/CSA-G12;
b) have a minimum diameter of 6.35 mm; and
c) not exceed the maximum allowable tensions specified in Table 34.

8.9 Trolley contact conductors

To allow for wear, trolley contact conductors shall be not smaller than No. 1/0 AWG when made of hard-drawn copper, or No. 4 AWG when made of silicon bronze, and shall be replaced when they have deteriorated to 60% of their original rated tensile strength.

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8.10 Supply cables — Grades 1, 2, and 3

There are no strength requirements for supply cables supported by messengers or by supporting conductors.

8.11 Communication conductors and cables — Grades 1, 2, and 3

8.11.1 Grade 3

There are no special requirements for size and strength of communication conductors where Grade 3 construction is used.

8.11.2 Material — Grades 1 and 2

Bare conductors shall be of a material or combination of materials that has a resistance to corrosion at least equivalent to that of steel line wire having a zinc coating of at least 0.25 kg/m^2.

8.11.3 Sags and tensions

Sags of all open-wire communication conductors shall be such that, under the assumed loading conditions for the applicable area, the tension of the conductor is not more than 60% of its rated tensile strength. In addition, the tension at 15 °C, without an external load, shall not exceed 20% of the conductor's ultimate rated tensile strength.

8.11.4 Communication conductors crossing over supply lines

Open-wire communication conductors crossing over open-wire supply conductors shall meet the size and strength requirements of supply conductors of the required grade. See Clauses 8.7 to 8.9.

8.12 Communication drops and cables — Grades 1 and 2

8.12.1 Communication drops and cables supported by messengers

Where communication drops or cables cross over supply conductors of more than 750 V line-to-ground, they shall be supported by a messenger that meets the requirements of Clause 8.13.2.

8.12.2 Communication drops or cables not supported by messengers — Grades 1 and 2

8.12.2.1

Communication drops or cables without supporting messengers shall not be used

a) in proximity to or crossing over supply lines greater than 750 V line-to-ground; or
b) in spans longer than 30 m in heavy and severe loading areas, or spans longer than 38 m in medium loading areas.

Note: *Item (b) applies to communication drops only.*

8.12.2.2

When communication drops or cables are used on Grade 1 and 2 construction they shall not be stressed beyond their rated tensile strength under the loads specified in Clause 7.3.

Note: *Appendix B of the Canadian Electrical Code, Part I ("CEC") includes comments on installations meeting the requirements of CEC Rule 6-112.*

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8.13 Communication cables and messengers

8.13.1 Communication cables

There are no strength requirements for communication cables supported by messengers.

8.13.2 Messengers

8.13.2.1 Maximum allowable tension — Grades 1, 2, and 3

Messengers shall not be stressed beyond 60% of their rated tensile strength under the loads specified in Clause 7.3.

8.13.2.2 Size — Grade 1

At crossings of facilities requiring grade 1 construction or in proximity to crossings, messengers supporting cables shall have a rated tensile strength not less than 12.4 kN. At crossings that are in joint use with, or in proximity to, supply conductors greater than 750 V line-to-ground, messengers shall have a rated tensile strength not less than 12.4 kN. At crossings or in proximity to crossings, communication drops supported by messengers or with integral strength members shall have a rated tensile strength not less than 5.3 kN.

8.13.2.3 Material — Grades 1 and 2

All steel stranded messengers shall meet the minimum requirements for zinc-coated steel strand specified in CAN/CSA-G12. Steel messengers consisting of a single wire shall meet the requirements for individual zinc-coated wires, before stranding, specified in CAN/CSA-G12. Messenger materials other than zinc-coated steel may be used, provided that these materials have a corrosion resistance at least equivalent to that required for zinc-coated steel of similar size and strength.

8.14 Supply and communications — Messenger clamps and fastenings: Grades 1, 2, and 3

Messenger clamps and fastenings shall meet the requirements of Clause 8.2.

8.15 Supply and communications — Cable attachments to suspension strand: Grades 1, 2, and 3

8.15.1 General

Cable attachments shall be spaced so that they will safely support the cable and prevent appreciable sagging of the cable.

8.15.2 Cable attachments at railway crossings

At railway crossings where lashing is employed, the cable shall be double-lashed using stainless steel lashing wire at least 1.14 mm in diameter.

8.16 Splices at crossings — Grades 1

8.16.1

Splices shall not be made in a crossing span except as specified in Clause 8.16.2, and should not be made in the adjoining spans. If making a splice in the adjoining span is unavoidable, the splice shall be of a type that resists corrosion and withstands the assumed loads specified in Clause 7.3.

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8.16.2
Full-tension compression splices may be used over crossings to effect repair or maintenance.

8.17 In-span taps at crossings — Grades 1 and 2

8.17.1 Supply conductors greater than 750 V
The use of in-span taps in railway crossing spans shall be avoided. Where their use is unavoidable, they shall withstand the assumed loads specified in Clause 7.3 and, except as allowed otherwise by Clause 8.17, they shall be installed using compression-type connections. Under no circumstances shall a tap be installed using bolted connections, except that bolted stirrups with hot line taps may be used where unavoidable. The restrictions of this Clause shall not apply to the conductor tail at dead ends or to taps in other spans.

8.17.2 Supply conductors of 0 to 750 V and communication conductors
In-span taps shall not be used in Grade 1 or 2 crossings unless they are attached to the conductor tail at dead ends.

8.18 Supply-line insulators

8.18.1 General
Insulators that operate on supply circuits greater than 750 V line-to-ground shall

a) meet the requirements of
   i) CSA C411.1, CAN/CSA-C411.4, CSA C411.5, CSA C411.6, and CSA C411.7; or
   ii) ANSI C29.1, ANSI C29.11, ANSI C29.12, and ANSI C29.13; or
b) be of another insulating material that provides comparable mechanical and electrical performance.

8.18.2 Selection of insulators
When selecting insulators to be used for line, consideration shall be given to the prevailing level of contamination. See IEC/TS 60815-1, IEC/TS 60815-2, and IEC/TS 60815-3 for guidance on the selection of insulators in polluted conditions.

8.18.3 Protection against arcing
When installing insulators and conductors, good practice shall be used to prevent, as much as is possible, an arc from forming, including those arcs due to weakening or burning of any parts of the structures, insulators, or conductors.

8.19 Communication and power line hardware
Hardware used in supply and communication lines shall be in accordance with CSA C83 as applicable.

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9 Grounding methods for supply systems less than or equal to 22 kV and communications facilities

9.1 Supply systems less than or equal to 22 kV

9.1.1 Conductors

9.1.1.1 Conductor composition

The grounding conductor shall be made of copper or another metal or combination of metals that will not corrode excessively during the expected service life of the conductor under existing conditions. Where joints are unavoidable, they shall be made and maintained such that they do not materially increase the resistance of the grounding conductor and they shall have appropriate mechanical and corrosion-resistant characteristics.

9.1.1.2 System grounding conductors

System grounding conductors, exclusive of grounds at individual services, shall have a short-time ampacity adequate for the fault current that can flow in the grounding conductors for the operating time of the system-protective device. Where the fault current cannot be readily determined, continuous ampacity of the grounding conductor or conductors shall be not less than the full-load continuous current of the supply transformer or other source of supply. In no case shall the ampacity of the grounding conductor be less than that of No. 4 AWG copper.

Notes:

1) *For bare grounding conductors, the short-time ampacity is the current that the conductor can carry for the applicable time without melting or affecting the design characteristics of the conductor.*

2) *For insulated grounding conductors, the short-time ampacity is the current that the conductor can carry for the applicable time without affecting the design characteristics of the insulation.*

3) *Where grounding conductors at one location are paralleled, the increased total current capacity may be considered.*

9.1.2 Ground resistance requirements

9.1.2.1 Multi-grounded neutral systems

9.1.2.1.1

The neutral shall be of sufficient size and ampacity for the intended use and shall be connected to a ground electrode at each piece of active electrical equipment and to a sufficient number of additional ground electrodes (not including grounds at consumers' services) to prevent electric shock hazards caused by the buildup of excessive steady-state neutral-to-earth voltage.

Where practicable, the resistance of the interconnected neutral system to ground shall not exceed 6 Ω.

Note: *Multi-grounded neutral systems that extend over a substantial distance depend more on the multiplicity of grounding electrodes than on the resistance-to-ground of one individual electrode. Therefore, no specific values are prescribed for the resistance of individual electrodes.*

9.1.2.1.2

Where there are isolated multi-grounded neutral systems that are bridged by telecommunication

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circuits, a continuous multi-grounded neutral system complying with Clause 9.1.2.1.1 shall be provided for single-phase and three-phase grounded systems supplied from the same three-phase feeder, unless

a) the distance between the multi-grounded neutral systems prevents electrical hazards from ground current flow on communication equipment; or
b) other methods are used to prevent electrical hazards from ground current flow on communication equipment.

9.1.2.2 Earth return systems

9.1.2.2.1

When designing the grounding of an earth return system, the following factors shall be considered:

a) soil resistivity;
b) step and touch potential under steady-state and fault conditions;
c) magnitude of fault currents; and
d) frequency and number of ground electrodes installed.

9.1.2.2.2

Where earth return systems are used, the following criteria shall be met:

a) The resistance-to-ground of any individual electrode shall not exceed 25 Ω and the resistance of the grounding installation without interconnection to the consumer's service grounding system shall not exceed 6 Ω. Where these readings cannot be achieved, an additional two electrodes connected in parallel or two deep-driven electrodes, shall be used. Where the required readings cannot be achieved with the two additional electrodes, the grounding system shall be extended into a multi-grounded system until the 6 Ω interconnected reading can be achieved.
b) Measures shall be taken to prevent electric shock hazard to persons caused by the buildup of steady-state neutral-to-earth voltage.
c) The grounding installation shall consist of a redundant grounding system with ground electrodes separated by a distance greater than their depth that are located on different sides of the pole or on separate poles.
d) The transformer primary neutral terminal, transformer case, lightning arrester, grounded conductor, and secondary neutral terminal shall
    i) be connected to the ground electrode using the appropriate ground conductor; or
    ii) have a suitable warning label or markings on the pole where primary and secondary neutrals are not connected.

9.1.3 Ground electrodes and connections

9.1.3.1

Ground electrodes, ground electrode couplers, and connections to ground electrodes shall

a) have sufficient ampacity to safely conduct the electric utility system fault current and any steady-state ground current that might flow on the electric utility grounding system;
b) be corrosion resistant; and
c) be enduring.

Note: *Where applicable, ground electrodes, couplers, and connectors may be manufactured in accordance with CSA C22.2 No. 41.*

9.1.3.2

Where practicable, ground electrodes shall be installed such that they extend below the frost level.

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9.1.3.3

Where used, rod-type ground electrodes shall be installed such that

a) the top ends of the ground electrodes are a minimum distance of 300 mm below final grade level; and
b) the horizontal distance between ground electrodes is greater than or equal to their length.

9.1.4 Mechanical protection of supply grounding conductors

Grounding conductors of supply systems shall be protected by a covering that provides suitable mechanical protection from at least 0.3 m below to at least 2.0 m above grade.

The minimum distance above grade level for the protective covering on grounding conductors of gang-operated switches may be reduced to the height of the operating handle.

Where metallic conduit is used to protect ground and neutral conductors on poles or structures, the metallic conduit shall be grounded.

9.1.5 Non-conducting covering for supply grounding conductors

On joint-use poles where the communication facility is not bonded to the supply ground, supply grounding conductors shall be covered with a suitable non-conductive protective covering. The non-conducting covering shall extend up to the communication plant. This covering need not be applied to supply grounding conductors on metal structures.

Note: *The non-conducting covering may also be used to provide the mechanical protection specified in Clause 9.1.4.*

9.1.6 Interconnecting ground electrodes and grids

Ground conductors used to interconnect ground electrodes or form ground grids underground shall be installed a minimum distance of 300 mm below grade.

9.1.7 Grounding metal and concrete poles

Metal poles or the reinforcing metal of concrete poles, where such poles carry supply-line conductors or are in contact with metal-sheathed supply cable or the metal cases of supply equipment, shall be effectively grounded, guarded, or isolated.

9.1.8 Non-current-carrying items of supply and communication equipment

Items such as metal conduit, cable sheaths, metal lamp posts, frames, cases, and hangers of equipment shall be effectively grounded where

a) they might become energized at more than 750 V;
b) they are less than 2.5 m above the ground or accessible surface; or
c) they are not guarded from accidental contact by anyone other than qualified persons.

9.1.9 Grounding lightning arresters

Lightning discharge currents contain high-frequency components that produce large voltage gradients on grounding conductors. The grounding connection between the arrester and the frames of equipment being protected shall be as short, straight, and free from sharp bends as practical in order to

a) avoid hazards due to flashover to nearby objects;
b) permit effective operation of the lightning arrester; and
c) minimize the impedance of the connection.

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The grounding conductor shall have adequate short-time ampacity to carry the excess current caused by or following a surge. Individual arrester grounding conductors shall be no smaller than No. 6 AWG copper or an electrical equivalent.

9.1.10 Grounding pole-mounted equipment

For equipment connected in a grounded-wye configuration, the neutral point(s) of pole-mounted transformers, capacitors, or regulators shall be grounded with a ground conductor connected to a ground electrode.

9.1.11 Gang-operated switches

The metallic frame and operating mechanism, including control boxes of gang-operated switches, shall be grounded with a ground conductor connected to a ground electrode or ground grid at the pole. Grounding of the control rod is not required where the control rod is isolated with a fully rated insulator and the switch frame above the insulator is bonded to the system neutral. In this case, any ground on the pole shall not extend below the insulator. Grounding is not required for hook-stick-operated gang switches.

9.1.12 Ground interconnections

Where different systems serve the same customer, the grounds of the different systems shall be bonded. A single grounding conductor may be used for both supply and communication grounding, provided that the ground connection is of sufficiently low impedance and of sufficient current-carrying capacity to prevent the buildup of voltages that can result in a hazard to persons or equipment.

9.1.13 Grounding of riser pipes and guards

Exposed metal riser pipes and guards in contact with supply cables shall be grounded.

9.1.14 Grounding conductors on joint-use structures

9.1.14.1 Grounding of supply attachments on joint-use structures

With the exception of the attachments specified in Clause 9.1.14.2 , where conductors, apparatus, or equipment such as transformer casings or luminaire brackets are constructed to the clearance and separation values for effectively grounded systems as specified in Clause 5.10, each grounding conductor and all connection points directly associated with such apparatus or equipment shall, in the case of wood structures, be constructed such that they are visible to a qualified person standing on the ground. As far as practicable, the requirements of this Clause shall also apply to metal, fibre-reinforced composite, and concrete structures.

9.1.14.2 Grounding of consumer service and equipment on joint-use structures

Where outdoor electrical equipment that is approved in accordance with the *Canadian Electrical Code, Part I*, manufactured in accordance with applicable Standards of the *Canadian Electrical Code, Part II*, and connected to a consumer's service at 750 V or less, is installed in accordance with the clearance and separation values for effectively grounded systems as specified in Clause 5.10, the attachments and service shall meet the requirements of the *Canadian Electrical Code, Part I*. The installation shall have a means of disconnect that is approved in accordance with the *Canadian Electrical Code, Part I* and located such that it can be operated safely by communications workers.

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9.2 Grounding and bonding of communications facilities

9.2.1 General

9.2.1.1 Grounding conductor

The grounding conductor shall be made of copper or another metal or combination of metals that will not corrode excessively during the expected service life of the conductor under existing conditions. Where joints are unavoidable, they shall be made and maintained such that they do not materially increase the resistance of the grounding conductor and they shall have appropriate mechanical and corrosion-resistant characteristics.

9.2.1.2 Grounding and bonding conductor size

For communications-only lines, the grounding conductor shall be a minimum of No. 6 AWG insulated wire. For joint-use and non-joint-use poles, the bonding conductor shall be a minimum of No. 6 AWG insulated wire.

9.2.1.3 Ground electrodes

Ground electrodes shall

a) be corrosion resistant;
b) be enduring; and
c) extend below the frost level, where practical.

Note: *Where applicable, ground electrodes may be manufactured in accordance with CSA C22.2 No. 41.*

9.2.1.4 Ground connections

Where ground electrodes are manufactured to CSA C22.2 No. 41, the ground connections used on grounding systems for communications systems shall also comply with CSA C22.2 No. 41, as applicable.

9.2.2 Grounding and bonding intervals — Aerial non-joint use

9.2.2.1

On communication-only lines, the support strand shall be grounded at each end of the pole line and at intervals not exceeding 1200 m. No point on the cable sheath shall be further than 600 m from a ground.

The cable sheath bonding interval shall not exceed

a) 300 m for copper cables; and
b) 1500 m for fibre cables.

9.2.2.2

Support strands that cross over one another at right angles shall be bonded at the point of intersection. Support strand shall be bonded to guys at corners.

9.2.2.3

Bonds shall be made as close as possible to dead-end poles on cable laterals.

9.2.2.4

A bond to the multi-grounded neutral shall be placed at all joint-use crossover poles.

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9.2.2.5
Where mechanical protection is installed on the vertical ground conductor, it shall be provided by means of a protective covering that runs from at least 2.0 m above grade to at least 300 mm below grade.

9.2.3 Grounding and bonding intervals — Aerial joint use: copper cables

9.2.3.1
Grounding and bonding intervals of communications plant (copper cables) shall comply with CSA C22.3 No. 5.1.

9.2.3.2
Support strands shall be bonded across poles with double dead ends (e.g., at corners). Bonds shall be made as close as possible to dead-end poles on cable laterals.

9.2.3.3
Where two or more suspension strands are located on the same pole line, they shall be bonded together at 300 m intervals. Where the strands are supported on the same suspension clamp, the clamp shall function as the bond.

9.2.4 Grounding and bonding intervals — Aerial joint use: fibre cables

9.2.4.1
Grounding and bonding intervals of communications plant (fibre cables) shall comply with CSA C22.3 No. 5.1.

9.2.4.2
On joint-use poles, the sheath bonding interval for fibre cables shall be 1500 m (i.e., no point on the sheath shall be farther than 750 m from a sheath bond).

9.2.4.3
Metallic strength members in fibre cables and any armouring shall be bonded to the cable sheath at each splice.

9.2.5 Grounding and bonding in the zone of influence of a substation
Grounding and bonding of communications facilities in the zone of influence of a substation shall comply with IEEE 487.

9.2.6 Bonding and grounding with joint use on earth return and ungrounded (delta) supply systems

9.2.6.1
There should be no joint use with earth return and ungrounded systems. If joint use with either of these supply systems is the only option, CSA C22.3 No. 5.1 should be consulted for special engineering considerations.

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9.2.6.2

Joint use with earth return or ungrounded systems may be considered where ADSS fibre optic cable is used.

10 Reliability-based design method

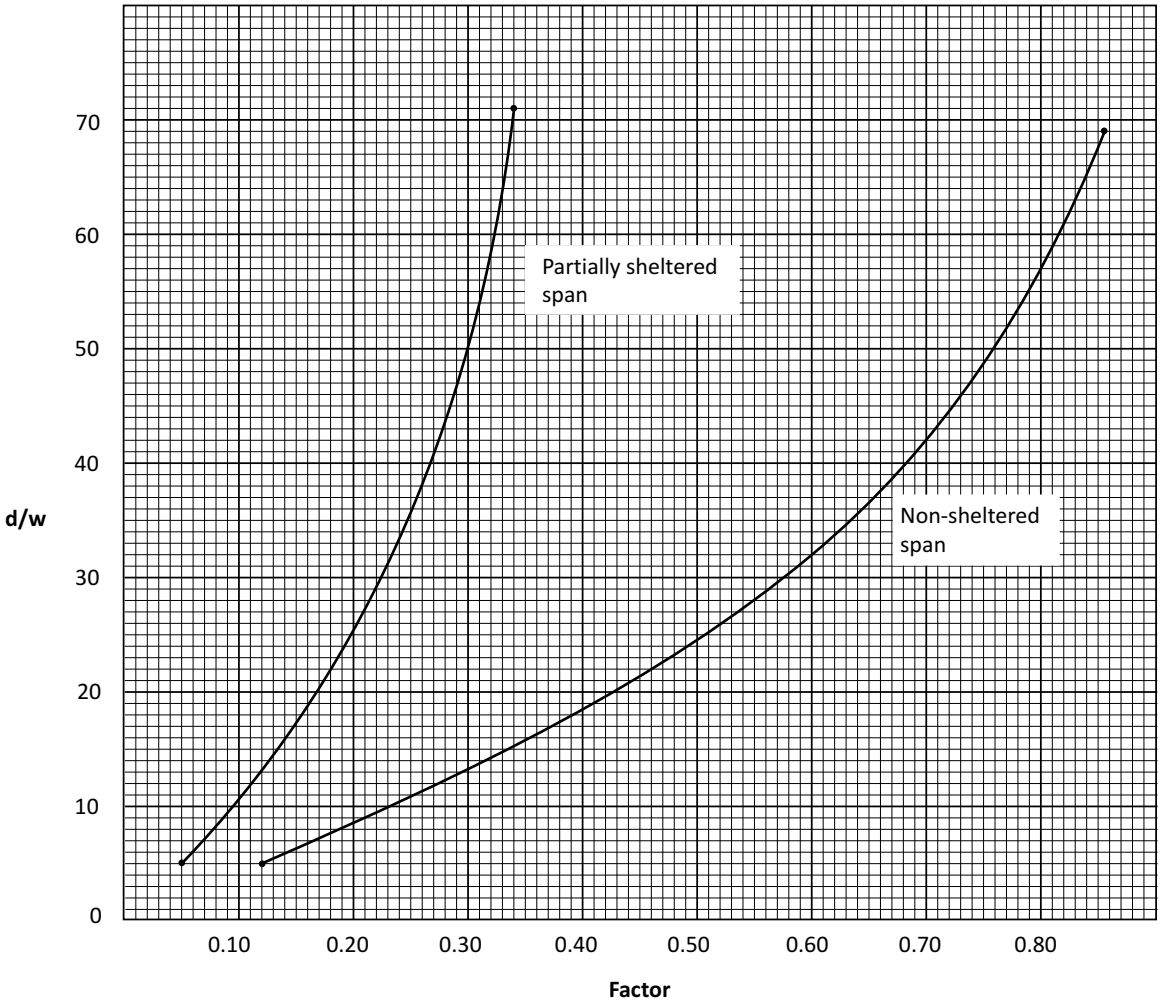
Supply and communication lines shall be designed using the deterministic design method or the reliability-based design method. Clauses 7 and 8 outline the deterministic method of calculating the loads to be applied to the design of the line and the strength of the structures needed to support these loads safely. The reliability-based design method is covered in CSA C22.3 No. 60826.

The reliability-based method should be used for supply lines greater than 70 kV phase-to-phase, in areas where significant amounts of meteorological data are readily available. This method may also be used for lines designed for specific climatic loads in accordance with previous experience or calibration with existing lines having a long history of satisfactory performance.

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Table 1
Wire and conductor swing for horizontal design clearance
(See Clauses 5.2.7, A.5.2.7, A.5.8.2 and A.7.6.1.)



Notes:

1) *Horizontal deviation = factor × conductor sag*
d = diameter of conductor, mm
w = mass per unit length, kg/m

2) *A partially sheltered span is one that is protected (from swinging transversely in the wind) by objects such as buildings whose height is greater than or equal to wire or conductor height and that are within an effective distance (e.g., 30 m) of the wire or conductor. Trees are not generally considered to provide shelter.*

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Table 2
Minimum vertical design clearances above ground or rails, ac
(See Clauses 5.3.1.1, 5.7.4.1, and A.5.3.1 and Tables 4, 9, and 11.)

	Minimum vertical clearance above ground or rails, m											
			Open supply conductors and service conductors, ac*, kV									
Location of wires or conductors	Guys; messengers; communication, span, and lightning protection wires; communication cables	Trolley contact conductors and associated span wires; trolley feeders 0–750 V when paralleled by trolley contact conductors	0–0.75	> 0.75 ≤ 22	> 22 ≤ 50	> 50 ≤90	> 90 ≤ 120	> 120 ≤150	> 150 ≤ 200	208 (360)	289 (500)	424 (735)
	Col. I	Col. II	Col. III	Col. IV	Col. V	Col. VI	Col. VII	Col. VIII	Col. IX	Col. X	Col. XI	Col. XII
Over land likely to be travelled by road vehicles (including highways, streets, lanes, alleys, and driveways other than those leading to residences or residence garages)	4.42	4.42	4.42	4.75	5.2	5.5	5.8	6.1	6.6	10.2†	15.4†	20.4†
Over the right-of-way of underground pipelines	4.42	4.42	4.42	4.75	5.2	5.5	5.8	6.1	6.6	6.8	9.9†	13.2†
Alongside land likely to be travelled by road vehicles§ or within the limits (with no overhang) of streets and highways	4.42‡	4.42	4.42	4.75	5.2	5.5	5.8	6.1	6.6	10.2†	15.4†	20.4†
Over or alongside farmland likely to be travelled by vehicles§	4.42**	N/A	4.42	4.75	5.2	5.5	5.8	6.1	6.6	6.8	9.9†	13.2†

(Continued)

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Table 2 (Concluded)

Location of wires or conductors	Minimum vertical clearance above ground or rails, m											
	Guys; messengers; communication, span, and lightning protection wires; communication cables	Trolley contact conductors and associated span wires; trolley feeders 0–750 V when paralleled by trolley contact conductors	Open supply conductors and service conductors, ac*, kV									
			0–0.75	> 0.75 ≤ 22	> 22 ≤ 50	> 50 ≤90	> 90 ≤ 120	> 120 ≤150	> 150 ≤ 200	208 (360)	289 (500)	424 (735)
	Col. I	Col. II	Col. III	Col. IV	Col. V	Col. VI	Col. VII	Col. VIII	Col. IX	Col. X	Col. XI	Col. XII
Over driveways to residences and residence garages	3.7	N/A	3.7	4.75	5.2	5.5	5.8	6.1	6.6	10.2†	15.4†	20.4†
Alongside roads and highways in areas unlikely to be travelled by road vehicles (with no overhang) and within 1.5 m of the limit of the road right-of-way††	3.0	N/A	3.4	4.15	4.6	4.9	5.2	5.5	6.0	6.2	7.2	8.4
Over walkways or ground normally accessible only to pedestrians, snowmobiles, and personal-use all-terrain vehicles	2.5	N/A	3.1	3.4	3.7	4.0	4.3	4.6	5.1	5.3	6.3	7.5
Above top of rail at railway crossings‡‡	7.3	6.7	7.3	7.6	8.1	8.4	8.7	9.0	9.5	9.7	10.7	11.9

* *For dc voltages below 750 V, use Columns II and III.*

† *Specifies clearances based on induced electrostatic steady-state currents and chosen line configurations and overvoltage levels – see Note (3). Other values may be used based on other line configurations, voltages, or assumptions regarding vehicle length and orientation. (See Clause A.5.3.1.) Provision of other mitigative measures such as shielding can reduce these clearances.*

(Continued)

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Table 2 (Concluded)

‡ *Where communication wires or communication cables run along alleys, this clearance may be reduced to 4 m.*
§ *Where a line runs parallel to land accessible to vehicles but is over land not requiring clearance for vehicles, the wire can swing out over the area accessible to vehicles or, at voltages over 200 kV, vehicles can be subjected to a hazard from induced voltages. These vertical clearances apply where the conductor (in the swing condition, where specified) is over, or within the following horizontal distances from the edge of, land accessible to vehicles:*

a) 0 m for wires in Columns I to V;
b) 1 m for wires in Column VI;
c) 1.4 m for wires in Column VII;
d) 1.7 m for wires in Column VIII;
e) 1.7 m + 0.01 m/kV over 150 kV, for wires in Columns IX and X; and
f) 18 m for wires at rest in Columns XI and XII.

These distances apply with the wire at the swing angle as calculated in Clause 5.2.7. Where the horizontal distances in Items (a) to (f) are exceeded, minimum vertical clearances are determined in relation to the ground over which the line passes.
** *On farmlands not likely to be travelled by high farm vehicles, these clearances may be reduced by 0.76 m. Examples include steep slopes, hillsides, and rocky ledges. This does not apply to swamps or other areas that can be crossed by vehicles in winter.*
†† *These areas are generally adjacent to fences and accessible to small vehicles, but are not likely to be travelled by high road vehicles or high farm machinery.*
‡‡ *See Clause 5.3.1.1(c).*

Notes:

1) *Voltages are rms line-to-ground. Voltages in parentheses are phase-to phase.*
2) *For elevations exceeding 1000 m, see Clause 5.3.1.1(b).*
3) *Seasonal conditions can warrant additional clearances (see Clause 5.3.1.1(e)).*
4) *For induced electrostatic governing clearances for voltages greater than 200 kV, the specified clearances are based on the following horizontal configurations:*

Nominal voltage, kV	***Overvoltage line-to-ground, kV***	***Phase-to-phase spacing, m***	***Number of conductors in bundle***	***Conductor diameter, mm/ name***
360	*220*	*8.5*	*2*	*28.1/Drake*
500	*318*	*10.7*	*4*	*25.4/Mica*
735	*442*	*12.8*	*4*	*35.1/Bersimis*

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Table 3
Minimum vertical design clearances above waterways*, ac and dc†
(See Clause 5.3.3.2.)

Crossing class	Type of waterways crossed: A = water areas D = water depth W = water width H = reference vessel height‡	Minimum clearance above OHWM, m					
		Guys; messengers; communication, span, and lightning protection wires; communication cables	Open supply conductors and service conductors, ac and dc, kV				
			0–22	> 22 ≤50	> 50 ≤90	> 90 ≤150	> 150
0	Minor waterways Minor waterway crossings do not require approvals from Transport Canada if they meet the definition of *Minor Works and Waters (Navigation Protection Act) Order.*	4.42	4.75	5.2	5.5	6.1	6.1 + 0.01 m/kV over 150 kV
1	Shallow or fast-moving waterways capable of being used by canoes and paddle boats in isolated areas where motor boats are not expected. Creeks and streams: W = 3–50 m and D < 1 m Ponds: A < 8 ha and D < 1 m H = 4.0 m	5.0	6.0	6.3	6.7	7.3	7.3 + 0.01 m/kV over 150 kV
2	Shallow or fast-moving waterways capable of being used by motorboats with antennas and unable to support masted vessels Creeks and streams: W = 3–50 m and D < 1 m Ponds: A < 8 ha and D < 1 m H = 6.0 m	7.0	8.0	8.3	8.7	9.3	9.3 + 0.01 m/kV over 150 kV
3	Small lakes and rivers used by masted vessels Rivers: W = 3–50 m and D > 1 m Ponds and lakes: A < 8 ha and D > 1 m H = 8.0 m	9.0	10.0	10.3	10.7	11.3	11.3 + 0.01 m/kV over 150 kV
4	Small resort lakes, medium-sized rivers and reservoirs, rivers connecting lakes, and crossings adjacent to bridges and roads Rivers: W = 50–500 m Lakes/reservoirs: 8 ha < A < 80 ha H = 10.0 m	11.0	12.0	12.3	12.7	13.3	13.3 + 0.01 m/kV over 150 kV
5	Large lakes, reservoirs, and main rivers in resort areas Rivers: W > 500 m Lakes/reservoirs 80 ha < A < 800 ha H = 12.0 m	13.0	14.0	14.3	14.7	15.3	15.3 + 0.01 m/kV over 150 kV

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Table 3 (Concluded)

Crossing class	Type of waterways crossed: A = water areas D = water depth W = water width H = reference vessel height‡	Minimum clearance above OHWM, m					
		Guys; messengers; communication, span, and lightning protection wires; communication cables	Open supply conductors and service conductors, ac and dc, kV				
			0–22	> 22 ≤50	> 50 ≤90	> 90 ≤150	> 150
6	Main lakes on main navigation routes and marinas A > 800 ha H = 14.0 m	15.0	16.0	16.3	16.7	17.3	17.3 + 0.01 m/kV over 150 kV
7	Federally maintained commercial channels, rivers, harbours, or heritage canals	§	§	§	§	§	§

* *The clearance over a canal, river, or stream normally used to provide access for sailboats to a larger body of water shall be the same as that required for the larger body of water.*

† *Clearances are applicable to both ac and dc circuits. For dc circuits, the clearance requirements shall be the same as those for ac circuits having the same crest voltage-to-ground.*

‡ *Reference vessel height refers to the overall height of the vessel, including the heights of antennas or other attachments.*

§ *In Canada, clearances are specified by the Transport Canada office responsible for the coastal regions, Great Lakes system, Red River–Lake Winnipeg system, Mackenzie River, and interior lakes of British Columbia. Where tide water has an effect on a body of water being crossed, the vertical design clearance shall be increased by an amount that takes into account peak tide.*

Notes:

1) *Voltages are rms line-to-ground.*

2) *The formula used to calculate minimum clearances is as follows:*

Minimum vertical clearance = reference vessel height + 1.0 m buffer + safe electrical clearance

where

1.0 m buffer = allowance for variation in boat height above water (due to different loading conditions) and for wave action

safe electrical clearance = safety margin + air gap flashover distance based on switching surge

= 0.78 m + (0.01 m/kV × max. kV in each column)

3) *Vertical design clearance is based on the clearance needed for the vessel height that a waterway can support. If a waterway cannot support the referenced vessel height, the vertical clearance requirement may be reduced in accordance with the appropriate requirement.*

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Table 4
Minimum vertical design clearances above ground or rails, dc
(See Clauses 5.3.1.1, 5.7.4.1, and A.5.3.1 and Tables 9 and 11.)

	Minimum vertical clearance above ground or rail, m								
	Open supply-line conductors, dc, kV								
	≥ 0.75 ≤100*	> 100 ≤150	> 150 ≤ 200	> 200 ≤ 250	> 250 ≤ 300	> 300 ≤ 350	> 350 ≤ 400	> 400 ≤ 450	> 450 ≤ 500
Location of conductors	**Col. I**	**Col. II**	**Col. III**	**Col. IV**	**Col. V**	**Col. VI**	**Col. VII**	**Col. VIII**	**Col. IX**
Over land likely to be travelled by road vehicles (including highways, streets, lanes, alleys, and driveways other than to residences or residence garages)†	5.8	6.0	6.3	6.5	6.7	6.9	7.2	7.4	7.7‡
Over the right-of-way of underground pipelines	5.8	6.0	6.3	6.5	6.7	6.9	7.2	7.4	7.7‡
Alongside land likely to be travelled by road vehicles or within the limits (with no overhang) of streets and highways	5.8	6.0	6.3	6.5	6.7	6.9	7.2	7.4	7.7‡
Over or alongside farmland likely to be travelled by vehicles	5.8	6.0	6.3	6.5	6.7	6.9	7.2	7.4	7.7‡
Over driveways to residences or residence garages	5.6	5.9	6.1	6.3	6.6	6.8	7.0	7.2	7.5‡
Alongside roads and highways in areas unlikely to be travelled by road vehicles (with no overhang) and within 1.5 m of the limit of the road right-of-way§	5.5	5.7	6.0	6.2	6.4	6.6	6.9	7.1	7.4‡
Over walkways or ground normally accessible only to pedestrians, snowmobiles, and personal-use all-terrain vehicles	4.6	4.8	5.0	5.3	5.5	5.7	6.0	6.2	6.5‡
Above top of rail at railway crossing**	8.8	9.1	9.3	9.5	9.8	10.0	10.2	10.4	10.7‡

* *For dc voltages below 0.75 kV, use Columns II and III of Table 2.*
† *See Clause 5.3.1.1(d).*
‡ *Add 0.005 m/kV for each kV over 500 kV to clearances in Column IX.*
§ *These areas are generally adjacent to fences and accessible to small vehicles but are not likely to be travelled by high road vehicles or high farm machinery.*
** *See Clause 5.3.1.1(c).*

Notes:

1) *Voltages are line-to-ground.*

2) *Seasonal conditions can warrant additional clearances (see Clause 5.3.1.1(e)).*

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Table 5
Minimum separations (heights) of supply equipment from ground
(See Clauses 5.3.2.1 and A.5.3.2 and Table 11.)

Type of equipment	Location of equipment	Minimum separation from ground, m			
		0–750 V	> 0.75 kV ≤22 kV	> 22 kV ≤50 kV	> 50 kV ≤90 kV
Live or exposed current-carrying parts of supply equipment (e.g., cable terminals, arresters, line switches) and ungrounded cases of supply equipment (e.g., transformers, regulators, capacitors)	Areas accessible to pedestrians only	3.0	3.4	3.7	4.0
	Areas likely to be travelled by vehicles	4.4	4.7	5.2	5.5
Effectively grounded cases of supply equipment (e.g., transformers, regulators, capacitors)	Areas accessible to pedestrians only	3.0	3.0	3.0	3.0
	Areas likely to be travelled by vehicles	4.4	4.4	4.4	4.4

Note: *Voltages are rms line-to-ground.*

Table 6
Minimum horizontal design clearances between wires and railway tracks
(See Clauses 5.4.2 and 5.4.3.)

	Wire closest to tracks	Minimum clearance, m	
		Main tracks	Siding
Guys; messengers; communication, span, and lightning protection wires; communication cables		2.5	1.9
Open supply-line conductors and service conductors of 0–750 V and effectively grounded continuous metallic sheathed cables of all voltages		2.5	1.9
AC	Open supply-line conductors and cables other than those having an effectively grounded continuous metallic sheath		
	> 0.75 ≤ 22 kV	2.7	2.1
	> 22 ≤ 50 kV	3.2	2.6
	> 50 ≤ 90 kV	3.5	2.9
	> 90 ≤ 120 kV	3.8	3.2
	> 120 ≤ 150 kV	4.1	3.5
	Supply conductors > 150 kV	4.1 + 0.01 m/kV over 150 kV	3.5 + 0.01 m/kV over 150 kV
DC	Open supply conductors		
	0–750 V	2.5	1.9
	> 0.75 ≤ 100 kV	2.8	2.2
	> 100 ≤ 150 kV	3.0	2.5
	> 150 ≤ 200 kV	3.3	2.7
	> 200 ≤ 250 kV	3.5	2.9
	> 250 ≤ 300 kV	3.7	3.1
	> 300 ≤ 350 kV	4.0	3.4
	> 350 ≤ 400 kV	4.2	3.6
	> 400 ≤ 450 kV	4.4	3.8
	> 450 kV	4.4 + 0.005 m/kV over 450 kV	3.8 + 0.005 m/kV over 450 kV

Note: *Voltages are line-to-ground.*

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Table 7
Minimum horizontal separations from supporting structures to railway tracks
(See Clauses 5.5.2 and 5.5.3.)

Tracks	Minimum horizontal separation, m
Main tracks (straight, level runs)	2.5
Sidings (straight, level runs)	1.9

Table 8
Minimum design clearances of supply conductors attached to buildings
(See Clause 5.7.2.)

Conductor attached to building*		Minimum clearance, m		
		Horizontal to surface	Vertical to normally inaccessible surface	Vertical to readily accessible surface
0–750 V	Insulated or grounded	1.0†	1.0	2.5
	Enclosed in effectively grounded metallic sheath	0	0	0
	Not insulated, grounded, nor enclosed in effectively grounded metallic sheath	1.0†	1.0	2.5
> 0.75 kV ≤ 5 kV	Enclosed in effectively grounded metallic sheath	0	0	0
	Not enclosed in effectively grounded metallic sheath	2.0‡	1.2	2.7
> 5 kV ≤ 22 kV	Enclosed in effectively grounded metallic sheath	0	0	0
	Not enclosed in effectively grounded metallic sheath	2.0§	1.5**	3.0**
> 22 kV	—	2.0 + 0.01 m/kV over 22 kV	3.6 + 0.01 m/kV over 22 kV**††	3.6 + 0.01 m/kV over 22 kV**††

* *Clearances are applicable to non-metallic buildings or buildings whose metallic parts are effectively grounded. For other buildings, an assessment might be needed to determine additional clearances for electrostatic induction.*
† *For inaccessible surfaces, this value may be reduced to 0.08 m. At the service attachment point, this value may be further reduced to 0.02 m.*
‡ *This value may be reduced to 1.2 m where the building does not have fire escapes, balconies, and windows that can be opened adjacent to the conductor.*
§ *This value may be reduced to 1.5 m where the building does not have fire escapes, balconies, and windows that can be opened adjacent to the conductor.*
** *Conductors of these voltage classes should not be carried over buildings where other suitable construction can be used.*
†† *Where it is necessary to carry conductors of these voltage classes over buildings, it should be determined whether additional measures, including increased clearances, are needed to ensure that the crossed-over buildings can be used safely and appropriately.*

Note: *Voltages are rms line-to-ground.*

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Table 9
Minimum design clearances from wires and conductors not attached to buildings, signs, and similar plant
(See Clauses 5.7.3.1 to 5.7.3.3.)

Wire or conductor		Minimum clearance, m			
		To buildings*†		To signs, billboards, lamp and traffic sign standards, and similar plant	
		Horizontal to surface‡	Vertical to surface	Horizontal to object‡	Vertical to object
Guys, communication cables, and drop wires		0	0.08	0	0.08
Supply conductors					
0–750 V	Insulated or grounded	1.0	2.5§	0.3	0.5
	Enclosed in effectively grounded metallic sheath	0	0	0	0.08
	Not insulated, grounded, or enclosed in effectively grounded metallic sheath	1.0	2.5§	1.0	0.5
> 0.75kV ≤22 kV	Enclosed in effectively grounded metallic sheath	0	0	0	0.08
	Not enclosed in effectively grounded metallic sheath	2.0‡‡	3.0**	2.0	2.5
> 22 kV**††		2.0 + 0.01 m/kV over 22 kV	3.6 + 0.01 m/kV over 22 kV	2.0 + 0.01 m/kV over 22 kV	3.6 + 0.01 m/kV over 22 kV

* *Clearances over or adjacent to portions of a building normally traversed by persons or vehicles are specified in Tables 2 and 4.*
† *Clearances are applicable to non-metallic buildings or buildings whose metallic parts are effectively grounded. For other buildings, an assessment might be needed to determine additional clearances for electrostatic induction (see Clause 5.7.3.3).*
‡ *Conductor swing shall be added to these values in accordance with Clause 5.7.3.1.*
§ *This clearance may be reduced to 1 m for portions of the building considered normally inaccessible.*
** *Conductors of these voltage classes should not be carried over buildings where other suitable construction can be used.*
†† *Where it is necessary to carry conductors of these voltage classes over buildings, it should be determined whether additional measures, including increased clearances, are needed to ensure that the crossed-over buildings can be used safely and effectively.*
‡‡ *This value may be reduced to 1.5 m where the building does not have fire escapes, balconies, and windows that can be opened adjacent to the conductor.*

Note: *Voltages are rms line-to-ground.*

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Table 10
Minimum design clearances from supply conductors to bridges
(See Clause 5.7.4.2.)

	Minimum design clearance from supply conductor to bridge, m											
	Readily accessible portions*						Inaccessible portions					
	Horizontal		Vertical				Horizontal		Vertical			
			Conductor attached to bridge		Conductor not attached				Conductor attached to bridge		Conductor not attached	
Supply conductor	Conductor attached to bridge	Conductor not attached	Over bridge	Under bridge	Over bridge	Under bridge	Conductor attached to bridge	Conductor not attached	Over bridge	Under bridge	Over bridge	Under bridge
0–750 V	1.2†	1.2†	2.5	0.3	2.5	0.3	0.15	0.3	0.9	1.5	2.5	0.3
> 0.75 ≤ 22 kV	1.5	1.5	2.7	0.9	2.7	0.9	0.5	0.6	1.2	0.5	2.7	0.9
> 22 ≤ 50 kV	1.8	1.8	3.0	1.2	3.0	1.2	0.9	0.9	1.5	0.9	3.3	1.2
> 50 kV ac	Clearances required at > 22 and ≤ 50 kV + 0.010 m/kV over 50 kV											
> 50 kV dc	Clearances required at > 22 and ≤ 50 kV + 0.005 m/kV over 50 kV											

** Readily accessible portions include portions of a bridge likely to be used by workers and portions readily accessible to the public. Bridge seats of steel bridges supported by masonry, brick, or concrete abutments that need periodic inspection should be considered readily accessible portions. Concrete, brick, or masonry portions that need to be passed by workers so that they can gain access to other areas shall be considered readily accessible portions.*

† This clearance may be reduced to 1 m where the portion of the bridge involved is readily accessible to workers but not to the public.

Notes:

1) *Voltages are rms line-to-ground.*

2) *Grounding conductors or conductors installed in effectively grounded conduit do not require clearance under a bridge.*

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Table 11
Minimum separation of equipment and clearances of conductors from swimming pools
(See Clauses 5.7.5 and A.5.7.5 and Figure 1.)

Equipment or conductors	Minimum separation or clearance, m	
	A — Measured in any direction from the water level, edge of pool, or diving platform	B — Measured vertically over land
Supply equipment	4.6	See Clause 5.3.2 and Table 5
Guys, messengers, span wires, communication circuits, secondary cable 0–750 V, and multi-grounded neutral conductors	4.6	See Clause 5.3.1 and Tables 2 and 4
Other supply conductors ≤22 kV	6.7	See Clause 5.3.1 and Tables 2 and 4
Supply conductors > 22 kV and ≤ 150 kV	6.7 + 0.01 m/kV above 22 kV	See Clause 5.3.1 and Tables 2 and 4
Supply conductors > 150 kV	*	See Clause 5.3.1 and Tables 2 and 4

* *Supply conductors greater than 150 kV shall not cross over swimming pools or be closer to the edge of a pool than 8.0 m + 0.01 m/kV above 150 kV.*

Notes:

1) *Voltages are rms line-to-ground.*

2) *See Figure 1 for an illustration of swimming pool clearance limits.*

Table 12
Minimum horizontal separations from supply equipment to buildings
(See Clauses 5.7.6 and A.5.7.6.)

Voltage	Equipment	Minimum separation from building, m			
		Attached equipment		Unattached equipment	
		Normally inaccessible	Readily accessible	Normally inaccessible	Readily accessible
0–750 V	Effectively grounded equipment	0	0	0.15	1.0
	Exposed live parts*	1.0	†	1.0	1.0
> 0.75 ≤ 22 kV	Effectively grounded equipment	0	†	0.15	1.0
	Exposed live parts*	1.5	†	1.5	2.0

* *Exposed live parts include any parts not effectively grounded.*

† *This equipment shall not be attached in any readily accessible location.*

Note: *Voltages are rms line-to-ground.*

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Table 13
Minimum design vertical clearances between wires crossing each other and supported by different supporting structures

(See Clauses 5.8.1.1, 5.8.1.2, A.5.8.1, and A.5.8.1.3.)

Type of line wire, cable, or other plant being crossed over		Line wire or cable at upper level, minimum clearance, m										
		Guys, span wires, aerial grounding conductors, and communication wires and cables	Open supply-line conductors and service wires, ac, kV									
			0–0.75	> 0.75 ≤ 22	> 22 ≤ 50	> 50 ≤ 90	> 90 ≤ 120	> 120 ≤ 150	> 150 ≤ 190	> 190 ≤ 220	> 220 ≤ 320	> 320 ≤ 425
Communication wires and cables		0.2	0.3	0.6	0.9	1.2	1.5	1.8	2.4	2.7	3.9	4.6
Open supply conductors, ac, kV	**0–0.75**	—	0.3	0.3	0.7	1.0	1.2	1.6	2.1	2.6	3.7	4.4
	> 0.75 ≤ 22	—	—	0.4	0.8	1.1	1.3	1.7	2.3	2.7	3.8	4.5
	> 22 ≤ 50	—	—	—	0.9	1.2	1.4	1.9	2.4	2.8	3.9	4.7
	> 50 ≤ 90	—	—	—	—	1.3	1.6	2.1	2.6	3.0	4.1	4.9
	> 90 ≤ 120	—	—	—	—	—	1.8	2.2	2.8	3.2	4.3	5.1
	> 120 ≤ 150	—	—	—	—	—	—	2.4	2.9	3.4	4.5	5.2
	> 150 ≤ 190	—	—	—	—	—	—	—	3.2	3.6	4.7	5.4
	> 190 ≤ 220	—	—	—	—	—	—	—	—	3.7	4.8	5.6
	> 220 ≤ 320	—	—	—	—	—	—	—	—	—	5.4	6.1
	> 320 ≤ 425	—	—	—	—	—	—	—	—	—	—	6.7
Guys, span wires, and aerial grounding wires		0.2	0.2	0.5	0.8	1.2	1.5	1.8	2.4	2.7	3.4	4.6

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Table 13 (Concluded)

Type of line wire, cable, or other plant being crossed over		Line wire or cable at upper level, minimum clearance, m									
		Guys, span wires, aerial grounding conductors, and communication wires and cables	Open supply-line conductors and service wires, dc, kV								
			0–100	> 100 ≤ 150	> 150 ≤ 200	> 200 ≤ 250	> 250 ≤ 300	> 300 ≤ 350	> 350 ≤ 400	> 400 ≤ 450	>450 ≤500
Communication wires and cables		0.2	0.6	0.8	1.1	1.2	1.4	1.7	1.8	2.0	2.2
Open supply conductors, ac, kV	0–0.75	—	0.3	0.6	0.8	1.0	1.3	1.5	1.7	2.0	2.2
	> 0.75 ≤ 22	—	0.5	0.7	0.9	1.1	1.4	1.6	1.8	2.1	2.3
	> 22 ≤ 50	—	0.9	1.0	1.2	1.3	1.5	1.7	1.9	2.2	2.4
	> 50 ≤ 90	—	1.2	1.3	1.4	1.6	1.7	1.9	2.1	2.3	2.5
	> 90 ≤ 120	—	1.4	1.5	1.6	1.8	1.9	2.1	2.2	2.4	2.6
	> 120 ≤ 150	—	1.8	1.9	2.0	2.2	2.3	2.5	2.6	2.8	2.9
	> 150 ≤ 190	—	2.3	2.4	2.6	2.7	2.9	3.0	3.2	3.3	3.5
	> 190 ≤ 220	—	2.8	2.9	3.0	3.2	3.3	3.4	3.6	3.7	3.9
	> 220 ≤ 320	—	—	4.0	4.1	4.3	4.4	4.5	4.7	4.8	5.0
	> 320 ≤ 425	—	—	—	4.9	5.0	5.2	5.3	5.4	5.6	5.7

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Table 13 (Concluded)

Type of line wire, cable, or other plant being crossed over		Line wire or cable at upper level, minimum clearance, m									
		Guys, span wires, aerial grounding conductors, and communication wires and cables	Open supply-line conductors and service wires, dc, kV								
			0–100	> 100 ≤ 150	> 150 ≤ 200	> 200 ≤ 250	> 250 ≤ 300	> 300 ≤ 350	> 350 ≤ 400	> 400 ≤ 450	>450 ≤500
Open supply conductors, dc, kV	Trolley contact conductors*, 0–0.75	0.6	0.9	1.2	1.4	1.5	1.8	2.0	2.1	2.4	2.6
	> 0.75–100	—	0.6	0.9	1.1	1.3	1.6	1.8	2.0	2.3	2.5
	> 100 ≤ 150	—	—	1.0	1.2	1.5	1.7	1.9	2.2	2.4	2.6
	> 150 ≤ 200	—	—	—	1.4	1.6	1.9	2.1	2.3	2.6	2.8
	> 200 ≤ 250	—	—	—	—	1.8	2.0	2.2	2.5	2.7	2.9
	> 250 ≤ 300	—	—	—	—	—	2.2	2.4	2.6	2.9	3.1
	> 300 ≤ 350	—	—	—	—	—	—	2.5	2.8	3.0	3.2
	> 350 ≤ 400	—	—	—	—	—	—	—	2.9	3.1	3.3
	> 400 ≤ 450	—	—	—	—	—	—	—	—	3.3	3.5
	>450 ≤500	—	—	—	—	—	—	—	—	—	3.7
Guys, span wires, and aerial grounding wires		0.2	0.5	0.7	0.9	1.1	1.3	1.5	1.7	1.9	2.1

** Where the trolley pole is not prevented from de-wiring, the clearance shall be increased so as to maintain the minimum specified clearance between the trolley pole in the de-wired position and the crossing conductor.*

Notes:

1) *Voltages are rms line-to-ground.*

2) *Refer to Clause A.5.8.1 for explanation of the calculation method used to produce the clearance values shown in this Table.*

3) *Refer to Table A.1 for typical nominal system voltages and overvoltage values.*

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Table 14
Minimum vertical clearances for crossings over aerial tramways
(See Clause 5.8.1.1.)

	Line wire or cable at upper level, minimum clearance, m											
		ac										dc
Type of plant being crossed over	Communication conductors and cables, span and grounding wires carried aerially	0–750 V	> 0.75 kV ≤ 22 kV	> 22 kV ≤ 50 kV	> 50 kV ≤ 90 kV	> 90 kV ≤ 120 kV	> 120 kV ≤ 150 kV	> 150 kV ≤ 190 kV	> 190 kV ≤ 220 kV	> 220 kV ≤ 320 kV	> 320 kV ≤ 425 kV	0–100 kV
Aerial tramways Gondolas and similar apparatus providing a roof over passengers	0.15	0.15	0.5	0.8	1.2	1.5	1.8	2.4	2.7	3.9	4.6	0.5
Chairlifts, T-bars, and similar apparatus or towers of any type of aerial tramway	1.7	1.7	2.0	2.3	2.7	3.0	3.4	3.9	4.3	5.4	6.0	2.0

Note: *Voltages are rms line-to-ground.*

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Table 15
Clearances between conductors supported by different structures but not crossing each other — Clearance increments to be added to horizontal displacement
(See Clauses 5.8.2 and A.5.8.2.)

Sum of voltages of conductors*	Clearance increment, mm
0–750 V	300
> 750 V ac	300 + 10 mm/kV over 750 V
> 750 V dc	300 + 6 mm/kV over 750 V

* *Communication conductors are assumed to operate at less than 130 V dc at 0.25 A or less per pair (1500 pairs or fewer).*

Note: *Voltages are rms line-to-ground.*

Table 16
Minimum design clearances between conductors of one line and supporting structures of another line
(See Clause 5.8.3.1.)

Voltage of line conductor*, kV	**Minimum clearance between conductor and supporting structure, mm**
0–5	1000 wherever practicable, but in no case less than (a) 150 for spans 0–6 m; (b) 230 for spans > 6 and ≤ 15 m; and (c) 300 for spans > 15 m
> 5 and ≤ 22	1000 wherever practicable, but in no case less than 500
> 22 and ≤ 50	1000
> 50	1000 + 10 mm/kV over 50 kV

* *Communication conductors are assumed to operate at less than 130 V dc at 0.25 A or less per pair (1500 pairs or fewer).*

Note: *Voltages are rms line-to-ground.*

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Table 17
Minimum horizontal separations of supply-line conductors attached to the same supporting structure

(See Clauses 5.9.1.1, 5.9.1.2, and 5.9.1.3.)

Line conductor	Minimum separation of conductors for spans ≤ 50 m, mm		Minimum separation of conductors for spans > 50 m and ≤ 450 m*, mm
0–5 kV ac†	Span length	Separation	300 + the following three increments: (a) 3 × the distance (in m) by which the span length exceeds 50 m; (b) 83 × the final unloaded sag (in m) at 15 °C conductor temperature for conductor (s) having the greatest sag; and (c) 10 mm/kV over 5 kV.
	0–6 m	225	
	> 6 m and ≤ 50 m	300	
> 5 kV ac†	300 + 10 mm/kV over 1 kV		
0–750 V dc, same polarity, different circuits‡	150		150 + increments (a) and (b) above
0–750 V dc, opposite polarity	300		300 + increments (a) and (b) above
> 0.75 kV and ≤ 22 kV dc, same polarity, different circuits‡	Span length	Separation	300 + increments (a) and (b) above
	0–6 m	225	
	> 6 m and ≤ 50 m	300	
> 0.75 kV and ≤ 22 kV dc, opposite polarity	375		375 + increments (a) and (b) above
> 22 kV dc	375 + 5 mm/kV over 22 kV		375 + 5 mm/kV over 22 kV plus increments (a) and (b) above

* *For spans longer than 450 m, the separation shall be based on best engineering practices, but shall be not less than the separations specified for spans of 450 m.*

† *Phase-to-phase voltages for the same circuit or the sum of phase-to-ground voltages for different circuits.*

‡ *There is no minimum separation for conductors of the same polarity in the same circuit.*

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Table 18
Minimum vertical separations between supply-line conductors attached to the same supporting structure, ac
(See Clause 5.9.2.1.)

	Minimum vertical separation, m				
	Conductors at higher level				
Conductors at lower level	**0–750 V**	**> 0.75 kV ≤ 5 kV**	**> 5 kV ≤ 22 kV**	**> 22 kV ≤ 50 kV**	**> 50 kV ≤ 90 kV**
0–750 V*	0.1	0.4	0.4	1.0	1.5
> 0.75 kV ≤ 5 kV	—	0.4	0.4	1.0	1.5
> 5 kV ≤ 22 kV	—	—	0.4†	1.0	1.5
> 22 kV ≤ 50 kV	—	—	—	1.0†	1.5
> 50 kV ≤ 90 kV	—	—	—	—	1.5†

* *This category also includes dc.*
† *This value does not apply to conductors on adjacent supports of the same circuit or circuits.*

Note: *Voltages are rms line-to-ground.*

Table 19
Vertical spacings between supply-line conductors of 0 to 750 V on vertical brackets
(See Clause 5.9.2.2.)

Maximum span length, m	Minimum average spacing at brackets, mm
20	100
45	150
60	200
75	250

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Table 20
Minimum in-span vertical clearances between supply conductors of the same circuit that are attached to the same supporting structure

(See Clause 5.9.2.3.)

	Minimum clearance, mm	
Maximum circuit line-to-ground voltage, kV	**Between multi-grounded neutral and circuit conductor**	**Between circuit conductors**
0.750	100	100
5	150	150
10	150	250
15	250	300
22	350	450
50	750	900
90	1000	1250

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Table 21
Minimum in-span vertical clearances between supply conductors of different circuits that are attached to the same supporting structure
(See Clause 5.9.2.4.)

Maximum lower conductor line-to-ground voltage, kV	Minimum in-span vertical clearance, mm						
	Maximum upper conductor line-to-ground voltage, kV						
	5	10	17	22	30	50	90
0.750	150	150	250	350	450	750	1000
5	150	200	250	350	450	750	1000
10	—	250	300	400	450	800	1050
17	—	—	300	400	500	800	1050
22	—	—	—	450	500	800	1050
30	—	—	—	—	550	850	1100
50	—	—	—	—	—	900	1150
90	—	—	—	—	—	—	1250

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Table 22
Minimum separations or clearances in any direction from supply conductors to other supply plant attached to the same supporting structure
(See Clauses 5.9.5 and A.5.9.5.)

	Separation or clearance, mm						
	Voltage of conductor(s), ac*				Voltage of conductor(s), dc		
Between	**0–750 V**	**> 0.75 kV ≤ 5 kV**	**> 5 kV ≤ 22 kV**	**> 22 kV ≤ 50 kV†**	**0–750 V**	**> 0.75 kV ≤ 22 kV**	**> 22 kV ≤ 50 kV†**
Supply lateral, vertical, or line conductors *and* supply lateral or vertical conductors of the same or different circuits but not connected together	75	100	300	‡	75	120	§
Supply lateral, vertical, or line conductors *and* surface of structure, crossarms, and other non-energized supply plant, including grounding conductors	75**	100	150	‡	75‡	120	§
Supply lateral, vertical, or line conductors *and* span or guy wire, except where conductors are supported by the span wire	100††	170†††‡‡	380	‡††	75††	200†††‡‡	§
Supply-line conductor *and* lightning protection wire parallel to the line	§§	§§	§§	§§	§§	§§	§§

* *Clearances from connecting wires to switches or arresters may be less than that from line wires to switches or arresters.*
† *For voltages exceeding 50 kV, clearances shall be based on best engineering practices, but shall be not less than the separations specified for 50 kV.*
‡ *Add 10 mm/kV over 22 kV.*
§ *Add 5 mm/kV over 22 kV.*
** *This clearance may be reduced to 0 mm for conductors and cables 0–750 V, provided that the conductors are attached to the surface of the structure and grounded or covered by material of adequate insulating and mechanical properties.*
†† *Where this clearance cannot be achieved, adequate insulation shall be applied to the supply conductor.*
‡‡ *This clearance shall be increased to 300 mm for voltages 0.75–8 kV for a span wire or span guy wire running parallel to the supply conductor.*
§§ *The clearance shall be not less than the separations specified in Clause 5.9.1.*

Note: *Voltages are rms line-to-ground.*

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Table 23
Minimum vertical separations at a joint-use structure
(See Clauses 5.10.1.1, 5.10.1.6, 5.10.1.7, 5.10.6.2, and A.5.10.1 and Figures A.6 and A.7.)

Between	Minimum vertical separation, m Voltage of supply conductors 0–750 V	 > 0.75 kV ≤ 22 kV	 > 22 kV ≤ 50 kV*
Live or current-carrying supply plant (including neutrals) *and* communication line plant	1.0†	1.2‡	1.5‡
Non-energized supply plant (excluding luminaire span wire and brackets) *and* communication line plant			
Option A§			
Ungrounded	1.0†	1.0	1.5
Effectively grounded	0.75†	0.75†	1.0
Option B§			
Ungrounded	0.75	1.0	1.5
Effectively grounded	0.10	0.30	0.30
Trolley span wires or brackets *and* communication line plant	0.30	N/A	N/A
Luminaire span wires or brackets *and* communication line plant			
Ungrounded	1.0	1.0	N/A
Effectively grounded	0.10	0.10	N/A
Point of attachment of combined communication drop and supply service conductor *and* communication line plant	1.0†	N/A	N/A
Housing containing communication power supply, communication compressor dehydrator, or other communication equipment (effectively grounded or insulated) *and* communication cable	0	N/A	N/A

* *For voltages greater than 50 kV, add 0.010 m/kV over 50 kV.*
† *On lateral communication drop wire plant, this separation may be reduced to 0.6 m.*
‡ *See Clause 5.2.6 for requirements for neutral conductors.*
§ *Option A or Option B shall be selected in accordance with Clauses 5.10.1.6 and 5.10.1.7.*

Note: *Voltages are rms line-to-ground.*

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Table 24
Minimum in-span vertical clearances between supply and communication conductors
(See Clauses 5.10.3.2 and A.5.10.3.)

Voltage of supply conductor	Minimum clearance of supply conductor above line of sight of points of support of highest communication wire or cable, mm
0–750 V with thermoplastic weatherproof covering	2*
0–750 V with other covering or bare	75*†
> 0.75 kV and ≤ 15 kV	300
> 15 kV and ≤ 22 kV	380
> 22 kV and ≤ 250 kV	380 + 10 mm/kV over 22 kV

* *For effectively grounded neutral, see Clause 5.10.3.3.*
† *While the use of the design limit of 75 mm can yield a minimum actual clearance of approximately 300 mm under the worst expected conditions, most situations will result in clearances in excess of 600 mm.*

Note: *Voltages are rms line-to-ground.*

Table 25
Minimum clearances from guys to plant of another system
(See Clauses 5.11.2.1 and A.5.11.2 and Figure A.8.)

Type of plant near which the guy passes	Minimum clearance, m
Communication line plant	0.6
Trolley contact conductors, dc	
0–750 V	1.2
> 0.75 kV and ≤ 22 kV	1.8
> 22 kV	1.8 + 0.01 m/kV over 22 kV
Current-carrying supply plant, ac	
0–750 V	0.6
> 0.75 kV and ≤ 22 kV	1.2
> 22 kV	1.2 + 0.01 m/kV over 22 kV
Supply guy wires and span wires	0.6

Note: *Voltages are rms line-to-ground.*

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Table 26
Minimum clearance or separation between guys and other plant attached to the joint-use structure
(See Clauses 5.11.2.2 and A.5.11.2 and Figure A.8.)

Type of plant over or near which the guy passes	Minimum clearance or separation from guy, mm	
	Guy not parallel to plant	Guy parallel to plant
Communication line plant	150*	75
Current-carrying supply plant		
0–750 V	150	150*
> 0.75 kV and ≤ 5 kV	150	230
> 5 kV and ≤ 15 kV	230	300
> 15 kV and ≤ 22 kV	300	380
> 22 kV	300 + 10 mm/kV over 22 kV	380 + 10 mm/kV over 22 kV

* *This clearance may be reduced to 75 mm where adequate insulation is provided.*

Note: *Voltages are rms line-to-ground.*

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Table 27
Minimum grades of construction for crossings
(See Clauses 6.3.1, 6.3.2.1, 7.8.1.1, 7.8.2.1, 7.8.2.4, 7.9.1, A.7.8.2, and A.8.1.)

	Minimum grade of construction where the conductors, messengers, or cables are at the upper level		
Item at lower level	**Communication**	**0–750 V**	**> 750 V**
Railway control facilities and tracks	1	1	1
Limited- or controlled-access highways	1	1	1
Roads and highways — General	3	3	3
Pipelines	3	3	3
Navigable waterways requiring permits	1	1	1
Aerial tramways	1	1	1
Other private or public property	3	3	3
Communications			
Cable	3	3	1*
Open wire — General	3	3	1*
Drop wire	3	3	3
Supply			
0–750 V	2†	3	2‡
> 750 V	1†	2	2

** The grade of construction may be Grade 2 where one of the following conditions exists:*
a) the supply and communication lines have coordinated electrical protection (see Clause 4.4);
b) where coordinated electrical protection is not practical, the supply conductors have a rated tensile strength of 12.4 kN or greater; or
c) the supply conductors are enclosed in effectively grounded continuous metallic sheathed cable.
† The communication line may be Grade 3 where the supply conductors are in effectively grounded continuous metallic sheathed cable.
‡ Grade 3 construction may be used where the supply conductors at the upper level are in effectively grounded continuous metallic sheathed cable.

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Table 28
Minimum grades of construction for proximities
(See Clauses 6.3.1, 6.3.3, 7.8.1.1, 7.8.2.1, 7.8.2.4, 7.9.1, A.7.8.2, and A.8.1.)

	Minimum grade of construction		
In proximity to	**Communication conductors, messengers, or cables**	**Conductors 0–750 V or effectively grounded continuous metallic sheathed supply cables of all voltages**	**Conductors > 750 V**
Railway control facilities and tracks	1	1	1
Limited- or controlled-access highways	1	1	1
Roads and highways — General	3	3	3
Pipelines and waterways that do not require permits	3	3	3
Navigable waterways requiring permits	1	1	1
Aerial tramways	1	1	1
Other private or public property	3	3	3
Communications			
Cable	3	2	1*
Open wire — General	3	2	1*
Drop wire	3	3	3
Supply			
0–750 V	2	3	2†
> 750 V	1	2	2

* *The line may be Grade 2 if the supply and communications lines have coordinated electrical protection (see Clause 4.4).*
† *Grade 3 construction may be used where the supply conductors at the upper level, or the supply conductors creating the proximity, are in effectively grounded continuous metallic sheathed cable.*

Table 29
Minimum grades of construction for joint use
(See Clauses 6.3.1, 6.3.4, 7.8.1.1, 7.8.2.1, 7.8.2.4, 7.9.1, A.7.8.2, and A.8.1.)

	Minimum grade of construction	
Conductor at lower levels	**Conductors 0–750 V or effectively grounded continuous metallic sheathed supply cables of all voltages**	**Conductors > 750 V**
Communications		
Cable	2	1*
Open wire in general	2	1*

(Continued)

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Table 29 (Concluded)

Conductor at lower levels	Minimum grade of construction	
	Conductors 0–750 V or effectively grounded continuous metallic sheathed supply cables of all voltages	Conductors > 750 V
Drop wire	3	3
Supply		
All voltages	3	3

* *The grade of construction may be Grade 2 where one of the following conditions exists:*
a) the supply and communications lines have coordinated electrical protection (see Clause 4.4);
b) where coordinated electrical protection is not practical, the supply conductors have a rated tensile strength of 12.4 kN or greater; or
c) the supply conductors are enclosed in effectively grounded continuous metallic sheathed cable.

Table 30
Deterministic weather loads*†
(See Clauses 7.2, 7.3.1, 7.3.5, 7.5.1, 7.6.2.1, and A.7.3.)

Loading conditions	Loading area‡			
			Medium	
	Severe	Heavy	A	B
Radial thickness of ice, mm	19	12.5	6.5	12.5
Horizontal wind loading, N/m²	400	400	400	300
Temperature	–20 °C	–20 °C	–20 °C	–20 °C

* *When large conductors are used in locations where high winds are prevalent, the loading on bare conductors should be assessed, as high winds can produce a transverse loading in excess of the wind and ice combinations.*
† *On small conductors, the ice effect is more significant than on large conductors, as it constitutes a greater percentage of the total mass.*
‡ *See Clause 7.2 and Annex C.*

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Table 31
Minimum load factors for non-linear analysis of structures
(See Clauses 7.4.2, 7.5.2, 7.6.4, 7.7.2, 7.8.1.2, 8.1.3, 8.3.10.1, 8.3.12.2, and A.7.4.2.)

		Minimum load factor		
		Material		
Type of load	Construction grade	COV ≤ 10%	10% < COV < 15%	COV ≥ 15%
Vertical	1	1.30	1.60	2.00
	2	1.15	1.30	1.50
	3	1.00	1.10	1.20
Transverse	1	1.20	1.50	1.90
	2	1.10	1.20	1.30
	3	1.00	1.10	1.10
Longitudinal with terminations or tension changes	1	1.20	1.50	1.90
	2	1.10	1.20	1.30
	3	1.00	1.10	1.10
Longitudinal without terminations or tension changes	1	1.10	1.20	1.20
	2	1.00	1.00	1.00
	3	1.00	1.00	1.00

Notes:

1) *Non-linear analysis, including stability (buckling) check, is the method of analysis of structures specified in this Standard.*

2) *Default COV values for pole and structure materials are:*
- a) *steel — ≤ 10%;*
- b) *concrete — 10% < COV < 15%; and*
- c) *wood — ≥ 15%.*

3) *The selection of appropriate minimum load factors for steel, concrete, fibre-reinforced composite, laminated wood, and wood poles is based on the coefficient of variation (COV), for the given pole material as verified by the pole manufacturer. The lowest minimum load factors (i.e., steel pole default values) may be used for concrete, laminated wood, and fibre-reinforced composite structures, provided that the manufacturers can verify through required testing that the COV of the poles in question is less than or equal to 10%. Similarly, the next lowest minimum load factors (i.e., concrete pole default values) may also be used for fibre-reinforced composite and laminated wood structures, provided that the manufacturers can verify through required testing that the COV of the poles in question is greater than 10% but less than 15%. The highest minimum load factors (i.e., wood pole default values) shall be used for naturally grown wood, fibre-reinforced composite or laminated wood poles in the event that the manufacturer is unable to verify that the COV of the poles in question is less than 15%.*

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Table 32
Minimum load factors for guy assemblies at installation
(See Clauses 7.4.2, 7.6.4, 7.7.2, 7.8.1.2, 7.8.2.5, 8.3.10.2, 8.3.12.3, and A.7.4.2.)

Application	Minimum load factor
Guy assembles including anchors and foundations	1.1

Table 33
Conductor tensioning — Recommended catenary parameter
(See Clauses 8.7.3 and A.8.7.3.)

Span length, m	Recommended initial maximum catenary*, m
> 400	2000
> 350 and ≤ 400	1900
> 300 and ≤ 350	1800
> 250 and ≤ 300	1700
> 200 and ≤ 250	1600
> 150 and ≤ 200	1500
> 100 and ≤ 150	1400
< 100	1300

* *The reduction of the catenary value in relation to the span is based on the principle that lower tensions are safer than higher ones from the point of view of aeolian vibrations. This reduction will not affect the sags significantly, but will reduce loads on angle structures.*

Table 34
Maximum tensions for galvanized steel wire strand lightning protection wires
(See Clause 8.8.2.)

Loading conditions	Maximum tension, % of rated tensile strength
Maximum allowable tension under appropriate loading conditions	60
Maximum recommended initial unloaded tension at the mean annual temperature	20
Maximum recommended final unloaded tension at the mean annual temperature	15

Note: *The unloaded tension is intended only as a guide. The unloaded tension may be increased when factors that produce conductor fatigue are at a minimum or when suitable steps are taken to suppress fatigue failure.*

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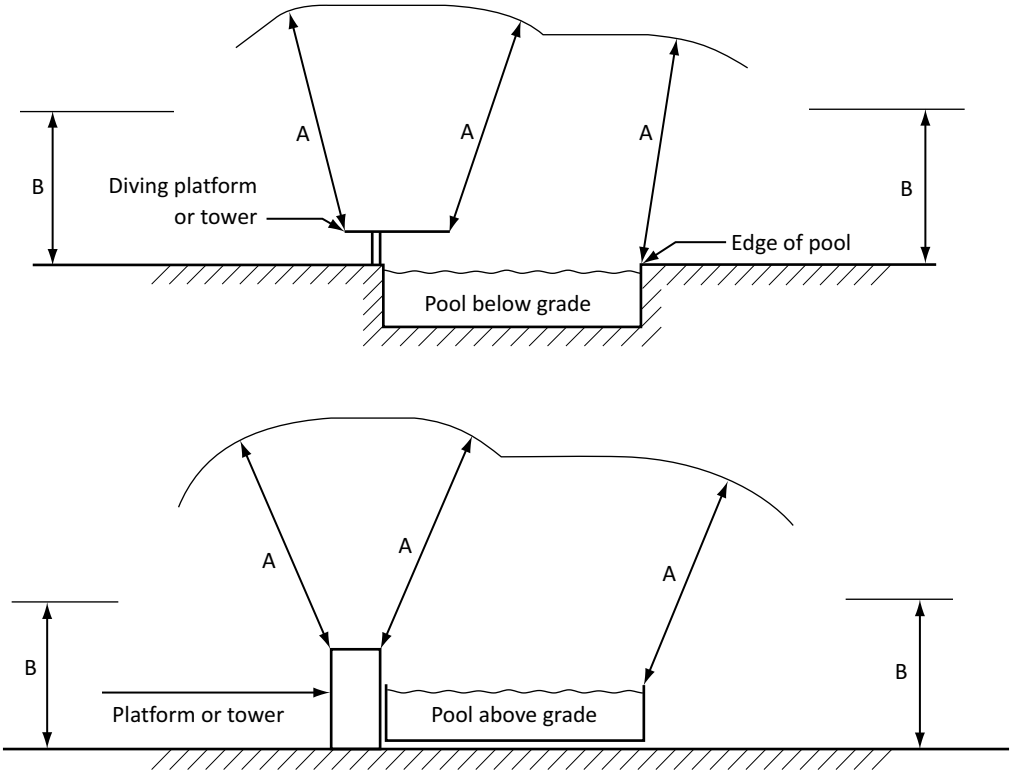
Table 35
Flashover distance for ac conductors for tree pruning
(See Clause 4.1.7.)

Maximum line-to-ground voltage, kV	Flashover-to-ground distance, mm
0.75	8
5	72
10	144
15	216
22	316
50	718
90	970
120	1185
150	1586
190	2129
220	2566
250	3000
318	3640
442	4600

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Figure 1
Swimming pool clearance limits
(See Clause 5.7.5 and Table 11.)



Note: *See Table 11 for the minimum clearance values for A and B.*

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Annex A (informative)
Commentary

Notes:
1) *This Annex is not a mandatory part of this Standard.*
2) *The clause numbers in this Annex correspond to the clauses in this Standard.*
3) *The tables and figures in this Annex do not correspond to the tables and figures in this Standard.*

A.1 Scope

A.1.2
The intent of this Clause is to permit overhead lines that have been constructed in compliance with a prior version of the Standard to remain in service in the event of a subsequent revision of this Standard, without being modified to comply with the revised Standard.

This Clause is not intended to permit the addition of new line sections, new conductors, new attachments, or new equipment to an existing line that do not comply with the current version of the Standard. Such new additions are intended to comply with the latest revision of the Standard.

Maintenance of an existing line, including repair, or replacement where necessary, of failed or failing components of the line (including poles, conductors and other equipment), is permitted without modifying the line to meet the latest revision of the Standard.

A.1.5
The requirements of this Standard are designed to provide a minimum acceptable class of construction for application over a broad area. Requirements more stringent than the specified minimum should be met where

- a utility's own experience indicates that more severe weather conditions are common in specific areas;
- it is necessary to obtain acceptable radio influence voltage (RIV); or
- the security level of a circuit warrants more stringent requirements.

A.3 Definitions and abbreviations

A.3.1 Definitions
The user of this Standard should fully understand the definitions and note that, in accordance with the definitions, specified voltages are voltages-to-ground unless otherwise noted.

Definition of rated tensile strength (RTS)
General
The rated tensile strength (RTS) of a conductor is a calculated value based on the nominal area and minimum tensile stress value of each of its component(s). The term RTS is preferred over ultimate tensile strength (UTS) because it is a tensile rating, a consistent number assigned to the conductor, whereas the ultimate tensile strength implies a variability inherent to both the material composing the conductor and the testing method used.

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The RTS is usually provided as a "stand alone" number, but it should be understood that it applies at a given temperature, for the following two reasons:

- the tensile properties of the conductor's components may vary with temperature. For example, aluminum's strength increases at low temperature, and decreases at high temperature, while steel strength is relatively constant over the temperature range experienced by overhead conductors; and
- the conductor's components are assumed to have the same length, which occurs at only one temperature (usually room temperature). A change in temperature changes the relative length of each component as well as the stress distribution within the conductor. This would change the *RTS*.

The *RTS* of a conductor composed of n components is calculated using the following equation:

$$RTS = \sum_{i=1}^{n} (f_{1i} f_{2i} \delta_i A_i)$$ **Equation 1**

where

RTS = Conductor's rated tensile strength, N

f_{1i} = Strength reduction factor related to elongation, for component *i*

f_{2i} = Strength reduction factor related to the lay, for component *i*

δ_i = Minimum tensile stress of component *i*, MPa

A_i = Nominal area of component *i*, mm^2

Strength reduction factor

Equation 1 contains two strength reduction factors (SRF). In most cases, the factors are equal to 1.0, and the RTS is simply the sum of the product of the area by the tensile stress of each component. The role of each SRF is explained below.

SRF f_{1i} is used to adjust the minimum tensile stress of a component that exceeds 1% elongation at rupture. As stated above, stress values used in Equation 1 assume that all components have the same elongation at rupture. Since A1 (1350-H19) wires usually break at, or around, 1% elongation, the tensile stress of any other component used in conjunction with these wires must be taken at 1% elongation as well.

Note: *For A1/Sxy constructions (ACSR), the steel core stress is taken at 1% elongation.*

When combined with A2 (6101-T81) or A4 (6101-T83) wires to form an A1/A2 or A1/A4 (ACAR), the tensile stress of the alloy wires must be reduced as their elongation at rupture is around 3.5%. It is the only *SRF* prescribed by CAN/CSA-C61089 and has a value of 0.95. The use of this *SRF* is illustrated in Example 1 below.

Example 1 — RTS Calculation for a A1/A2 conductor according to CAN/CSA C61089:

Conductor: 400-A1/A2-30/7		
	A1 Component	**A2 Component**
f_{1i},	1.0	0.95 (CAN/CSA-C61089)
f_{2i},	1.0	1.0
δ_i, MPa	165 (CAN/CSA-C60889)	305 (CAN/CSA-C60104)

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Conductor: 400-A1/A2-30/7		
	A1 Component	**A2 Component**
A_i, mm²	332.6	77.60
$RTS = (1.0 \times 1.0 \times 165 \times 332.6) + (0.95 \times 1.0 \times 305 \times 77.60) = 77360$ N		

SRF f_{2i} is used to take into account the spiraling (the *lay*) of the wires in the conductor. ASTM (American Society for Testing and Materials) is the only standards organization prescribing them. Presumably, it assumes that the stranding operation changes the mechanical properties of the wires, reducing their tensile strength.

Note: *Most national and international conductor Standards recommend to decrease the mechanical requirements of wires if tested after stranding because of the manipulation required to straighten the samples.*

The lay produces radial reactions from each wire when the conductor is subjected to a pure tensile force, increasing the stress on each wire. This could adversely affect the value of the RTS.

ASTM has a set of standard SRFs as a function of the number of layers in a conductor. The Table below is an example of the factors to be applied to steel cores. Similar values of SRFs exist for all types of constructions.

Example of ASTM SRFs for steel cores (Reference ASTM B500/B500M):

Number of wires in core	Strength reduction factor
1	0.96*
7	0.96
19	0.93
37	0.91
61	0.90

**From ASTM B232/B232M.*

Example 2 below illustrates the use of both of those SRF applied to the calculation of the RTS for the same conductor as Example 1. Note that this calculation is made in accordance with ASTM.

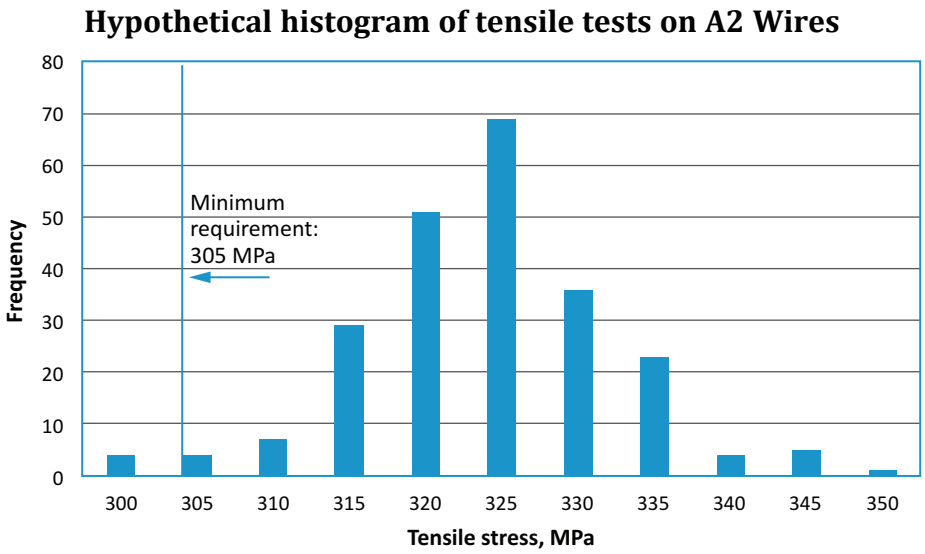
Example 2 — RTS Calculation for an ACAR according to ASTM B524/B524M:

Conductor: 400mm² 30/7 ACAR		
	1350-H19 Component	**6201-T81 Component**
f_{1i},	1.0	0.95 (ASTM B524/B524M)
f_{2i},	0.93 (ASTM B524/B524M)	0.96 (ASTM B524/B524M)
δ_i, MPa	170 (ASTM B230/B230M)	315 (ASTM B398/B398M)
A_i, mm²	324.3	75.67
$RTS = (1.0 \times 0.93 \times 170 \times 324.3) + (0.95 \times 0.96 \times 315 \times 75.67) = 73{,}010$ N		

Minimum tensile strength

The minimum tensile stress is a value determined at the time when the material is developed. It is selected low enough to allow most of the samples to be above its limit. The Figure below is a hypothetical histogram of a series of tensile tests on a production run of A2 wires.

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The choice of the minimum tensile stress limits the number of non-conforming samples that would otherwise inflate the cost of the conductor for the user.

A.4 General requirements

A.4.1 General design and maintenance

Owners of lines should be aware of the specific terrain, exposure to climate, traffic, accessibility, and other relevant matters concerning their lines and equipment, and should have the competence to design, build, and maintain the lines accordingly.

The requirements of this Standard are the minimum considered to be adequate for the expected conditions. These minimum requirements are based on reasonable electrical and mechanical criteria and do not include additional amounts for special circumstances. Where conditions are expected to be more severe, the owner is responsible for exercising judgment and for making appropriate adjustments in design to meet the general intent of this Standard or the objectives of good practice.

A.4.1.4 Climbing space

For examples of climbing space involving drop wires, see Figure A.1.

Additional requirements for tall structures should be based on good practice. Individual safety authorities might necessitate more specific measures.

A.4.2 Structures and attachments

During the selection of a site, consideration should be given to the protection of the new structure from damage, as well as to any hazard the new structure might present. The possibility of unauthorized access by means of the structure to adjacent property, or any easily accessible means allowing the public to climb the structure, should be eliminated to the greatest extent practicable.

These considerations also apply to the maintenance of existing structures. Attention is needed to ensure correction of any unsatisfactory conditions arising after the plant has been constructed.

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A.4.2.4.2 Mechanical protection of supply cables

The protective covering of riser cables of supply systems in Clause 4.2.4.2 is not intended to protect against vehicular accidents, acts of vandalism, severe weather events, or similar incidents, etc.

During the selection of material strength and location of the covering, consideration should be given to provide protection suitable for the situation and hazards to which the riser cables are reasonably expected to be exposed. Examples of potential hazards include casual contact from sources such as pole climbing by trades, use of hand tools in the vicinity of the riser, or incidental contact by the public.

A.4.2.7 Grounding and insulating of guys

A.4.2.7.1 General

Clause 4.2.7.1 is intended to protect the public and workers from contacting an energized guy wire. Where a guy is located above an energized conductor, the guy needs to be insulated or separately grounded so that, if the guy is broken at the ground line, its upper portion will remain insulated or grounded. The separate ground may be provided by connection to a multi-grounded neutral, another grounded guy, or any other effective ground.

Where the guy wire is located below the conductor, the only hazard is contact between a broken conductor and the guy wire. The metal anchor rod may be considered an effective ground where the soil resistance is suitably low. When using a guy as an effective ground where the guy is not continuously under tension, it might be necessary to ensure the electrical conductivity of the mechanical connection between the guy wire and the anchor by means of a jumper or similar device.

A.4.4 Coordinated electrical protection

Every attempt should be made to achieve coordinated electrical protection between power and communication circuits; a stronger grade of construction is not an equivalent (see CSA C22.3 No. 5.1). However, stronger grades of construction may be used where they are the only option.

A.5 Clearances, separations, and spacings

Clause 5 covers clearances, separations, and spacings involving overhead line components and their relationships to each other, to the ground, and to other plant.

The following basic factors were considered in the preparation of Clause 5:

- a protected zone that accounts for such factors as the height of a vehicle, the reach of a pedestrian, and the access area around a building or equipment;
- a clearance (separation or spacing) between grounded or non-energized supply plant and other grounded or non-energized plant, persons, or objects that offers a protected zone and provides for the swing of wires or the swing of conductors and insulators;
- a clearance from an energized supply plant to other energized or non-energized objects or persons that offers a protected zone and provides for the greater of the distance needed to withstand an electrical flashover and the distance needed to prevent an unsafe level of induced current in a vehicle under or adjacent to an extra-high-voltage line; and
- a small buffer added to the items specified above. For example, for road crossings, a rounding of the clearance requirement results in a buffer of about 0.3 m.

The electrical flashover distance used when calculating clearances is based on the voltage under switching surge conditions. To arrive at distances considered safe, adequate values of surge overvoltages were chosen, based on systems being designed at the time of drafting of this Standard.

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The clearances in the tables, however, apply an increment for voltage to the normal (rms) operating voltage. The values assumed for the switching surge overvoltage factors, the surge voltage, and the resulting air gap distance are given in Tables A.1 and A.2.

The examples for Tables 2 and 4 given in Clause A.5.3.1 show the justification for applying a voltage increment of 0.01 m/kV to the rms voltage by calculating the clearance in two ways: using the test data and using the simplified approximation of 0.01 m/kV.

This increment for voltage of 0.01 m for each kilovolt of normal (rms) operating voltage has been commonly used in both the *Canadian Electrical Code* and ANSI/IEEE-C2. This is a derived value applied to the normal (rms) operating voltage, but calculated so that it accounts for switching overvoltage conditions.

The following air gap flashover data for ac and dc systems are used in the basic calculations of Clause 5.2:

- AC systems — Full-scale wet switching surge tests on ac conductor-to-ground plane configurations have shown that the flashover strength of air gaps is relatively uniform up to a switching surge voltage of approximately 500 kV. Beyond this, the strength of air gaps is reduced (i.e., a given air gap withstands a smaller voltage). Using this data, the following values are derived for use in this Standard:
  - for 500 kV or less, a safe approximation for the air gap withstand is 2.54 mm/kV; and
  - for greater than 500 kV, a safe approximation for the air gap withstand is 3.81 mm/kV.
- DC systems — From full-scale switching surge tests on dc conductor-to-ground configurations, the following values are used for this Standard:
  - for 144 kV or less, the air gap withstand is 2.0 mm/kV; and
  - for greater than 144 kV, the air gap withstand is 2.92 mm/kV.

A.5.2 General application

A.5.2.2 Vertical design clearances

When the maximum sag is determined by ice and wind conditions, it is recognized that the plane of the conductor is in a position of swing; however, for simplicity, it is assumed that the maximum sag is the sag in the vertical plane. Designers must check the specified vertical clearance-to-ground for hillside terrain. See Annex B for an explanation of stress-strain characteristics and conductor sag.

A.5.2.3 Horizontal design clearances

Horizontal clearances for energized plant are based on the following:

- For conductors that are free to swing and are in the swing position as specified in Clause 5.2.7, the safe electrical component of the total clearance is taken as the switching surge withstand voltage of the air gap.
- For energized utility plant that is fixed (i.e., not free to move in the wind), the separation between live parts or between a live part and ground must withstand the switching surge voltage of the air gap.

A.5.2.5 Conductor temperature for thermal loading conditions

The thermal loading requirements are divided into three categories. In the first two cases, a specified wire or conductor temperature is used to determine maximum sag. The other case allows a utility engineer to calculate the conductor temperature for the purpose of determining maximum sag.

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The maximum conductor temperature may be calculated using ANSI/IEEE-738. Also, it is intended that utilities be permitted to structure their maximum current to fit a variety of ambient conditions, provided that the resulting conductor temperature does not exceed the value that ensures that specified minimum clearances are met. It follows that short-term currents or emergency currents exceeding the maximum planned summer operating current may be carried, provided that they do not heat the conductor sufficiently to reduce the clearance below the specified minimum at the prevailing ambient conditions. Designers should ensure that, under emergency current conditions, the clearances specified in this Standard will be maintained.

Curves are included to show the conductor temperature rise above 40 °C ambient for various sizes of conductors (see Figure A.2).

Clause 5.2.5.2 does not preclude the planned summer operating of lines at currents exceeding rated ampacity, provided that the specified standard minimum clearances are met for the conductor temperatures calculated in accordance with Clause 5.2.5.2.

A.5.2.7 Wire or conductor swing for horizontal design clearances

The wire or conductor swing is based on a wind pressure of 230 N/m^2 on the projected area of the span of conductor. This represents a low-velocity wind of 18 m/s, considered to occur frequently enough to coincide with a switching surge overvoltage. (An air gap adequate to withstand the assumed value of switching surge voltage is added to the distance required for conductor swing.) The 18 m/s wind velocity is an empirical value; therefore, the 230 N/m^2 is used over the broad range of conductor sizes as a minimum value. In open terrain where severe winds occur, there can be situations where clearances might need to be increased for more severe wind conditions, but this is left to good practice.

The general formula for velocity pressure on wires is as follows:

$P = C_d \times 1/2p \times V^2$

where

P = velocity pressure, N/m^2

C_d = drag coefficient or shape factor

p = air density, kg/m^3

V = wind speed, m/s

Note: *Table 1 assumes a value of $C_d \times 1/2p = 0.719$.*

The curves shown in Table 1 are derived by using the following formula:

$$Factor = \sin\left\{a \tan\left[\left(\frac{P}{1000G}\right)\left(\frac{d}{w}\right)\right]\right\}\{SSF\}$$

where:

Factor = the values shown on the abscissa of Table 1

P = velocity pressure, 230 N/m^2

G = gravitational acceleration, 9.806 m/s^2

d/w = the values shown on the ordinate of Table 1

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SSF = Sheltered span factor, a value that adjusts the values from a non-sheltered span to the values for a partially sheltered span. For non-sheltered span, *SSF* = 1 and for a partially sheltered span, *SSF* = 0.40

A.5.2.8 Insulator string swing for clearance calculations — Transmission line structures

Tangent suspension insulator strings commonly used on transmission lines are subject to swing due to wind pressure. Sufficient clearances between the live end of an insulator string and any grounded component of a support structure (including guy wires) must be maintained to provide for safe and reliable operation of a transmission line.

The following three wind conditions should be considered when designing supporting structure top geometry, as shown in the Figure below.

Case 1: High wind condition

In this case, an insulator string is exposed to a high wind that is expected to occur very rarely, while the system is under power frequency voltage only. High wind is defined as the 5-year return design wind.

If wind statistics are not available, 75% of the 50-year return design wind speed (56% of the wind pressure) may be used in calculations.

Other wind values can be used based on local experience and satisfactory performance.

Case 2: Moderate wind condition

In this case, an insulator string is exposed to moderate wind, frequent enough to coincide with a switching surge overvoltage. Moderate wind is defined as the daily maximum wind averaged over a number of years of data available.

If wind statistics are not available, 230 Pa (18 m/s) wind value may be used in calculations.
Note: *This wind value is also used in calculation of conductor swing for horizontal clearances in Clause A.5.2.7.*

Other wind values can be used based on local experience and satisfactory performance.

Case 3: Low wind condition

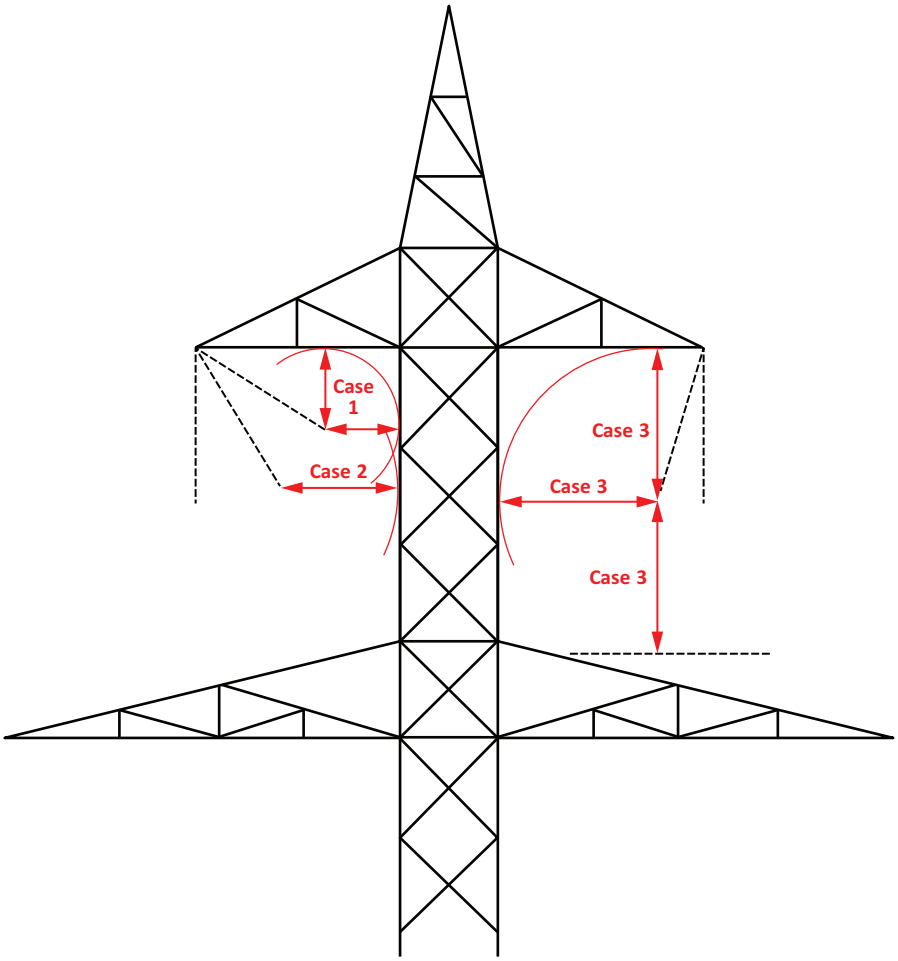
In this case, an insulator string is exposed to low wind, while the system is under switching surge overvoltage. Low wind is defined as the wind during which live-line procedures can be safely executed by qualified personnel.

This requirement may be critical in determining tower top geometry of a structure with vertical conductor configurations where a worker may be standing on a crossarm below an energized conductor. This requirement may also govern the length of the tower arms required to provide horizontal clearance to the vertical climbing space off the face of the tower, or the vertical clearance to the bottom of the arms.

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Clearances between insulator string and supporting structure



Case 1 (for operation):	High wind + power frequency voltage
Case 2 (for operation):	Moderate wind + overvoltage surge
Case 3 (for live-line maintenance):	Low wind + overvoltage surge

When calculating an insulator string position under wind, the following factors should be considered;

- length of an insulator string;
- weight of an insulator string and associated hardware;
- wind-to-weight span ratio;
- effect of a conductor bundle (sub-conductor spacing and configuration);
- route angle affecting insulator swing of a running angle tangent tower; and
- route angle affecting clearance of a jumper string of a dead-end tower.

In case of running angle tangent towers, the following conditions should also be taken into account:

- summer and winter conductor tensions, and
- forward and backward wind direction.

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A.5.3 Vertical design clearances and separations

A.5.3.1 Vertical design clearances of wires and conductors above ground or rails and Tables 2 and 4

Particularly for transmission lines, utility Standards should provide margins both for the minimum clearance specified in this Standard and for the accuracy of field measurements (e.g., accuracy of surveying related to ground profile, accuracy of sagging and, where applicable, movement or settling of structures with time, as well as conductor creep).

The phrase “likely to be travelled by road vehicles” means areas over which road vehicles are normally expected to travel. While vehicles are physically able to travel over such locations as sidewalks, it is not normal for them to do so. It is not necessary, therefore, to provide clearances for vehicles over sidewalks, except at special locations such as sidewalks in front of loading platforms or driveways that cross sidewalks.

The phrase “normally accessible only to pedestrians” means walkways or other ground (such as parks) across which pedestrians normally travel. This category does not include areas of a bridge or building on which it would not be normal to expect a vehicle or person, or to swamps and areas that can only be crossed in winter.

The basic vertical clearance provided over a pedestrian way is 2.5 m for non-energized wires and grounded cables and 3.1 m for an energized open wire at a voltage of 750 V or less. An increment for voltage is added for conductors operating at greater than 750 V.

The basic vertical clearance over a roadway is 4.42 m at zero voltage. This clearance provides an additional 0.27 m over the maximum vehicle or load height permitted by most provinces. For energized ac and dc conductors less than or equal to 200 kV to ground, an increment for voltage is added. For ac conductors at voltages-to-ground greater than 200 kV, clearances are determined by calculated electrostatic induction or flashover requirements.

For energized conductors operating at 60 Hz voltages greater than 200 kV to ground and located so that they pass over or are adjacent to vehicles with large metallic surfaces, the clearance may be determined using calculated electrostatic induction requirements. This Standard is based on worst-case conditions, which assume that the vehicle is insulated from ground voltage by rubber tires and that a person contacting the vehicle is touching or standing on a well-grounded surface. Under these conditions, the minimum clearances specified in this Standard protect the public from harmful effects due to induced current that could flow from the vehicle to ground. This Standard is simplified to the extent that a fixed clearance value is specified for a given voltage level.

In actual practice, the tires on vehicles are conductive, and there is considerable leakage over the tires. In addition, the person contacting a vehicle is unlikely to be well grounded because of the resistance in his or her footwear and the fact that perfect grounding conditions are rare. The induced current in all practical cases is below the “5 mA let go” values that are documented in technical literature.

The clearance is evaluated as follows:

- For 200 kV or less to ground, the customary value for the increment is 0.01 m/kV.
- For greater than 200 kV to ground, electrostatic induction clearances are based on a large farm vehicle that is 7.6 m long, 2.4 m wide, and 4.15 m high, and a tractor-trailer that is 23.0 m long, 2.6 m wide, and 4.15 m high. Flashover calculations for low-induction vehicles are based on 0.01 m/kV for the increment for a voltage greater than 200 kV. The electrostatic induction calculations are derived from “Field Effects of Overhead Transmission Lines and Stations”.

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Under normal line operating conditions, IEEE C95.6 requires that human exposure to electric fields be limited to 10 kV/m. This requirement is based on comfort rather than safety and assumes normal line operational conditions rather than maximum sag conditions, as used for Table 2. For voltage classes 200 kV and greater, the electric field under maximum sag conditions specified in Table 2 can exceed 10 kV/m. Consideration should be given to increasing the clearance where normal line operating conditions approach maximum conditions.

Sample calculations of clearances specified in Tables 2 and 4

The following examples illustrate the calculation of clearances in Tables 2 and 4, using the increment of 0.01 m/kV, where appropriate.

Example 1 — 230 kV phase-to-phase ac line

Switching surge voltage = 583 kV (from Table A.1)

Electrical flashover air gap (from Clause A.5)

= (2.54 mm/kV × 500 kV) + [3.81 mm/kV × (583 kV – 500 kV)]

= 1270 mm + 316 mm

= approximately 1.6 m

Applied to roads, the clearance requirement = 1.6 m + 4.1 m (truck height) + 0.3 m (added buffer) = 6 m.

The minimum clearance specified in Table 2 for a 230 kV phase-to-phase ac line is 6.1 m.

Example 2 — 345 kV phase-to-phase ac line

Switching surge voltage = 840 kV (from Table A.1)

Electrical flashover air gap (from Clause A.5)

= (2.54 mm/kV × 500 kV) + [3.81 mm/kV × (840 kV – 500 kV)]

= 1270 mm + 1295 mm

= 2.56 m

Applied to roads, the clearance requirement = 2.6 m + 4.1 m (truck height) + 0.3 m (added buffer) = 7 m.

Table 2 places a 345 kV nominal (380 kV maximum) line in the 220 kV rms line-to-ground category. The simplified calculation is as follows:

Clearance = 6.1 m + 0.01 m (220 – 150) = 6.1 + 0.7 = 6.8 m

While this value is less than the 7 m value calculated above, it is considered acceptable because the assumed surge overvoltage is a conservative value. Electrostatic induction at this voltage level for a tractor-trailer requires 10.2 m of clearance.

Example 3 — 735 kV phase-to-phase ac line (765 kV maximum)

$$\text{Switching surge voltage} = \frac{765}{\sqrt{3}} \times \sqrt{2} \times 2.2 = 1374 \text{ kV (from Table A.1)}$$

Electrical flashover air gap (from Clause A.5)

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= (2.54 mm/kV × 500 kV) + [3.81 mm/kV × (1374 kV – 500 kV)]

= 1270 mm + 3330 mm

= 4.6 m

Applied to roads, the clearance requirement = 4.6 m + 4.1 m (truck height) + 0.3 m (added buffer) = 9 m (clearance based on switching surge flashover).

Table 2 places 765 kV (442 kV voltage-to-ground) clearance at 20.4 m (clearance based on electrostatic induction), which is greater than the 9 m calculated value. Therefore, the 9 m clearance does not apply to clearances over roads. It applies to situations where electrostatic induction is not a factor.

Example 4 — 900 kV pole-to-pole (± 450 kV) dc line

Switching surge voltage = 720 kV (from Table A.2)

Electrical flashover air gap (from Clause A.5)

= (2.0 mm/kV × 144 kV) + [2.92 mm/kV × (720 kV – 144 kV)]

= 288 mm + 1682 mm

= 2 m

Applied to roads, the clearance requirement = 2 m + 4.1 m (truck height) + 0.3 m (added buffer) = 6.4 m.

This value is considered low and, in view of the fact that the number of years of experience with, and testing of, dc lines at high voltages is small, 1.0 m has been added for a more conservative value.

The minimum clearance specified in Table 4 is 7.4 m (6.4 m + 1.0 m).

The clearance over a driveway to a residence for an ac conductor of 0 to 750 V to ground is specified as 3.7 m at the maximum sag position. This is considered the minimum practical clearance, and does not include clearances for maximum truck height (4.1 m).

The clearance over a road for an ac conductor of 0.75 to 22 kV to ground is 4.1 m + 0.3 m + 0.3 m (electrical requirement) = 4.7 m. Table 2 specifies a minimum clearance of 4.75 m.

Historical railway standards specify a minimum vertical clearance requirement of 6.706 m (22 ft) above the top-of-rail level for beams, members, or portions of any bridge, tunnel, erection, or structure constructed for non-electrified lines after February 1, 1904. When the requirement was first established, one of the considerations was to provide safe clearance for a train worker standing on top of a freight car and signaling by hand or lantern, or both. Subsequent developments in railway operating practices, signaling methods, and sizes and types of rolling stock have made this practice obsolete, and the original minimum clearance requirement of 6.858 m (22 ft, 6 in) for rigid structures has been reduced to 6.706 m (22 ft) and is now relied upon to accommodate larger loads and special-purpose freight cars. The same minimum clearance requirement has been used for the *Canadian Electrical Code, Part III*, as the basic height to be protected for flexible structures such as conductors crossing over railway tracks, with additional allowances for safety and voltage.

The clearance over a railway track for an ac conductor of 0.75 to 22 kV to ground is calculated as follows:

- Basic height to be protected = 6.706 m
- Clearance = 6.706 m + 0.45 m (safety) + 0.3 m (electrical requirement) = 7.46 m.

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Table 2 specifies a minimum clearance of 7.6 m.

An extra 0.3 m should be allowed for ballast where required (see Clause 5.3.1.1(c)).

The clearance for 0 to 750 V ac conductors over land, alongside roads and highways in rural areas, and within 1.5 m of the limit of the right-of-way, is calculated as follows:

- Clearance = 3 m (the protected height for a vehicle, e.g., a grass cutter, that is likely to be found on the unpaved portion adjacent to a fence) + 0.3 m (buffer) + 0.0001 m (electrical requirement) = 3.3 m.

Table 2 specifies a minimum clearance of 3.4 m.

At 22 kV, the specified clearance is less than that for roads that accommodate a 4.1 m high truck, but provides a larger buffer than that provided at 750 V.

- Clearance = 3 m (vehicle height) + 0.76 m (buffer) + 0.3 m (electrical requirement) = 4.1 m.

Table 2 specifies a minimum clearance of 4.15 m.

The same buffer is provided for conductors at higher voltages, which are not as likely to appear under these circumstances as are conductors of 0 to 750 V.

A simplified increment of 0.005 m for the voltage of dc conductors is used in Table 4. Using the air gap flashover data from Clause A.5 and two lines, one designated ± 200 kV and the other ± 450 kV, the following clearances are calculated:

- Switching surge voltage of ± 200 kV line = 320 kV to ground
  Clearance = (2.0 mm/kV × 144 kV) + [2.92 mm/kV × (320 kV – 144 kV)] = 802 mm
- Switching surge voltage of ± 450 kV line = 720 kV to ground
  Clearance = (2.0 mm/kV × 144 kV) + [2.92 mm/kV × (720 kV – 144 kV)] = 1970 mm

The difference in the two clearances = 1970 mm – 802 mm = 1168 mm or 1.17 m.

Using the increment of 0.005 m/kV, the difference in the two clearances = (450 kV – 200 kV) × 0.005 m/kV = 1.25 m.

A.5.3.2 Vertical separations (heights) of supply equipment from ground

Table 5 specifies minimum separations from ground of supply equipment mounted on poles. It does not refer to equipment in an enclosed substation. The specified separations of supply equipment from ground are minimum values. Utilities may increase the height; especially in areas such as farmyards, where hay racks or equipment of great height are commonly used.

A.5.7 Clearances of wires, conductors, and equipment from buildings, signs, bridges, swimming pools, and similar plant

The phrase “normally inaccessible”, when applied to buildings, refers to those surfaces that cannot be reached easily without special aid (e.g., a ladder).

“Readily accessible surfaces” are those surfaces to which a person could normally gain access with no special effort (whether or not they are intended to be used by persons), such as windows, balconies, and flat roofs.

The horizontal clearance between a wire at zero voltage and a surface on a building should be such that, under a heavy wind, the wire will not be abraded by contact with the building. The clearances for

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energized conductors should be such that no flashover will occur under switching surge conditions. A buffer is added to provide for, among other things, the reach of a person located at a readily accessible point of the building.

Where an assessment of a particular condition calls for additional caution, greater clearances should be provided. Where extreme wind gusts are anticipated, extra clearances can be necessary, particularly in the lowest voltage categories. The clearances specified in Clause 5.7 are the minimum and should be increased where feasible.

Under maximum sag conditions, which are extreme conditions that arise infrequently, specified minimum vertical clearances provide a vertical space of 2.4 m or more to ensure that persons will have a reasonable degree of freedom to reach above head level. The electrical flashover component of clearance is generous, as it provides a switching surge withstand distance.

Clearances between oil and gas wells and associated drilling equipment from overhead supply lines are addressed in provincial regulations and are not covered in this Standard.

A.5.7.5 Minimum separation of equipment and clearances of conductors over and adjacent to swimming pools and Table 11

In general, conductors over swimming pools are undesirable for new construction. However, due to various conditions (e.g., limited lot size), it can be difficult to avoid this situation.

Clearances used in Table 11 have been determined based on the assumption that cleaning tools (up to 4.6 m in length) might be manoeuvred by inexperienced and/or young people.

A.5.7.6 Horizontal separations from supply equipment to buildings and Table 12

Horizontal separations for equipment attached to normally inaccessible surfaces of buildings provide a safe flashover distance. The separation for a readily accessible surface, where equipment is not attached to the surface, provides approximately 1 m for the possible reach of a person, plus a voltage increment.

A.5.7.8 Clearances from other flammable hazards

Clause 5.7.8 recognizes that open link fuse elements or other devices that can emit hot particles or glowing embers should not be installed over gas filling stations, fuel storage tanks, or other such installations.

A.5.7.9 Clearances for other wire facilities

The requirements of Clause 5.7.9 are derived from CSA Z98.

A.5.7.10 Clearance from irrigation systems

Introduction

Irrigation systems operating in proximity to energized supply lines can pose safety hazards which can result in injuries to personnel or equipment and power outages. Potential hazards include

- leakage current from the water either contacting or being in close proximity to energized conductors, which can cause hazardous or nuisance shocks, if irrigation equipment is touched;
- flashovers from water jet contacting energized conductor;
- induced voltage to conductive objects electrically insulated from the ground, including metal irrigation pipes, especially when they are parallel to energized conductors and not well grounded, causing nuisance shocks to personnel;

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- flashovers from direct contact with energized conductors during storing, handling, installation and operation of irrigation equipment inside overhead line right-of-way, particularly in dry conditions; and
- line faults where voltage can be transferred to an irrigation system in close proximity to an energized conductor or during the ground potential rise situation.

Severity of shock to personnel will depend on many factors, including proximity to energized conductor, operating line voltage, level of fault, orientation of irrigation equipment, adequacy of grounding, and weather conditions.

Common types of irrigation systems

While there are numerous types of commercial irrigation systems, it is the larger systems that have the most potential for conflicts with supply lines. The two most common types of large systems are

- centre pivot systems; and
- side-roll or “wheel move” systems.

The centre pivot system has a single lateral pipe which is supported on A-frame towers on wheels. The lateral pivots about a swivel joint in the middle of the field. The lateral pipe is typically supported between 3 to 5 m off the ground. Total length of the lateral is typically just less than 400 m (one quarter mile) so that a quarter section can be irrigated with one unit. Small and full section units are also available. Several approaches for watering the corners have been devised, including equipping the outer end with a single long range sprinkler unit having a large diameter nozzle or providing an additional lateral span which “fans-out” at the corners.

Side-roll or “wheel move” systems have lateral pipes which also act as axles for 1.2 to 2 m diameter steel wheels, thereby supporting the pipe 0.6 to 1.0 m off the ground. Sprinklers are mounted directly on the pipe. While irrigating the system remains stationary. To move the system, irrigation stops and the water is drained from the pipe. A small drive unit applies torque to the pipe and the system rolls forward. The lateral is then reconnected to the main pipe, which lies along the edge of the field perpendicular to the lateral, and irrigation can again resume.

Water stream contact with supply lines

The following two safety issues are associated with water streams from irrigation systems being sprayed toward supply lines:

- **First:** The possibility of an electrical fault from the water spray coming very near or making contact with the conductor.
- **Second:** Capacitive coupling where there is sufficient leakage current in the water spray to cause hazardous or nuisance shocks.

The first issue can be addressed by the following operating procedures:

- When possible, irrigation systems should be operated at distances sufficient to avoid direct contact between the water stream and conductors.
- A solid and continuous stream of water should never be directed on or near the energized line conductors. If there is a possibility that the water stream from an irrigation system can contact a power line, the nozzle design must be such as to break the water stream into a spray.

Test results and operating experience in the industry indicate that direct water spray contact with power line conductors is permissible when irrigation nozzles have discharge characteristics that cause water stream breakup due to normal water turbulence and nozzle or system design. Once the water

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stream has been broken up and becomes a spray, the probability of line conductor-to-nozzle flashover is greatly reduced or eliminated.

The second issue can be addressed by maintaining minimum distances (clearances) between energized power line conductors and the irrigation nozzle.

Minimum conductor to nozzle distances are specified in IEEE 1542, Table 2. The Table is based on the formulae given in the IEEE Guide, with irrigation system resistance of 10 ohms, body current 5 mA rms, and body resistance 1000 ohms. Water conductivity is 1200 µS/cm and water pressure is 552 kPa. Values apply for water spray (no solid and continuous stream).

Protection from nuisance shocks

Metallic parts of irrigation systems located near or under supply lines can become a potential shock hazard when insulated from ground or improperly grounded. This potential hazard can be minimized or eliminated with grounding and bonding of the equipment.

Wheel move systems in operation have metal wheels in contact with wet ground and this grounding is adequate to drain any induced voltages. When not in operation, wheel move systems should not be stored near supply lines, in particular parallel to the lines. This precaution will limit the possibility of nuisance shocks from the irrigation system, which can be insulated from ground, or the risk of electrocution from moving a section of pipe into contact with energized conductors.

Centre pivot systems are much larger than wheel move systems and move on rubber tired wheels. To minimize the risk of shocks from these systems operating near supply lines, the following steps should be taken:

a) Ensure a good electrical ground at the centre pivot point. If the water supply pipe is metal, with no insulating flanges or gaskets, the ground should be adequate. Otherwise, a proper ground should be installed.
b) Personnel should not touch the sprinkler pipe or its supporting structures when the system is in operation under, parallel to, or near a power line.
c) If it is necessary to carry out maintenance on a centre pivot system while the system is parallel to or near a power line, the irrigation system should be grounded before and during the work.

Direct contact with energized conductors

For wheel move systems, the risk of direct contact comes from people moving sections of pipe in the immediate vicinity of energized conductors. To minimize this risk, wheel move systems should not be stored near supply lines and appropriate limits of approach should always be maintained when moving irrigation pipe near energized lines.

For centre pivot systems, the risk of direct contact comes primarily from the height of the equipment being such that it is not able to pass safely under an energized power line. If it is not possible to prevent a centre pivot system from operating under a given power line, the height of the irrigation equipment and powerline must be such as to allow safe operation of the equipment.

Conductive coupling of power line fault current to irrigation equipment

When single phase-to-ground faults occur on the power system, fault current can flow down the legs of metal towers or the ground wire on wood structures. The fault current flowing into the ground causes a rise in the voltage at the ground level, at the grounding points near the fault (called ground potential rise — GPR). Where an irrigation system is very close to the power line structure, in some instances the

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GPR might be sufficient to pose a hazard for a person who is standing within the zone of influence of the GPR and in contact with the irrigation system at the same instant that the fault occurs.

Most power systems are multi-grounded and with the multi-grounded system intact, calculations show that even for very high fault currents (greater than 20 kA) hazardous voltages do not extend more than 10 to 12 m from the grounding point and the fault location. Thus for a hazardous situation to exist, a person would need to be in contact with the irrigation system, be standing within the 10 to 12 m of the structure grounding point, have a single line-to-ground fault occur at the structure which the person is closest to, and for the available fault current on that line to be at least 20 kA. The likelihood of these events occurring simultaneously seems extremely remote.

To minimize the risk due to conductive coupling, power line structures should be located at least 15 m from irrigation systems, if possible.

A.5.8 Clearances between wires and conductors of one line and wires, conductors, and structures of another line

A.5.8.1 Vertical clearances at crossings of line wires and conductors supported by different structures and Table 13

Where conductors cross each other, clearance has been provided to withstand the sum of the voltages between conductors, assuming a switching overvoltage on one line, normal operating voltage on the other line, and no sag in the lower conductor. Assuming no sag in the lower conductor provides for conductor jump (from ice shedding or galloping), which in short spans can reach the line of sight between structures. In rare instances, conductors have been reported to jump above the line of sight; however, the probability of a switching surge occurring on either line when this occurs is remote.

Furthermore, the probability of a switching surge occurring on both lines coincident with the upper conductor at maximum sag and the lower conductor at the line of sight is also considered extremely low.

Example 1 illustrates the method used to arrive at the clearances specified in Table 13:

Example 1

Assume an ac conductor in the 150 to 190 kV line-to-ground category crossing an ac conductor in the 50 to 90 kV line-to-ground category, with the switching surge voltage in the higher-voltage conductor.

- Line voltage of first line at 190 kV rms switching surge voltage
  = 190 × 1.414 (crest factor) × 2.7 (switching surge factor) = 725 kV
- Crest of peak voltage of second line = 90 × 1.414 = 127 kV
  Sum of voltages: 725 + 127 = 852 kV
- The air gap requirement for 852 kV (725 + 127 kV) is 2.54 mm/kV up to 500 kV plus 3.81 mm/kV over 500 kV = (2.54 × 500) + (3.81 × 352) = 1270 + 1341 = 2611 mm or 2.61 m.

Table 13 specifies a minimum clearance of 2.6 m.

Example 2

Assume an ac conductor in the 220 to 320 kV line-to-ground category crossing a dc conductor in the 150 to 200 kV line-to-ground category. Two scenarios are considered:

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Scenario A — ac line is under switching surge and dc line in normal operation:

- Line voltage of ac line at 320 kV rms switching surge voltage
  = 320 × 1.414 (crest factor) × 2.5 (switching surge factor) = 1131 kV
- Operating voltage of dc line = 200 kV
- The air gap requirement is:
  ac voltage air gap is 2.54 mm/kV up to 500 kV plus 3.81 mm/kV over 500 kV = (2.54 × 500) + (3.81 × 631) = 1270 + 2404 = 3674 mm, plus
  dc voltage air gap is 2.0 mm/kV up to 144 kV plus 2.92 mm/kV over 144 kV = (2.0 x 144) + (2.92 x 56) = 288 + 164 mm = 452 mm
  combined air gap = 3674 + 452 = 4126 mm or 4.13 m

Scenario B — ac line is in normal operation and dc line in under switching surge:

- Line voltage of ac line at 320 kV rms
  = 320 × 1.414 (crest factor) = 452 kV
- Line voltage of dc line at 200 kV under switching surge
  = 200 kV x 1.6 (switching surge factor) = 320 kV
- The air gap requirement is:
  ac voltage air gap is 2.54 mm/kV up to 500 kV = (2.54 × 452) = 1148 mm, plus
  dc voltage air gap is 2.0 mm/kV up to 144 kV plus 2.92 mm/kV over 144 kV = (2.0 x 144) + (2.92 x 176) = 288 + 514 mm = 802 mm
  combined air gap = 1148 + 802 = 1950 mm or 2.0 m

The more severe requirement is from Scenario A: 4.13 m air gap.

Table 13 specifies a minimum clearance of 4.1 m.

Example 3

Assume a dc conductor in the 450 to 500 kV line-to-ground category crossing a dc conductor in the 200 to 250 kV line-to-ground category, with the switching surge voltage in the higher-voltage conductor.

- Line voltage of first line at 500 kV under switching surge
  = 500 × 1.6 (switching surge factor) = 800 kV
- Operating voltage of second line = 250 kV
  Sum of voltages: 800 + 250 =1050 kV
- The air gap requirement for 1050 kV is 2.00 mm/kV up to 144 kV plus 2.92 mm/kV over 144 kV = (2.0 × 144) + (2.92 × 906) = 288 + 2646 = 2934 mm or 2.93 m.

Table 13 specifies a minimum clearance of 2.9 m.

Situations can arise whereby a line crosses over a long span of another line. Where the sag of the lower conductor exceeds 6 m at the point of crossing, it is unlikely that ice shedding or galloping will lead the lower conductor to reach the line of sight of the conductor.

The following examples show a 230 kV line (voltage-to-ground in the range of 120 to 150 kV) crossing over a 115 kV line (voltage-to-ground in the range of 50 to 90 kV) at two locations (see Figure A.3). The lower span is 300 m with a HAWK conductor.

To simplify the calculations in the examples, the ice- and wind-loaded sag of the bottom conductor is assumed to be in a vertical plane. This is in accordance with the definition of maximum sag.

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When crossing over inclined spans, an additional check should be performed to ensure that the clearances specified in Table 13 are achieved with the upper conductor in a swing position and the lower conductor at rest.

Example 4 (see Clause 5.8.1.1)

In Figure A.3, the sag of the lower conductor is less than 6 m, so Clause 5.8.1.1 applies. Table 13 specifies a clearance of 2.1 m from the nearest upper conductor at maximum sag to the line of sight between the lower conductor supports.

Example 5 (see Clause 5.8.1.2)

The sag of the lower conductor for a HAWK conductor in a 300 m span is 8.94 m. The electrical clearance specified in Table 13 is 2.1 m. The reduction in clearances allowed by Clause 5.8.1.2 is calculated as follows:

$$\frac{(8.94 - 6)}{2} = 1.47 \text{ m}$$

Subtract 1.47 m from 2.1 m to obtain a clearance of 0.63 m. This is the distance by which the upper conductor must be above the line of sight of the lower conductor. In excessively long spans, the sag of the lower conductor can be much larger, and the values subtracted from Table 13 can yield a negative result. In such cases, the upper conductor can be below the line of sight of the lower conductor.

A.5.8.1.3 In-span vertical clearances between supply and communication conductors

Where supply and communication conductors cross each other, clearance is provided to ensure that they will not come in contact. Providing a midspan support structure with both the supply and communication conductors contacting the common structure results in

- ensuring that minimum in-span vertical clearances between the supply conductor and communication conductors, under conditions of maximum sag in the supply conductors as specified in Table 13, are met; and
- increasing the vertical clearance between the communication circuit and electric circuit under maximum load conditions, where required for additional safety (e.g., in flat-line construction).

In the event of contact between supply and communication cables, coordinated electrical protection should be used, with damage to either the supply or communication plant being kept to a minimum. This can be accomplished by one of the following two methods:

- Method 1 — prompt operation of protective devices provided in the supply plant. This is the preferred method.
- Method 2 — burn-off. Although burn-off may be used in certain situations, dependence on this method is not recommended. Where this method is used, safety must be enhanced and damage to the communication plant must be minimized.

In order for the electrical supply system to de-energize upon contact with the communication plant, the following conditions must be met or exceeded:

- Condition 1 — the maximum fault current on the supply system is equal to at least twice the minimum operating value of the supply system's protective devices controlling the sections where the supply and communication systems are in joint use (i.e., minimum melting current for fuses and minimum pick-up current for relays and reclosers).
- Condition 2 — the communication plant is bonded to a multi-grounded neutral of the supply system at 300 m intervals and at common crossing structures, using a minimum conductor size of No. 6

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AWG copper or equivalent. In situations where the communication plant cannot be bonded to the supply multi-grounded neutral, the communication plant must be grounded with ground rods installed on both poles that support the communication wire crossing. Generally, a resistance of 25 Ω or less is recommended. If this cannot be achieved with a single rod, the rod may be coupled with another rod and driven deeper, or two ground rods may be installed, spaced at least 3 m away from each other, and aligned with the pole line.

Where supply and communication conductors cross each other, there are three options for safe installation:

- Option 1 — a common structure is installed, where practicable, at the point of crossing so that the supply and communications cable can be attached to the common crossing structure.
- Option 2 — where it is not practical to install a common structure, coordinated electrical protection is used.
- Option 3 — where coordinated electrical protection cannot be achieved due to the location of the wire crossings, a higher grade of construction is used to prevent future contact between supply and communication cables.

A.5.8.2 Clearances between conductors supported by different structures but not crossing each other and Table 15

The following examples illustrate clearances between conductors on different structures but not crossing one another. See also Figure A.4.

The minimum horizontal clearance between adjacent conductors A and B for a specific span in the line is calculated as follows:

- For line A, assume that the conductor sag plus insulator length at the 40 °C final unloaded condition is 3 m and that the d/w ratio is 22.4. From Table 1, the factor is 0.46 and the horizontal deviation is (0.46 × 3) = 1.38 m.
- For line B, assume that the conductor sag plus insulator length at the 40 °C final unloaded condition is 3.3 m and that the d/w ratio is 17.3. From Table 1, the factor is 0.38 and the horizontal deviation is (0.38 × 3.3) = 1.25 m.

Example 1

Where the structure of one line is opposite to the conductor of an adjacent line that is free to swing,

- the greater of the horizontal deviations = 1.38 m;
- the maximum voltage of line A $= \frac{127}{\sqrt{3}}$ = 73 kV to ground;
- the maximum voltage of line B $= \frac{253}{\sqrt{3}}$ = 146 kV to ground;
- the sum of these two voltages-to-ground = 73 kV + 146 kV = 219 kV;
- from Table 15, the clearance increment to be added to the horizontal deviation = 300 mm + (10 mm × 218.25) = 300 + 2183 mm = 2483 mm = 2.48 m; and
- the minimum horizontal clearance = 1.38 + 2.48 m = 3.86 m.

Example 2

Where the structures of each line are approximately adjacent to one another, the minimum horizontal clearance = (1.38/2) + 2.48 m = 3.17 m.

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A.5.9 Separations of supply-line conductors and conductor supports on the same supporting structure

A.5.9.2 Vertical separations and clearances of supply-line conductors attached to the same supporting structure

A.5.9.2.3 In-span vertical clearances between supply-line conductors of the same circuit

Conductors of the same voltage class and belonging to the same circuit generally consist of 2- or 3-phase conductors of a primary voltage (i.e., greater than 750 V) or 2, 3, or 4 conductors of a secondary voltage (i.e., 0 to 750 V). Because they are of the same circuit, their sag at a given time is assumed to be similar.

The neutral of a primary voltage circuit is not normally at the same conductor temperature as the phase conductors, even where the neutral conductor size is reduced. This is usually due to the nature of the effectively grounded neutral and to the electric utility's efforts to maintain a balanced load on the circuit. Similarly, secondary voltage circuit conductors will not be at the same conductor temperature as their related primary voltage circuit conductors. Where the maximum sag of the upper primary voltage conductors is due to thermal loading conditions, experience suggests a neutral/secondary conductor design temperature of 40 °C. Where the maximum sag of the upper conductors is due to ice-loading conditions, experience suggests a neutral/secondary conductor temperature of 0 °C.

A.5.9.2.4 In-span vertical clearances between supply-line conductors of different circuits

Where conductors of one circuit are above those of another, it is reasonable to assume that the two circuits will not be at maximum sag at the same time. Experience suggests that the initial unloaded sag at the mean annual temperature is an appropriate worst-case design condition.

A.5.9.5 Separations or clearances in any direction from supply conductors to other supply plant attached to the same supporting structure

The highest voltage level in Table 22 is 50 kV, because it is recognized that at higher voltage levels there can be significant differences in switching surge levels, depending on system design. The inclusion of conservative increments above 50 kV could result in clearances that would require structural dimensions greater than those necessary for safe operation.

A.5.10 Joint-use clearances and separations — Supply and communication plant

A.5.10.1 Vertical separations at the structure — Normal level arrangement and Table 23

Figure A.5 illustrates the relative locations of supply and communication attachments on a joint-use pole where the normal level arrangement is used.

The minimum vertical separation required between supply and communication attachments on joint-use poles consists of a communication safety space, which varies with voltage and isolates the communication worker from electrical hazards, and a workspace to permit movement of a worker's body, tools, and work materials while working on the communication plant.

The minimum separations necessary to provide communication safety space for the voltage under consideration and to provide a workspace are given in Table 23. However, a company or utility may

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increase these separations where it believes additional workspace is necessary to ensure the safety of workers. Safety space is necessary for communication personnel safety.

It should be remembered that safety requires more than just space; it also depends on safe working practices. For example, stringing a communication strand under wires requires special precautions to ensure that the strand does not flip up into the communication safety space.

Live or current-carrying supply plant, including neutrals (see Figure A.6)

The separations specified in Table 23 provide the communication worker with a 0.75 m workspace below the appropriate communication safety space for the voltage under consideration. This workspace is adequate for normal operations, including the erection of a platform or tent on the communication strand. However, it is incumbent upon the communication worker to restrict work operations to locations below the bottom of the communication safety space.

Except where the voltage of the neutral conductor can be classified as 0 to 750 V in accordance with Clause 5.2.6, neutrals need the same clearance from communication plant as do their phase conductors because, during ground faults, neutrals that are not effectively grounded can approach phase voltage.

Lateral communication drops on line or run-off poles may be attached 0.6 m below 0 to 750 V supply plant. This location, which is above the uppermost longitudinal communication attachment, permits the communication worker to provide increased clearance over roadways as a public safety measure, since insufficient clearance could produce serious injuries or damage if a vehicle hits a drop and breaks a pole or breaks the drop and causes it to flip over supply circuits. When using this location, the worker must place his or her drive hook at eye level so as to remain within the normal workspace below the communication safety space. Workers also need to ensure that they do not obstruct the climbing space (see Clause A.4.1.4).

Non-energized supply plant

Option A (see Clause 5.10.1.6) applies where the communication utility necessitates normal workspace, usually to permit a tent or other form of covering to be used as protection against exposure to the weather during work operations. Where non-energized supply plant is effectively grounded, there is no provision for communication safety space. When it is not effectively grounded, however, it can become energized without warning, so the specified separation includes a communication safety space.

Option B (see Clause 5.10.1.7) — applies where communication workers are trained to work with not more than head and shoulders above the longitudinal communication plant. As with Option A, the separation specified for plant that is not effectively grounded includes a communication safety space. Where 0 to 750 V supply plant is effectively grounded, a separation of 0.10 m from communication plant is considered adequate for Option B. At voltages greater than 750 V, a separation of 0.30 m is considered necessary because the supply plant involved usually consists of large, heavy equipment such as a transformer, which needs a greater separation to facilitate installation and removal. This value should be increased wherever the minimum clearance between the conductors serving the transformers and the communication plant governs the design.

Trolley span wires or brackets (see Figure A.7)

The separation specified in Table 23 is provided to facilitate installation and removal. The requirements of Clause 4.2.11 ensure that the communication worker is isolated from electrical hazard.

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Luminaire span wires or brackets (see Clause 5.10.1.2 and Figure A.7)

This includes traffic lights and associated brackets (see Clause 4.2.10). Because the exposed metallic surfaces of ungrounded luminaires and associated brackets can become energized without warning, the vertical separation required from communication plant provides a communication safety space supplementary to normal communication workspace.

Point of attachment of combined communication drop and supply service conductor (see Figure A.6)

A combined service drop needs the same separation from longitudinal communication plant as would the supply service conductor by itself. The communication drop needs therefore to be separated from the supply service drop and extended down the surface of the structure into the communication workspace at any location where the communication worker has to work on it.

A.5.10.2.3 Wireless communication antenna

The separations specified in Clause 5.10.2.3 are based on the workspace needed by power line workers to safely install and maintain wireless antennas without having to de-energize the power lines. Occupational exposure to radiofrequency radiation from wireless communication antennas is addressed in Health Canada's *Safety Code 6*.

A.5.10.3 In-span vertical clearances and Table 24

Table 24 specifies a minimum 75 mm clearance under conditions of maximum sag in the upper conductor with little or no sag in the lower wire. The lower wire can be galloping, so that the highest point in the span is level with the points of support. An effectively grounded neutral needs less clearance than do the energized conductors.

A.5.11 Guys and guy attachments

A.5.11.2 Guys attached to joint-use structures and Tables 25 and 26

Where one utility places a guy on a joint-use pole for support of its attachments and where the guy runs near other systems not attached to the joint-use pole, the clearances in Table 25 apply, but clearances between this guy and circuits on the same structure are specified in Table 26. (See Figure A.8.)

A.5.11.4 Guys attached above current-carrying supply plant

Where guys can fail and come into contact with any current-carrying supply plant, electrical hazards can be brought to lower levels of the structure and to normally accessible locations. The general public, workers, and equipment on the pole can be exposed to these hazards, of which an appropriately rated guy insulator, or guy grounding, can mitigate. Decisions on where the guys are attached on the pole can have an impact on the most appropriate location for a guy insulator, if grounding is required, or remove the need for hazard mitigation altogether.

The phrase "failed guy" is used in this Standard to represent any condition where the guy becomes disconnected from its anchoring location (other than the structure itself), resulting in most of its length falling against the face of the structure. The guy then becomes vertical across the face of the structure in an uncontrolled manner, enabling it to come in contact with any source of exposed electrical energy and transmit that energy to any point along its length. The failing of guys, although not that common, is something that has been observed by utilities over time. It is generally accepted that the probability of guys failing justifies risk mitigation measures to protect workers and the public.

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A.7 Weather loads and assumed loads according to deterministic design methods

A.7.2 Weather loads

Heavy loading and medium loading A ice and wind combinations were adopted from earlier CSA Standards. Medium loading B was added to the 1976 edition of this Standard as an alternative to medium loading A for areas where greater ice loads and lower wind force are experienced. Severe loading was added in the 2001 edition.

The loading maps in Annex C are based on Environment Canada data and the experience of utilities across Canada. Designers are cautioned that these maps should be treated only as a guide and that local areas can have higher icing and/or wind forces. Conversely, there are local areas, such as laneways between tall buildings or along mountain valleys, where lines are sheltered from high winds. In these areas, transverse wind forces can be low, but “funneling” winds tend to be high in the longitudinal direction.

Designers unfamiliar with local weather conditions should acquaint themselves with the terrain and obtain local information from weather records and regional meteorologists. Where weather data are unavailable, local residents can be consulted.

Further data on ice and wind loadings can be obtained from Environment Canada.

A.7.3 Assumed loads for wire and cable attachments

The wind pressures specified in Table 30 (i.e., 400 N/m2 and 300 N/m2) correspond approximately to steady-state hourly wind velocities of 25 m/s and 21 m/s, respectively; these values are based on the formula provided in Clause A.5.2.7.

Maximum wind speeds are not generally associated with maximum icing. Greater wind speeds can occur on bare conductors, resulting in a greater transverse load than is incurred from the ice and wind combinations specified. This can occur when short spans and large-diameter conductors are employed. However, high winds generally occur over relatively narrow fronts. Where spans exceed 200 m, a reduction factor is frequently used to determine the resultant wind force over the entire span. This reduction factor varies with wind speed and terrain.

A.7.3.3

The 1.6 load factor approximates the rated tensile strength of conductors when the maximum allowable tension of 60% is used.

A.7.3.4

Figure A.11 provides an illustration of determining the diameter of a cable bundle used in calculating wind and ice loads.

A.7.3.6

All metallic conductors, wires, messengers, and even cables with dielectric support members are not perfectly elastic in their stretching behavior when subjected to tension forces. In other words, their modulus of elasticity is not a constant. There is a measurable amount of permanent stretch that occurs which increases with increasing maximum tension or stress. If this permanent stretch is not calculated nor considered using the maximum tension condition, the amount of sag will be underestimated and the amount of tension and assumed loads on the support structures will be over-estimated. The stress-

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strain curve of these wires/cables is curved and said to not behave in a linear manner; i.e., it is non-linear. This behavior has led to the historical creation of initial tension, final tension under load, and final tension after creep tables of many wires for utilities. (See Appendix B).

A.7.4 General requirements — Loads on supports

A.7.4.2

The load factors take into account the grade of construction and the type of material being used and the fact that weather loadings cannot be accurately determined. However, it should not be assumed that applying such load factors will completely eliminate the possibility of failure; therefore, these should be considered minimum values.

The load factors specified in Table 31 take into account the varying strength and COV of various materials and, to some extent, deterioration of the poles with age.

Table 32 exists to ensure that load factors are applied to guy assemblies (guy wires, anchors, and pole braces).

Table A.3 specifies the strength of various pole classes. The wood pole strengths are taken from CAN/CSA-O15 and the concrete pole strengths are taken from CAN/CSA-A14.

A.7.5 Assumed vertical load on supports

A.7.5.3 Assumed vertical load on crossarm assemblies, pins, posts, insulators, and fastenings

The additional load of 1.0 kN specified in Clause 7.5.3.1 is intended to represent the weight of a line worker.

A.7.6 Assumed transverse load due to wind pressure

A.7.6.1 Assumed transverse load due to wind pressure on wire and cable attachments

Each structure that supports wire and cable attachments between them is expected to equally support 50% of the total transverse wire load due to wind. In this way the total loads due to wind on these items is supported by the supporting structures.

In cases where wire and cable attachments are sheltered from the full effects of wind load, a reduction multiplication factor of 0.40 can be applied. This is equivalent to a wind speed reduction of 63%. This reduction applies only when calculating the assumed loads to be applied to the structure from the wire and cable attachments, not to the structure itself. In addition, the wires and cable attachments themselves need to be designed, without this reduction, in accordance with Clause 7.3.

The concept of partially wind-sheltered spans is also referenced in Clause 5.2.7 and Table 1 for determining the horizontal swing of wires and conductors. In locations where the span of wire or cables are partly sheltered from the full effect of transverse winds by buildings or other construction, as in built-up commercial or residential areas, the assumed wind pressure can be reduced as stated in this Clause. A partially sheltered span is one that is protected by objects such as buildings whose height is greater than or equal to wire or conductor height and that are within an effective distance (e.g., 30 m) of the wire or conductor.

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Effective wind pressure over the span distance between structures is influenced by any obstacles that the wind must pass over or around. The greater the number, height, and size of these obstacles, the more turbulent and slower is the wind. Slower wind speed results in less wind pressure. Turbulent wind results in different portions of a span seeing a different speed and direction of wind. When the effect is combined over the entire span, the resulting average force per unit length of span is less than if there were no turbulence. The greater the turbulence, the less overall loads result from the wind. The presence of obstacles impacting wind pressure is a concept related to terrain roughness and is discussed in more depth in CAN/CSA-60826.

A.7.6.2 Assumed transverse load on a structure

The loads specified in Clause 7.6.2 apply to structures not exceeding 30 m in height. Where structures are higher than 30 m, the wind forces on the structure should be determined by a more detailed assessment.

The 100% increase in wind pressure takes into account the approximate difference between round and flat surfaces due to the aerodynamics of wind flow. The designer is cautioned that the transverse wind forces on certain types of lattice structures can be greater when the wind direction acts at 45° to the transverse direction.

A.7.6.2.1

Experience has shown that ice tends to collect on the down-wind side of the pole and thus does not add to the cross-sectional area upon which the wind acts. In addition, ice on the pole tends to lower the drag coefficient. In essence ice on the structure is not a governing case.

A.7.8 Assumed longitudinal loads on supports

The intent of this Standard is to ensure that there is sufficient longitudinal restraint built into tangent structures to minimize failures and to prevent cascade failure of adjacent structures in the line. In addition, dead-end structures should be designed to be capable of withstanding a theoretical unbalanced pull, independently, on each side of the structure.

With longitudinal restraint built into the dead-end structures, and with some degree of inherent longitudinal strength in most structures, there is no requirement to apply an assumed longitudinal load at tangent and angle structures in which the conductors are pin-connected or suspended, except where Grade 1 construction is required (see Clause 7.8.3).

A.7.8.2 Assumed longitudinal load — Without terminations or tension changes

Figures A.9 and A.10 and the following examples illustrate the application of loads and strength requirements for structures with pin-type construction at railway crossings.

For the following examples, it is assumed that

- N = 3 conductors, each with rated tensile strength greater than 12.4 kN;
- T = 7.63 kN (maximum loaded tension per conductor, heavy loading);
- W = 14.59 N (horizontal wind force per m of conductor length); and
- S = 76 m (average span length)

Example 1 — Pole A, with a line angle of 20°

$$\text{Transverse and angle force} = \left(N \times 2 \times T \times \sin\frac{A}{2}\right) + (N \times W \times S) + \text{wind on pole}$$

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where
A = line angle

$$\left(N \times 2 \times T \times \sin\frac{A}{2}\right) = 3 \times 2 \times 7.63 \times \sin 10^\circ = 45.78 \times 0.173 = 7.92 \text{ kN}$$

$N \times W \times S = 3 \times 14.59 \times 76 = 3.33$ kN

Wind on pole (assumed) = 0.51 kN

Resultant force = 11.76 kN

The strength requirements are calculated as follows:
- Angle — from the formula, the resultant transverse load at the angle structure is 11.76 kN. With a load factor of 2, a pole stronger than Class 1 is needed, so guying is necessary. The assumed horizontal loading on the guy(s) is 1.6 × 11.76, or 18.82 kN.
- Longitudinal — where the supply or communication lines adjacent to the crossing span are built to Grade 2 criteria or less, the assumed longitudinal load is taken as an unbalanced pull in the direction of the section requiring the stronger grade, in accordance with 33-1/3 percent of the sum of all the tensions, under loaded conditions [i.e., 33-1/3% x (3 × 7.63 kN) = 7.63 kN]. See Clause 7.8.2.3.

The centre phase conductor, which is approximately 0.3 m above the pole top, provides the maximum stress in the support. Where this load is transferred to 0.6 m below the pole top and a load factor of 1.5 is applied for an unguyed pole, the force is calculated as follows:

$$7.63 \times \frac{10.7}{9.8} \times 1.5 = 12.50 \text{ kN}$$

Table A.3 specifies that a Class 3 pole is needed to withstand this force. Where the pole is weaker than Class 3, it must be guyed away from the crossing span. Thus, where a pole weaker than Class 3 is used, it must be guyed in two directions, as shown in Figure A.10, for angle forces and longitudinal requirements.

Example 2 — Pole B

Transverse force= ($N \times W \times S$) + wind on pole
= 3.33 + 0.51
= 3.84 kN

The strength requirements are calculated as follows:
- Transverse — from the formula, the resultant wind force multiplied by a load factor of 2 is 7.68 kN. As specified in Table A.3, a Class 5 pole or better is needed, without guying. A lower class of pole needs to be guyed on each side in a transverse direction, as shown in Figure A.10.
- Longitudinal — the assumed longitudinal load calculations are the same as those determined for Pole A.

A.8 Strength of supporting systems according to deterministic design methods

A.8.1 General

Clause 8.1.2 explains the general application of load and strength requirements.

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There are three categories or situations covered in Tables 27, 28, and 29: crossings, proximities, and joint use, respectively. A crossing can be formed quite simply by a single span (two structures) over a railway or road. In other cases, it can take many structures or spans to form a crossing (e.g., crossings over private or public property). Grade requirements apply to the entire section that forms the crossing, proximity, or joint-use situation. The only exception to this is the longitudinal load of Clause 7.8.2, which is applicable only where Grade 1 is required and in which case the additional strength requirements are supplied at each end of the Grade 1 section. Where Grade 2 or 3 is called for in Tables 27, 28, and 29, there are no longitudinal strength requirements, except where wire and cable attachments are terminated (i.e., dead-ended).

A.8.1.3

Clause 8.1.3 mandates that geometric non-linear analysis is the structural analysis method required by this Standard. It also enables the use (non-mandatory) of load non-linearities which might exist as the structure deflects. If load non-linearity is considered, all other poles inter-connected with it should be assumed to be fixed, unless all parties agree with the methods used.

In simplest terms, a non-linear structural analysis is one that considers other secondary effects that may have an impact on the structure's response to the loads applied and which are not normally considered with a linear structural analysis. By considering these additional secondary effects in the analysis of a structure, the result will more closely reflect structures in reality. This is the primary reason why load factors for a non-linear analysis have always been lower than for a linear analysis in previous editions of this Standard.

Geometric non-linearity refers to any changes in the stiffness of the structure to resist the assumed loads which may occur when the structure deflects. In other words, if a structure's stiffness changes due to its deformed state, this is a geometric non-linear consideration. If the structure's stiffness changes to resist the assumed loads, it must be re-estimated and used to analyze the structure's response to the assumed loads once more. This may require several iterations in order for the final structural response to be determined.

Stiffness of the structure can change in a few ways. Individual members that make up the structure can begin to behave in a manner that is not linear, such as localized buckling or stress failure or they fail to provide the stiffness initially assumed for other reasons (e.g. cable or wire goes slack). Stiffness can also change strictly due to structure deflections. For instance, the direction of a member could change, which could change the stiffness this member provides to the overall structure. In a non-linear analysis, all significant changes to a structure's stiffness should be included as part of the structure's geometric non-linearity. Since structure support mechanisms such as guy wires and pole braces offer stiffness to the structure which must be considered as additional support members in the structure's analysis, any change to their stiffness as the structure deflects must be taken into account as well.

Another way a structure's stiffness can change is due to a change in where the loads are applied, since the attachment points on the structure have moved during its deflection. This effect has been traditionally called the P-Delta effect. As a structure deflects in response to the loads applied to it, the fact that the loads are essentially moved to a different absolute location will change the real loads being applied to the structure. This is said to "soften" the stiffness of the structure and reduce its ability to support the assumed loads. If the P-Delta effect is significant, the structure will collapse. If the effect is moderate, the structure will deflect more than initially expected, but not severe enough to cause complete failure. Estimating the structure's deflection as accurately as possible is important in order to have the best prediction of actual structure and member stresses.

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If the assumed forces (loads) that are applied to the structure in the analysis can change in magnitude or direction due to movement or deflection of the structure, this can be an important secondary effect. The primary source of load non-linearity comes from attachments that are connected between structures (e.g., wires, cables). If one or both of the structures deflect, the distance between the structures and the associated attachment's angle could change. The change in angle changes the direction of the loads being applied to the structure. The change in distance between the structures will change the magnitude of the loads through either slackening or tightening of the span attachments.

A.8.7 Supply conductors, supporting conductors, and messengers

A.8.7.3 Wire tension limits

In the process of designing overhead lines, tension limits of conductors are traditionally specified as a percentage of the conductor RTS. Invariably, these limits are intended to protect conductors from damaging aeolian vibrations and to provide enough safety margins to prevent conductors from breaking or overstretching under design ice and/or wind loads.

The percentages of RTS limits vary among countries and even among different provinces in Canada. The following limits are typical of many practices in various jurisdictions:

a) Maximum conductor tension < 50 to 75% of RTS under design wind/ice loads/temperature. The purpose of this limit is to prevent conductor failure under extreme weather conditions as well as the occurrence of excessive plastic stretch when stresses are applied to conductors in the range of 80 to 95% of their RTS.
b) Final bare conductor tension (without ice or wind loads) < 18 to 25% of RTS after the conductor has been subjected to long-term creep or permanent plastic deformation due to load. This and the following condition were meant to protect the conductor from fatigue failure due to aeolian vibration. The values of 18 to 25% of RTS were based on the assumption that conductors would be equipped with vibration dampers.
c) Initial bare conductor tension < 20 to 35% of RTS when a new conductor is strung and sagged (prior to any long-term creep). This condition is similar to the previous one, except that it tends to protect the conductors from fatigue damage at the initial stage, a few months following the construction, when the conductor tension is high and long-term creep has not yet developed.

The limits specified above have been found to be acceptable in many countries and lines using mostly Aluminum Conductors Steel Reinforced (ACSR or A1/S1A in accordance with CAN/CSA-C61089) with a steel-to-aluminum ratio lower than 13%, as well as in the case of all aluminum conductors.

In past decades, many other conductor types and strandings have been used and it has been found that the above tension limits specified might not provide sufficient vibration fatigue protection for mixed conductors having a high steel ratio or for aluminum alloy conductors.

There is now a consensus among experts that the tension limit for controlling aeolian vibrations should not be based on a percentage of RTS, as in past practice, but rather on the value of the conductor catenary parameter C (m).

Note: *In case the conductor is loaded with wind and/or ice, the unit weight is replaced by the resultant load per unit length of the conductor.*

Recent overhead line design standards such as CSA C22.3 No. 60826 recommend the use of two principal conductor tension limits. The first limit applies to unloaded conductors and aims to prevent conductor fatigue due to aeolian vibration, while the second limit aims to prevent excessive conductor stretch due to the plastic deformation of the ice/wind-loaded conductor.

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Because conductor tension varies with time due to long-term creep and the plastic stretch of the conductors related to ice/wind loads, there is also a debate about which conductor condition or catenary parameter should be used to control damaging aeolian vibrations. The recent study published in CIGRE Technical Brochure 273 proposes using the initial conductor tension at average temperatures of the coldest month as a basis for establishing the limits of the catenary parameter C.

Based on CIGRE Technical Brochure 273, limiting the initial conductor tension at average temperatures of the coldest month to a maximum value of catenary parameter C of 2000 m should be sufficient to reduce the chances of conductor fatigue due to aeolian vibrations when vibration dampers are used. Without vibration dampers, the same limit drops to between 1000 and 1400 m depending on terrain types. Because CIGRE Technical Brochure 273 does not propose any limit for the maximum conductor tension, the limit of 70 to 80% RTS specified in CSA C22.3 No. 60826 should be used. In this case, the tension should be calculated under the limit design loads used for the overhead line.

Note: *For design carried out in accordance with the working loads method, where a safety factor is imposed by the relevant standard/practice, the conductor tension under the combination of working loads, increased by the safety or overload factor, should not exceed a limit of 75 to 80% RTS.*

In addition to the above, it should be recognized that lines with short spans (i.e., 100 to 150 m, or less) do not benefit significantly from high conductor tensions or catenary parameters as do long spans (i.e., greater than 300 m). The table below provides the sags for a 1033.5 kcmil 54/7 “Curlew” ACSR conductor and two span lengths, both strung at C values ranging from 1000 to 2000 m.

Catenary parameter at –5 °C, initial conditions, m	Sag at 65 °C for 400 m span, m	Sag at 65 °C for 100 m span, m
2000	13.90	1.86
1800	14.59	1.96
1500	16.58	2.13
1200	19.29	2.32
1000	22.34	2.48

As specified in this table, the sag of the short span (e.g., distribution line) will increase only by 0.6 m when C is reduced from 2000 m to 1000 m, whereas the same reduction in C will increase the sag of the long span (e.g., transmission lines) by 8.5 m.

While the limit of C < 2000 m specified above provides a safe limit for vibration purposes, it might not be economical to use this limit for lines having spans less than 100 m. It should be recognized that, while increasing conductor tension reduces conductor sag, it will also increase loads on angle structures.

The following table illustrates the initial conductor tension as a percentage of RTS corresponding to the recommended catenary specified in Table 33 for various span lengths:

Span length, m	Recommended initial maximum catenary, m	Corresponding RTS for a 54/7 A1/S1A (ACSR) conductor, %	Corresponding RTS for a 45/7 A1/S1A (ACSR) conductor, %	Corresponding RTS for a 37 A1 (ASC) conductor, %	Corresponding RTS for a 37 A2 (AASC) conductor, %
> 400	2000	24	26	33	18
> 350 and ≤ 400	1900	22	25	31	17
> 300 and ≤ 350	1800	21	23	30	16
> 250 and ≤ 300	1700	20	22	28	15

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Span length, m	Recommended initial maximum catenary, m	Corresponding RTS for a 54/7 A1/S1A (ACSR) conductor, %	Corresponding RTS for a 45/7 A1/S1A (ACSR) conductor, %	Corresponding RTS for a 37 A1 (ASC) conductor, %	Corresponding RTS for a 37 A2 (AASC) conductor, %
> 200 and ≤ 250	1600	19	21	26	14
> 150 and ≤ 200	1500	18	20	25	13
> 100 and ≤ 150	1400	16	18	23	12
≤ 100	1300	15	17	21	11

A.8.12 Paired communication conductors — Grades 1 and 2

A.8.12.2 Paired conductors not supported by messengers — Grades 1 and 2

Clause 8.12.2 accounts for new communication conductors that incorporate tensile members as an integral part of the jacket.

A.9 Grounding methods for supply systems less than or equal to 22 kV and communication facilities

A.9.1 Supply systems less than or equal to 22 kV

A.9.1.2 Ground resistance requirements

A.9.1.2.1 Multi-grounded neutral systems

A.9.1.2.1.1

Due to the vast differences between utility practices, customer load conditions, and geographical soil conditions, it is often impossible to establish a level of steady-state neutral-to-earth voltage that can satisfy the safe let-go current value specified in IEEE 80 at all locations within a distribution system. Different measures or practices may be adopted by utilities to prevent possible electric shock hazards to persons, rather than trying to lower the neutral-to-ground voltage to a specific value at all points in the distribution system. For example, isolating the primary neutral from the secondary neutral prevents the passage of excessive voltage to a consumer's premises.

A.9.1.2.1.2

This Clause is intended to address the potential hazard arising from isolated sections of multi-grounded neutral systems situated in the same general vicinity. In some regions, it is common practice that two separate subdivisions (of the same municipality) are developed at different times but are fed by the same earth return 3-phase feeder. Each of the subdivisions is served by its own multi-grounded neutral system, but the neutral conductor does not extend beyond the subdivision boundary.

Over time, these two subdivisions will expand and eventually meet. As the physical distance between the subdivisions is reduced, the probability of the two subdivisions being bridged by a telecommunication conductor increases. It is thus important to ensure that the system neutral conductors of the two subdivisions are connected to prevent damage to the communication sheath and communication equipment due to a line-to-ground fault. Although the system neutral is not connected between the two subdivisions, bonding of the system neutrals might already be in place through the communication cables. During a line-to-ground fault, the fault current might flow back to the source through the sheath of the communication wire, potentially causing damage to communication

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equipment. It is thus crucial to ensure there is a continuous low-impedance path for the flow of fault currents.

A.9.1.2.2 Earth return systems

For an earth return system, the fact that all neutral return current flows through the ground electrode (as compared with neutral conductors for a multi-grounded neutral system) means that the heating effect of the ground electrode by the return current warrants special attention. A maximum of 5 A return current may typically be used for each deep-driven electrode or set of multi-rod electrodes under steady-state conditions.

Neutral return current is calculated in accordance with the following:

- For a single-phase load of 50 kVA on a 24940GrdY/14400 V system, neutral return current is 3.5 A. One set of ground electrodes and one set of redundant ground electrodes are necessary.
- For a balanced 3-phase load, there is no neutral return current. The neutral return current for 3-phase transformers depends on the amount of unbalanced load.
- For regulator banks, the neutral return current depends on the tap changer positions of all three phases. For example, a 100 A single-phase regulator with ±10% tap changer will have 10 A neutral return current at maximum boost position.

  Note: *The polarity of the neutral current reverses when the tap changer changes from buck to boost position. As a result, the neutral return current of a 100 A 3-phase regulator bank with ±10% tap changer under the worst condition can be 20 A (one phase under full buck position and the other two phases under full boost position).*
- For 3-phase capacitor banks without unbalanced protection, the neutral return current is the phase current. For 3-phase capacitor banks with unbalanced protection, the neutral return current is the trip setting of the unbalanced protection.

The ground electrode is surrounded by earth that is made up of concentric shells of equal thickness. Those shells closest to the ground electrode have the smallest amount of area, resulting in the greatest degree of resistance. Each subsequent shell incorporates a greater area, resulting in lower resistance, so that, eventually, additional shells offer little resistance to the ground surrounding the ground electrode.

A deep-driven ground electrode is generally more effective at lowering the ground resistance due to the long cylindrical shape of the shells. Where several ground electrodes are used to lower ground resistance, each ground electrode has its own sphere of influence. Without proper spacing of the ground electrodes, the spheres of influence intersect and the lowering of ground resistance is minimal. Thus, for additional electrodes to be effective, the spacing of additional rods needs to be at least equal to the depth of the driven rod.

Redundancy of ground electrodes should be ensured for earth return systems. If the connection of one ground electrode is broken for any reason, the transformer primary neutral floats at the phase voltage. To avoid this potentially hazardous situation, redundant ground electrodes and ground wire connections should be installed as close as practicable to the transformer primary neutral connection. Redundancy of ground electrodes is provided by installing an extra set of ground electrodes at a minimum distance of 10 m from the first ground electrode or, alternatively, at the adjacent pole. The 10 m distance is used to minimize the heating effect of the neutral return current through the ground electrode. The two sets of ground electrodes are interconnected by overhead conductors or buried ground wire.

Redundant ground wire connection can be provided by installing separate ground wires on separate poles, or on opposite sides of the same pole where the ground wires are on the same structure. The intent of this measure is to reduce the likelihood of damaging both wires at the same time.

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Table A.1
Switching surge values for ac conductors
(See Clause A.5.)

Nominal system voltages, kV phase-to-phase*	Maximum line-to-ground voltage, kV	Equivalent phase-to-phase voltage, kV	Switching surge factor	Switching surge crest voltage, kV	Flashover-to-ground distance, mm
—	0.75	—	—	—	8
4, 8	5	8.7	4	—	72
12, 13.2	10	17.4	4	—	144
25	15	26	4	—	216
34	22	38	4	—	316
66, 69	50	87	4	283	718
115, 132	90	155	3	382	970
161	120	208	2.75	467	1185
230	150	260	2.75	583	1586
287, 300, 315	190	329	2.7	725	2129
345, 360	220	380	2.7	840	2566
400	250	433	2.7	954	3000
500	318	550	2.5	1122	3640
735	442	765	2.2	1374	4600

* *There can be more than one nominal voltage.*

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Table A.2
Switching surge values for dc conductors
(See Clause A.5.)

Maximum line-to-ground voltage, kV	Maximum system voltages, kV	Switching surge voltage (line-to-ground voltage × 1.6), kV	Flashover-to-ground distance, mm
200	± 200	320	802
250	± 250	400	1036
300	± 300	480	1269
350	± 350	560	1503
400	± 400	640	1736
450	± 450	720	1970
500	± 500	800	2204

Note: *Note: The following examples illustrate the calculation of switching surge voltages in Tables A.1 and A.2:*

a) *ac nominal system voltage = 230 kV*
Maximum phase-to-phase voltage = 230 kV + 10%(230 kV) = 253 kV
Equivalent line-to-ground voltage = 253 kV ÷ $\sqrt{3}$ = 146 kV
This falls in the range of > 120 kV and ≤ 150 kV.
Switching surge factor = 2.75
Switching surge crest voltage = 146 kV × $\sqrt{2}$ × 2.75 = 568 kV

b) *ac nominal system voltage = 500 kV*
Maximum phase-to-phase voltage = 500 kV + 10%(500 kV) = 550 kV
Equivalent line-to-ground voltage = 550 kV ÷ $\sqrt{3}$ = 317.5 kV
Switching surge factor = 2.50
Switching surge crest voltage = 317.5 kV × $\sqrt{2}$ × 2.5 = 1122 kV

c) *dc voltage = ±450 kV*
Assume that 450 kV is the maximum pole voltage under steady-state conditions.
Maximum phase-to-ground voltage = 450 kV
Switching surge voltage = 1.6 × 450 kV = 720 kV

Table A.3
Pole classes and strengths
(See Clauses A.7.4.2 and A.7.8.2.)

Class		Ultimate strength horizontal force (applied 0.6 m from top of pole), kN
Wood	Concrete	
1	J	20.0
2	H	16.5
3	G	13.3
4	F	10.7
5	E	8.5
6	D	6.7
7	C	5.3

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Table A.4
Damage and failure limits of structures
(See Clauses 8.3.8.1, 8.3.9, 8.3.10.1, and 8.3.12.3.)

Supports				
Support type	**Material or elements**	**Loading mode**	**Damage limit**	**Failure limit**
Lattice tower (including guyed towers)	All elements except guys	Tension	Yield (elastic) stress	Ultimate tensile (breaking) stress
		Shear	90% (elastic) shear stress	Shear (breaking) stress
		Compression (buckling)	Non-elastic deformation* from L/500 to L/100	Collapse by instability
	Steel guys	Tension	Lower value of (a) yield stress (70% to 75% UTS); and (b) deformation corresponding to a 5% reduction in the tower strength	Ultimate tensile stress
Poles	Steel	Moments	1% non-elastic deformation at the top, or elastic deformation that can impair clearances	Local buckling in compression or ultimate tensile stress in tension
		Compression (buckling)	Non-elastic deformation from L/500 to L/100	Collapse by instability
	Fibre-reinforced composite	Moments	Elastic deformation that can impair clearances	Local buckling in compression
		Compression (buckling)	Deflection at pole mid-section	Collapse by instability
	Wood	Moments	3% non-elastic displacement at the top	Ultimate tensile stress
		Compression (buckling)	Non-elastic deformation from L/500 to L/100	Collapse by instability
	Concrete	Permanent loads and non-permanent loads	Crack opening† after release of load or 0.5% non-elastic deformation	Collapse of the pole

* *The deformation of compression elements is the maximum sag from the line joining end points. For elements subject to moments, it is the displacement of the free end from the vertical.*
† *The width of crack opening is subject to agreement between manufacturer and purchaser.*

Note: *L = length of the element.*

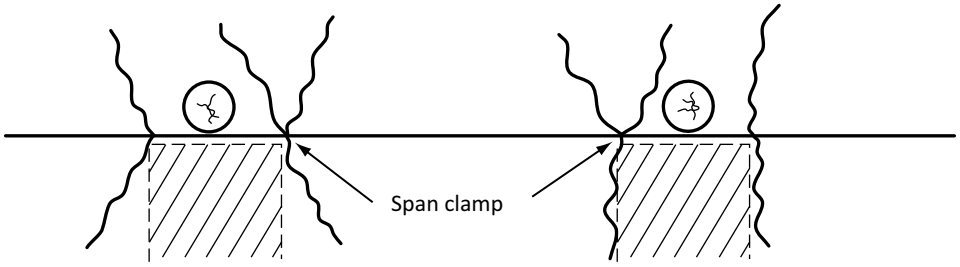
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Figure A.1
Examples of climbing space involving drop wires
(See Clause A.4.1.4.)

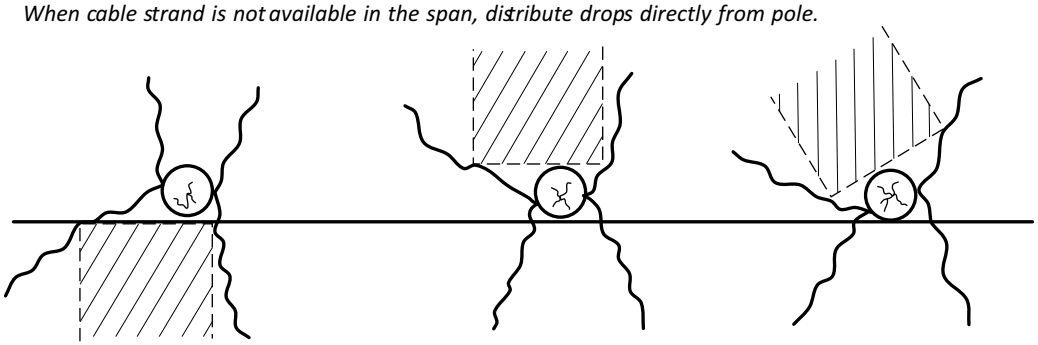
Method 1

When cable strand is available in the span, distribute drops from span clamps.



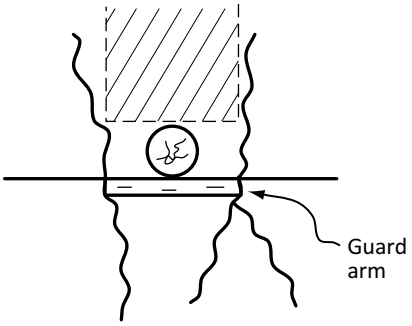
Method 2

When cable strand is not available in the span, distribute drops directly from pole.



Method 3

When climbing space cannot otherwise be achieved on either the field or street side of the pole, place a guard arm on the pole and distribute drops from the guard arm.



Legend:

= joint-use pole

= line conductor

= lateral drop wire

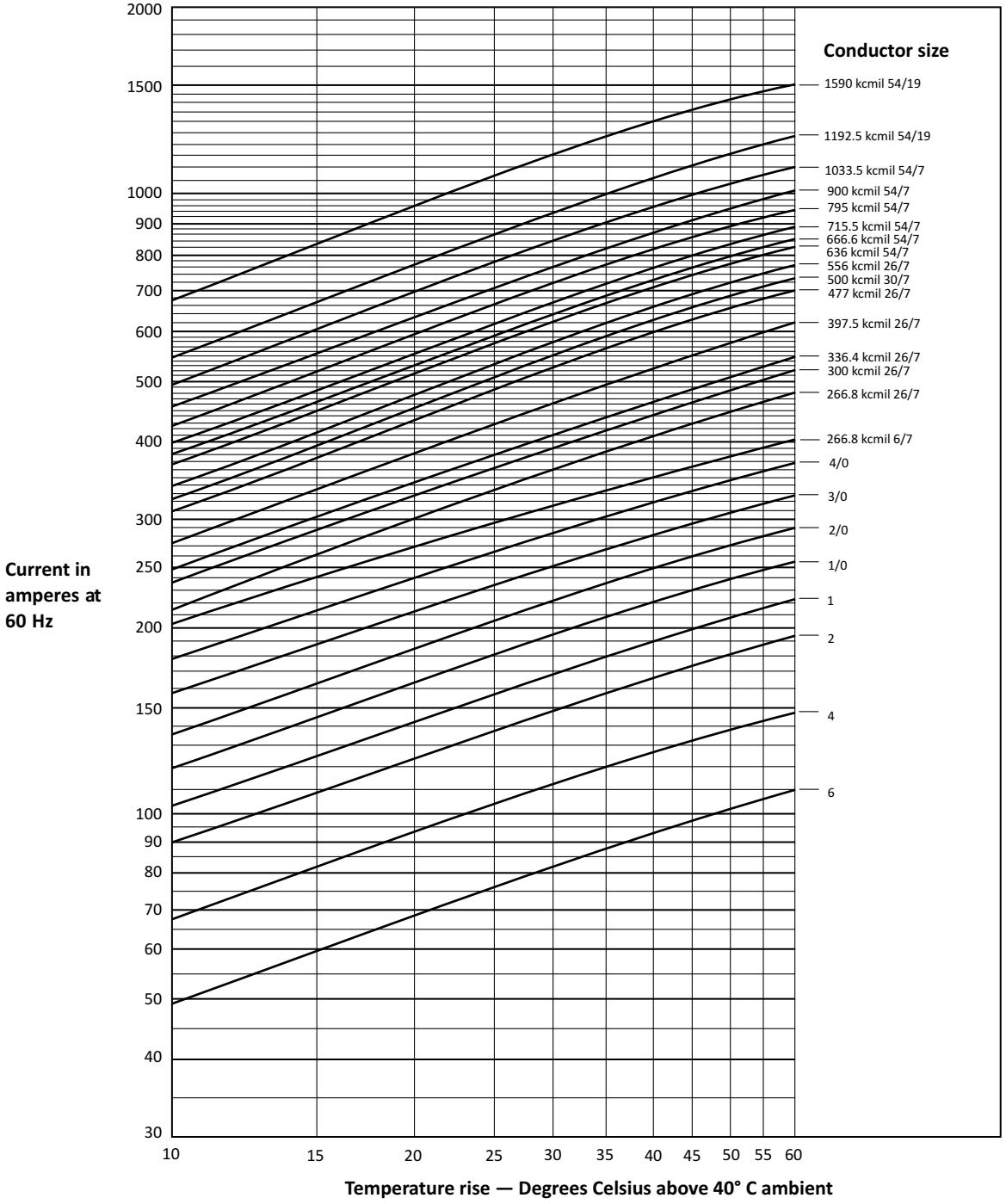
= minimum climbing space (0.75 m × 0.75 m)

Note: *Climbing space should be on the road side or field side of the pole.*

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Figure A.2
Example of temperature rise curves of Canadian standard sizes ACSR — Bare
(See Clause A.5.2.5.)



Notes:

1) *Air velocity = 0.61 m/s crosswise to conductor.*

2) *Emissivity = 0.5.*

3) *No sun; at sea level.*

4) *Ambient temperature = 40 °C.*

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Figure A.3
Vertical clearances at conductor crossings
(See Clause A.5.8.1.)

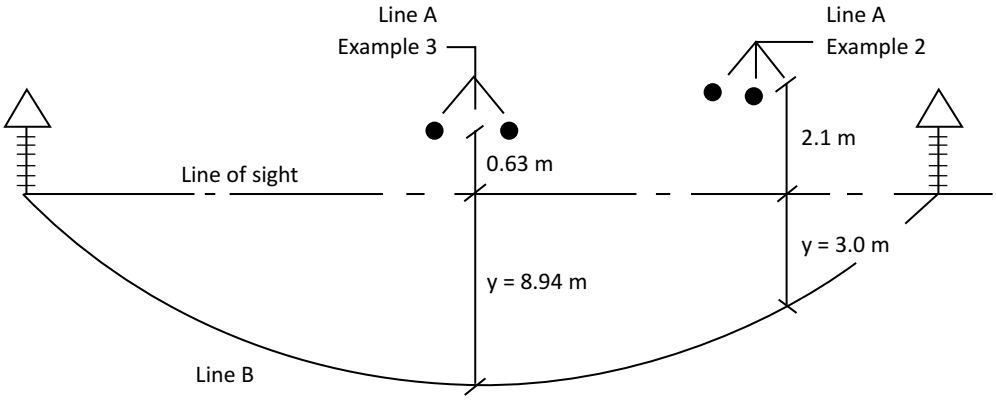
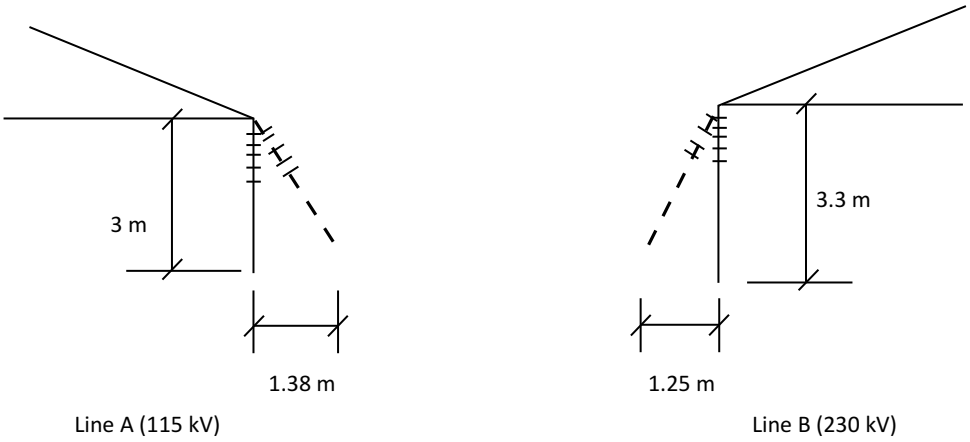
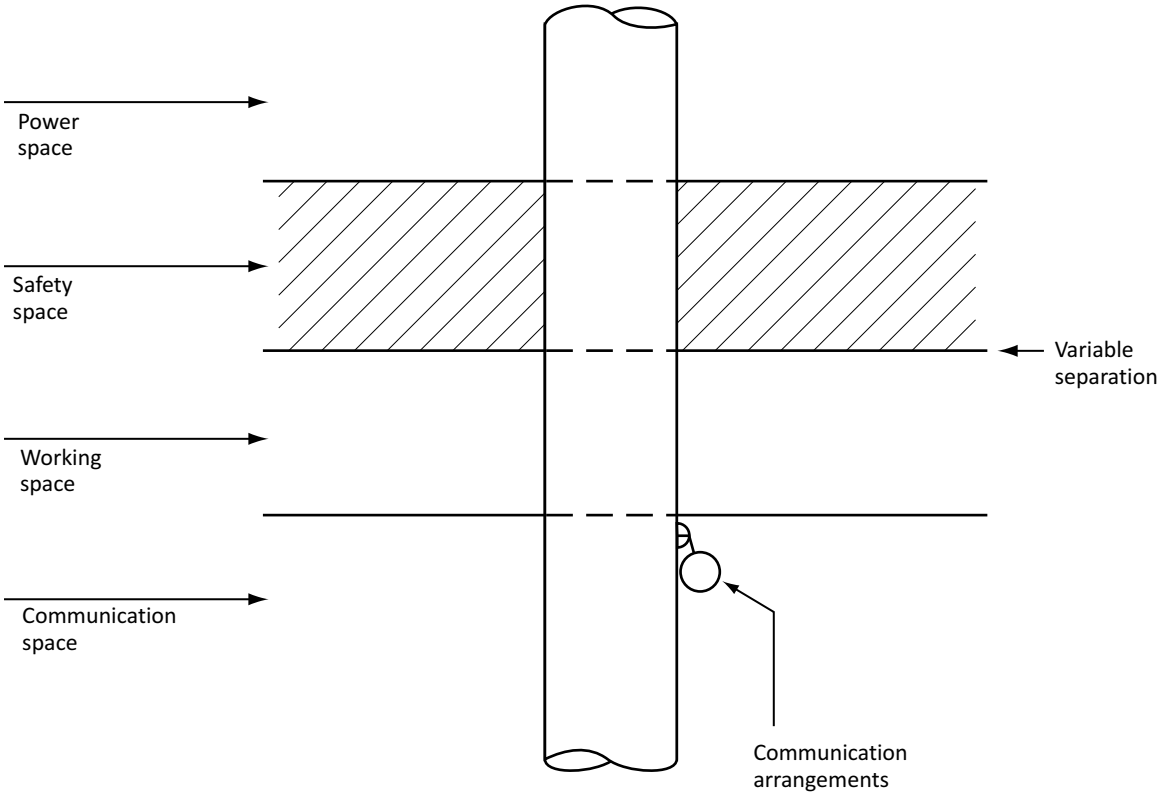


Figure A.4
Horizontal clearance between conductors on different structures
(See Clause A.5.8.2.)



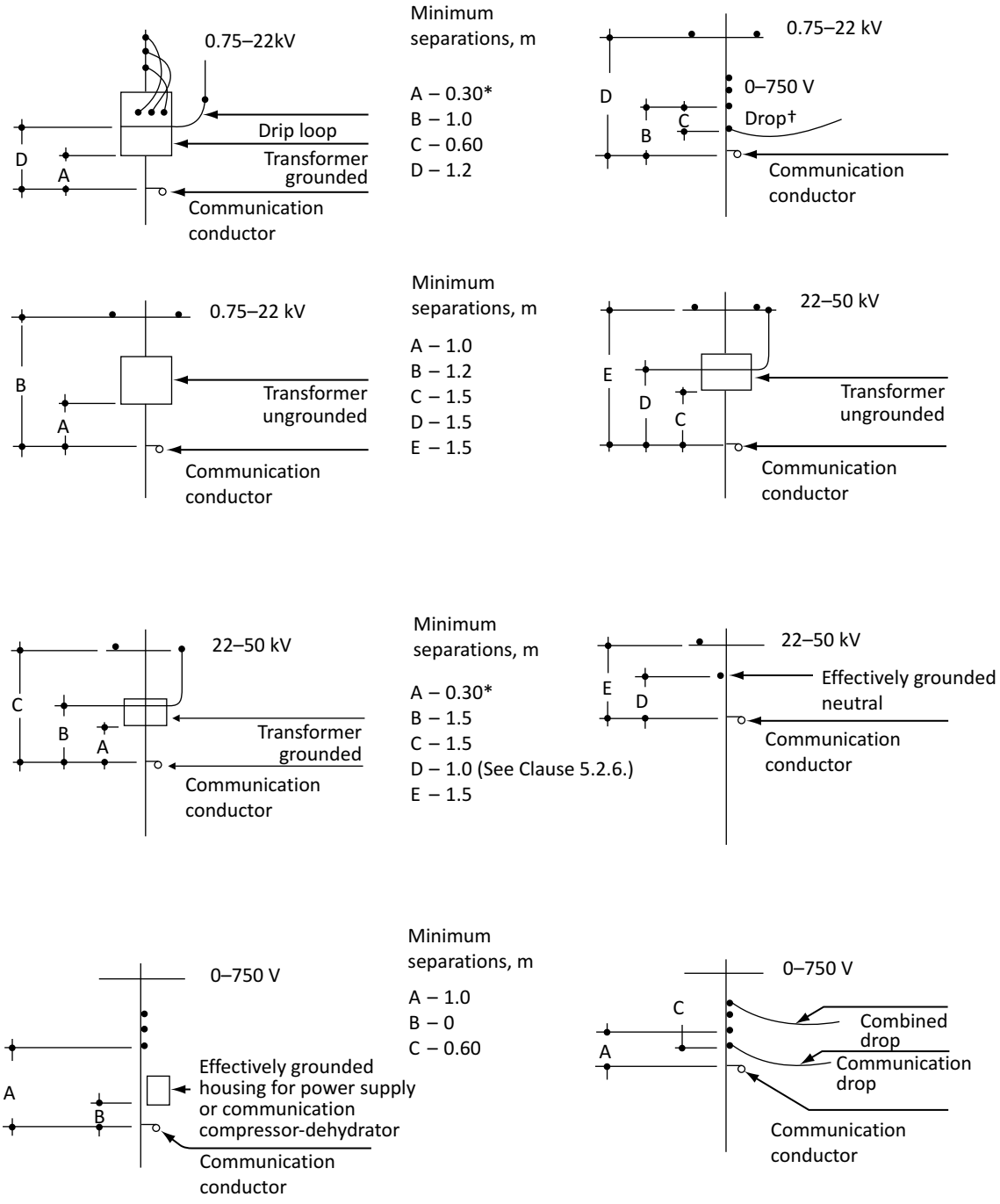
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Figure A.5
Joint-use pole spaces
(See Clause A.5.10.1.)



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Figure A.6
Illustrations of the minimum vertical separations specified in Table 23
(See Clause A.5.10.1.)

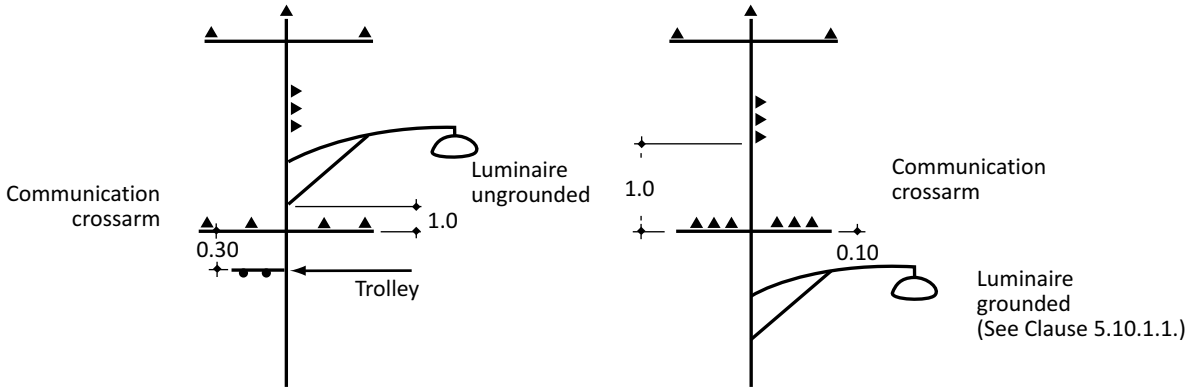


* *Where working space is needed for a tent (Option A), the minimum separation is increased to 0.75 m.*

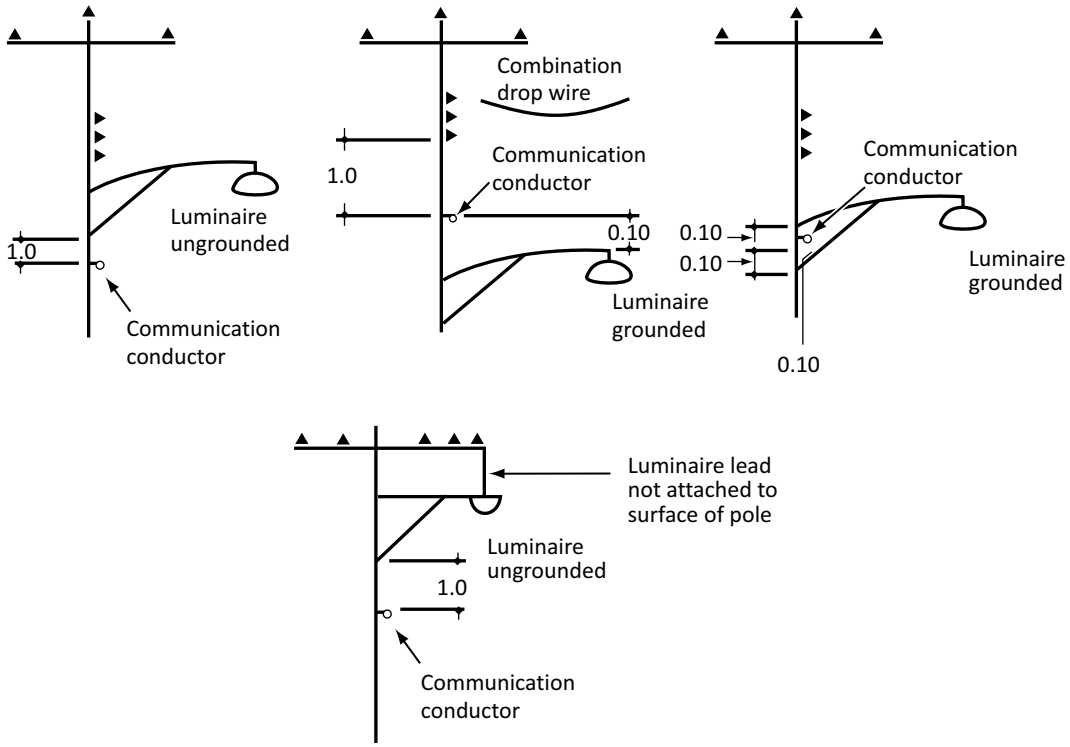
† *Working space needed for communications drop.*

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Figure A.7
Illustrations of the minimum vertical separations specified in Table 23
(See Clause A.5.10.1.)



(a) Separations to communication crossarms for luminaires operating at 0–750 V — Luminaire leads in cable on surface of pole



(b) Separations to cable or paired wire from luminaires operating at 0–750 V — Luminaire leads on surface of pole

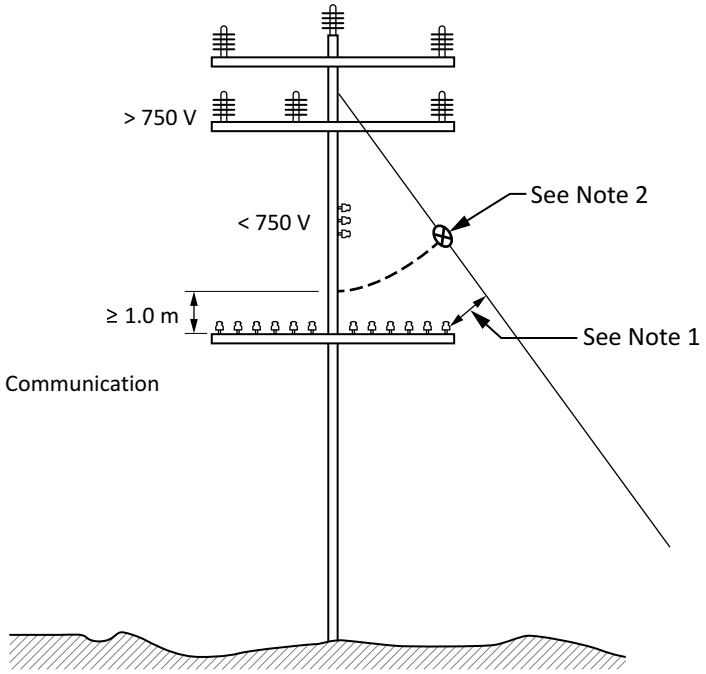
Notes:

1) *All values given are in metres unless otherwise specified.*

2) *All values given are minimum separations.*

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Figure A.8
Illustrations of minimum clearances for guy insulators as specified in Tables 25 and 26
(See Clause A.5.11.2.)



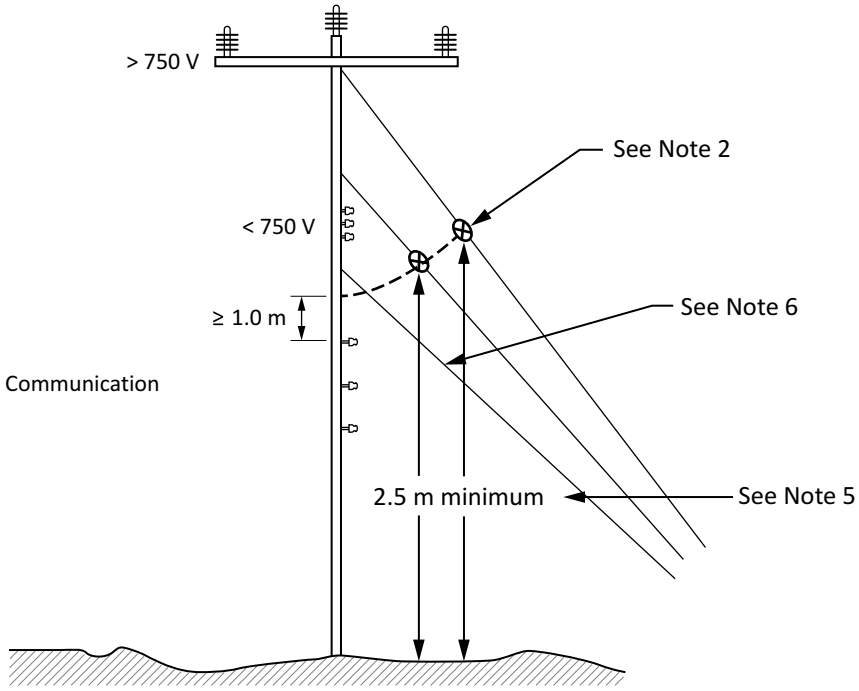
a) Insulated guy

(Continued)

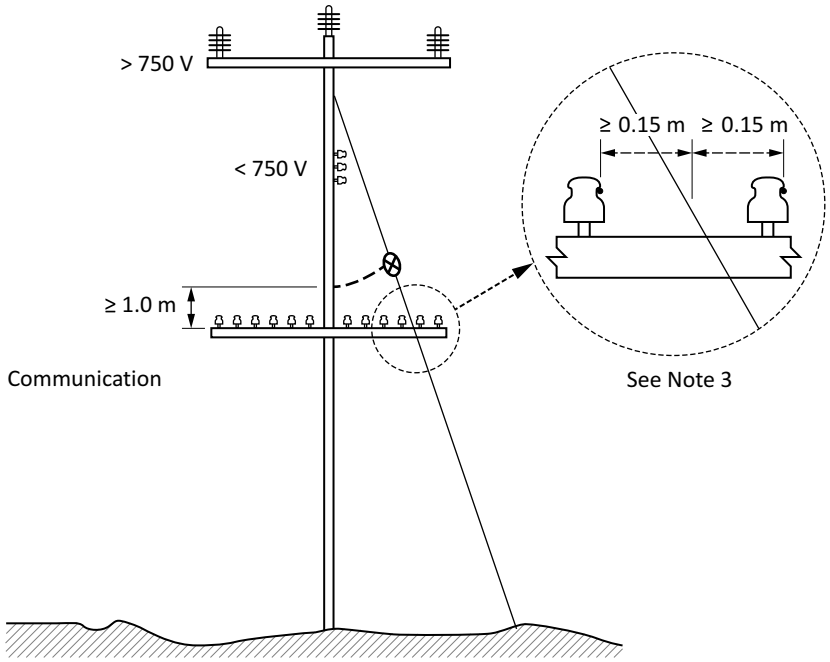
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Figure A.8 (Continued)



b) Guy insulator locations

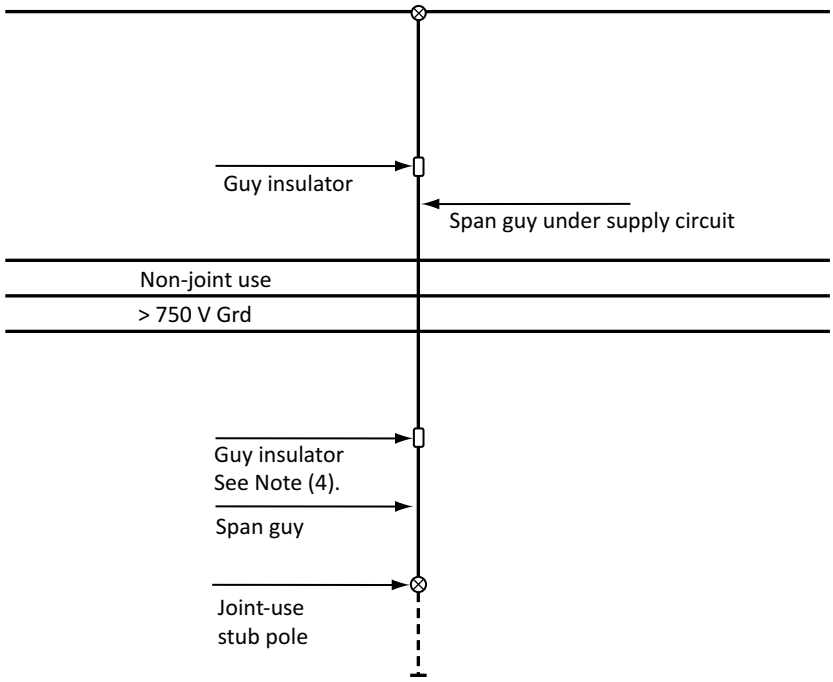


c) Wire clearance

(Continued)

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Figure A.8 (Concluded)



d)

Notes:

1) *This Standard does not specify a distance between communication plant and a guy insulator other than to require the minimum working space is not violated. A minimum clearance of 0.3 m otherwise should be used as a guide to provide safe clearance when working on communication plant (see Clause 4.1.5).*

2) *Where a guy is attached above supply conductors, an insulator is required above communication line plant and equipment (see Clause 5.11.4).*

3) *Communication pin space might have to be rearranged.*

4) *Guy insulators need to be appropriate for the voltages on the pole (See Clause 4.2.7.3).*

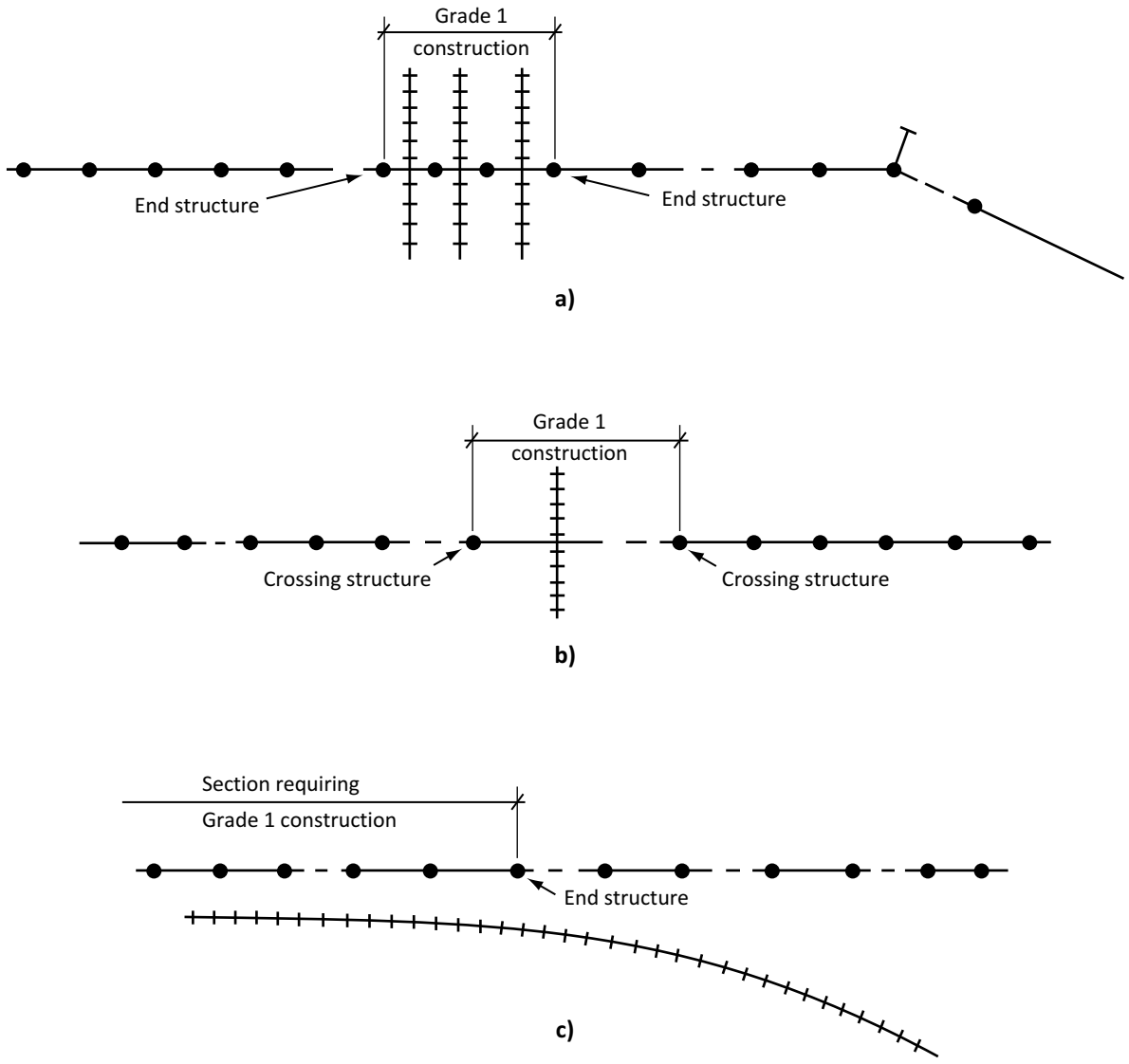
5) *Guy insulators should be no closer than 2.5 m to the ground under failed guy conditions (see Clause 4.2.7.2).*

6) *No guy insulator or grounding required for this guy as there are no supply wires below it on the pole.*

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Figure A.9
Crossing or end structures
(See Clause A.7.8.2.)



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Figure A.10
Railway crossing
(See Clause A.7.8.2.)

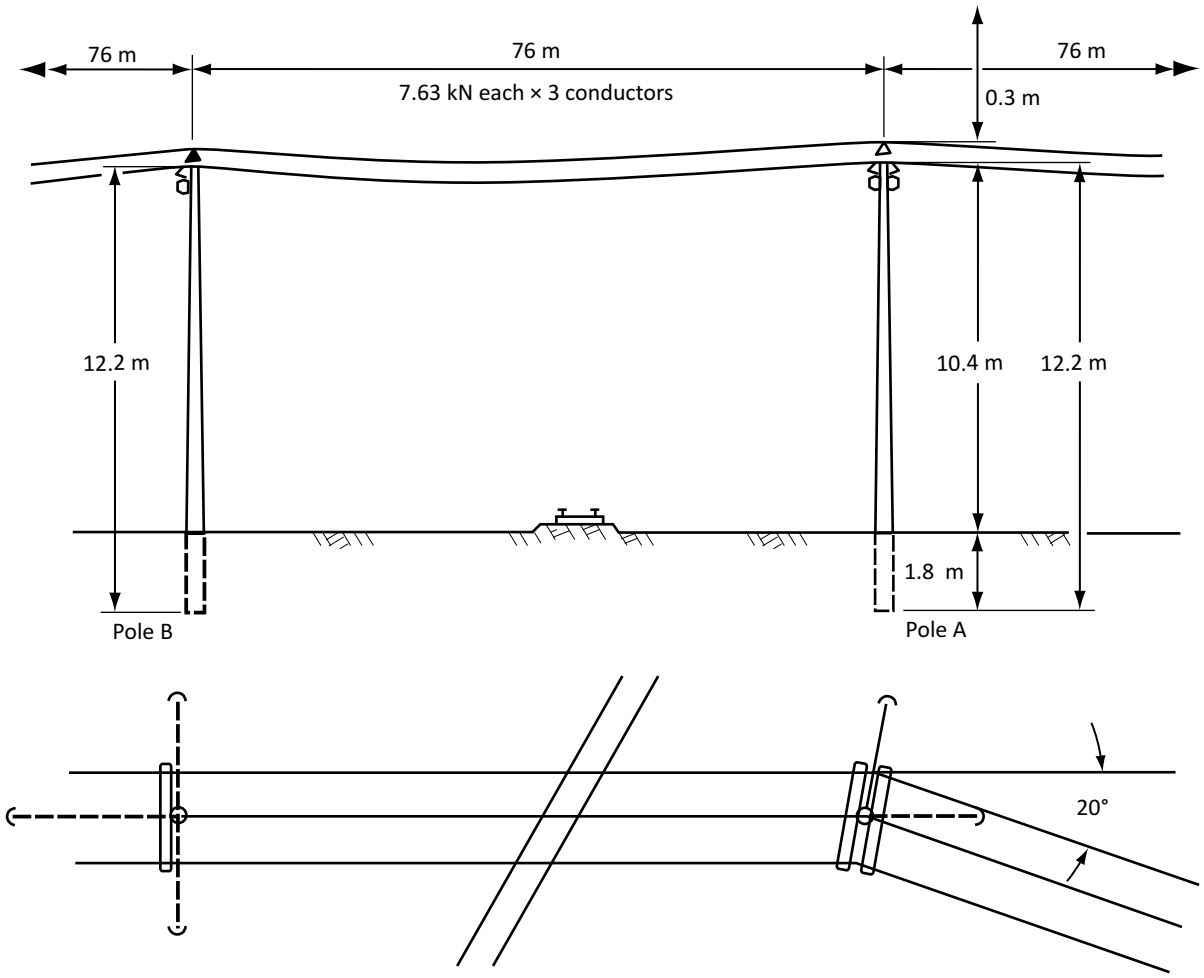
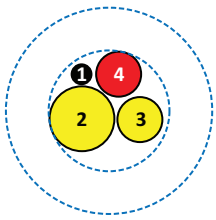
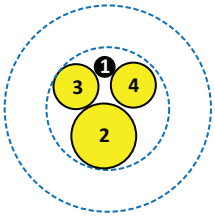


Figure A.11
Cable bundle diameter for calculating wind and ice loads
(See Clause A.7.3.4.)

Calculated Approximate Cable Bundle
Bundle Diameter = 35.38252441109248 mm



Interactive Cable Bundle Estimator
Bundle Diameter = 36.25516196522574 mm
1 = 1/4 in Grade 180 7 Strands Steel — adapted from 1970s publicly available data
2 = FOTS 156-216
3 = FOTS 12 to 72
4 = FOTS 12 to 72



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Annex B (informative)
Stress-strain characteristics and conductor sag

Note: *This Annex is not a mandatory part of this Standard.*

B.1 General
The purpose of this Annex is to illustrate the various sag and tension definitions used in this Standard.

B.2 Stress-strain characteristics

B.2.1
Conductor material such as hard-drawn copper is partially elastic. Thus, when an unstressed length of the wire is subjected to a certain tension, it will elongate to a length greater than the original unstressed length. When the tension is removed, the wire will contract to a length that is between the stressed and original unstressed lengths. The difference between the original unstressed length and the intermediate length is due to non-elastic elongation, whereas the difference between the stressed length and the intermediate length is due to elastic elongation. In other words, the elastic stretch is recovered when the tension is removed but the non-elastic stretch is not. If, upon removal of the tension, the wire contracts to the original unstressed length, there is no non-elastic elongation or permanent stretch, and the material is perfectly elastic.

B.2.2
The stress-strain characteristics of a conductor are illustrated in Figure B.1, which assumes that the conductor is of one metal only. A composite conductor such as ACSR would have a somewhat more complicated stress-strain characteristic, owing to the presence of two metals with dissimilar elastic properties.

Figure B.2 shows a conductor of the original unstressed length, *L*, strung up between supports A and B, which in this example are assumed to be at equal elevations. This conductor is subjected to the initial or stringing tension, T_1. In order to avoid the necessity of considering temperature changes, which would introduce further expansions and contractions of the conductor, it is assumed that the temperature remains constant at –18 °C.

Figure B.1 shows that tension T_1 is associated with point “a” on the curve, and that the wire has stretched from its original length, *L*, to a new length equal to *L* plus the stretch represented by the per cent elongation e_2. If the wire were taken down again and measured, it would be found to be longer than *L* by an amount corresponding to the per cent elongation e_1 (i.e., by the amount that has been referred to as non-elastic stretch).

Similarly, the loading of the conductor with ice increases its tension to a new value, T_2, resulting in the position indicated by the lowest of the three catenary curves in Figure B.2. This tension corresponds to point “b” in Figure B.1, and the stressed length of the conductor is now equal to *L* plus the stretch corresponding to per cent elongation e_3. If the ice loading is removed from the conductor, its tension will reduce to T_3, and it will take up a position indicated by the intermediate catenary curve of Figure B.2. The length of this catenary is smaller than that of the lowest catenary and is given by *L* plus the stretch associated with per cent elongation e_4. This corresponds from point “b” to point “c” along a line generally parallel to the line connecting point “a” with per cent elongation e_1. If the wire is taken down again and measured, it will have an unstressed length equal to *L* plus the stretch associated with

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per cent elongation e_5. In other words, the application of the ice load has introduced additional non-elastic stretch in the conductor over the amount that resulted from the stringing tension T_1; that is, the non-elastic elongation has been increased from e_1 to e_5.

If the wire is left in place when it reaches point “c”, subsequent loadings will move it up and down along the b–c line, provided that no tension greater than T_2 is experienced. This b–c line is approximately straight, indicating that once the wire is subjected to its maximum load, it is considered to be perfectly elastic.

B.3 Horizontal spans

Changes in temperature are accompanied by expansions or contractions in the length of a conductor. Figure B.3 shows a conductor strung by the “Initial (a) 16 °C” catenary, which is then subjected to code loading that brings it down to the catenary marked “Final loaded (heavy)”. Removal of this load and return to a temperature of 16 °C bring the conductor back to the catenary marked “Final unloaded 16 °C”, which lies between the initial and final loaded positions. If the temperature is varied, the conductor will assume other positions, such as those indicated for final unloaded sag at –18 °C, 49 °C, or 93 °C. In certain cases, the final unloaded sag for 93 °C will be greater than the final loaded sag, as shown in Figure B.3. Similarly, final unloaded sag for –18 °C can sometimes be less than the initial sag at 16 °C, as indicated by the dotted catenary marked “Initial (b) 16 °C”.

Assuming that Figure B.2 shows the upper catenary in the initial position at 16 °C, the lower catenary in the final loaded position (either medium or heavy loading), and the intermediate catenary in the final unloaded position at 16 °C, the three centre-span sags (assuming A and B to be at equal elevations) may be termed, reading from top to bottom,

a) initial unloaded sag at 16 °C;
b) final unloaded sag at 16 °C; and
c) final loaded sag (heavy or medium).

The final loaded sag minus the final unloaded sag is called the sag increase. Since the final unloaded sag can be at various temperatures, it is convenient to designate the sag increase by the temperature at which the final unloaded sag is given, as follows:

sag increase = final loaded sag (°C) – final unloaded sag (°C).

Note that where the final unloaded sag is greater than the final loaded sag, as for 93 °C in Figure B.3, the sag increase calculated using this formula will be negative. This situation occurs only with large conductors, but is possible for all lines. The minimum clearance values specified in this Standard are intended to accommodate this situation.

In this Annex, each span is assumed to be dead-ended, for the sake of simplicity. Situations involving a line containing spans of different lengths are ignored in order to avoid a discussion of the problem of ruling spans. The sag phenomena of dead-ended spans also apply to ruling spans, though the details will vary.

When a specific conductor is strung in a dead-ended span such that the tension in the conductor under the final loaded condition is 60% of the rated tensile strength, the sag increases vary according to the span length. A typical graph of sag increase versus span length is shown in Figure B.4. Each curve in Figure B.4 is calculated for a different percentage of rated tensile strength under the final loaded condition, and all curves have a characteristic shape. The sag increase peaks at a different span length for each curve. For span lengths greater than those associated with the peak, the sag increase drops off in value. The peak value of sag increase is usually termed the “maximum sag increase” and should be

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associated with a given size and type of conductor, span length, temperature for final unloaded sag, percentage of rated tensile strength in the final loaded condition, and final loaded condition (e.g., medium or heavy).

Figure B.1
Stress-strain graph
(See Clause B.2.2.)

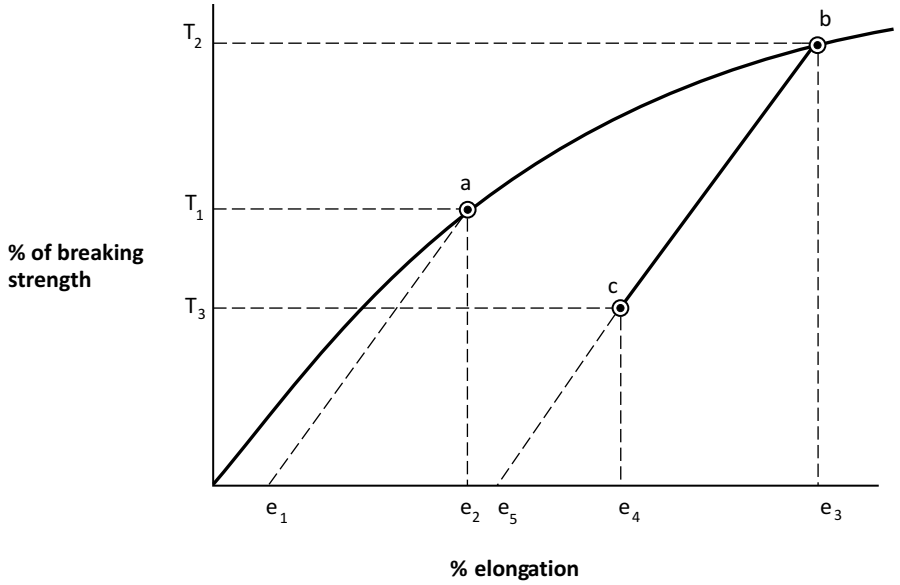
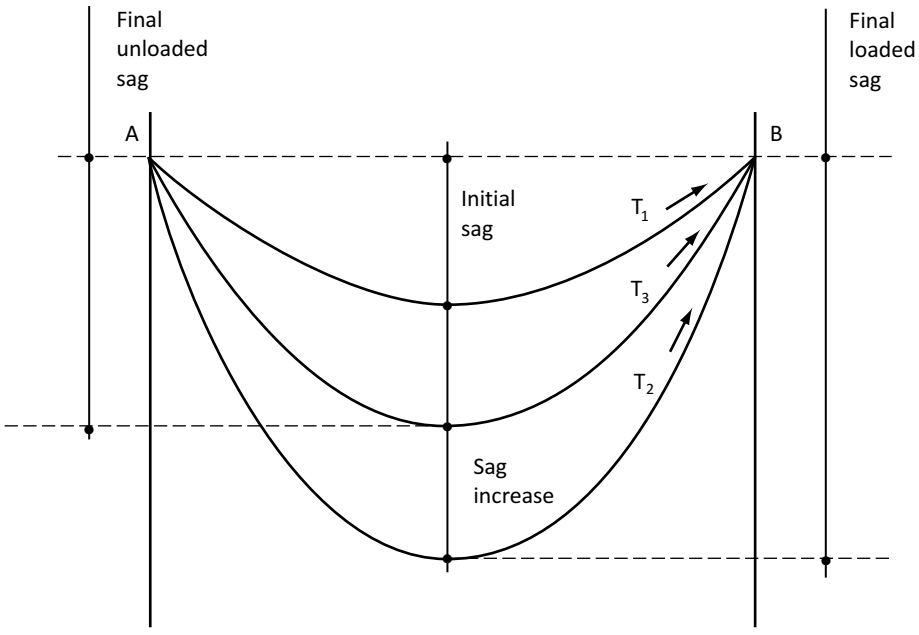


Figure B.2
Sag diagram — Constant temperature
(See Clauses B.2.2 and B.3.)



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Figure B.3
Sag diagram — Variable temperature
(See Clause B.3.)

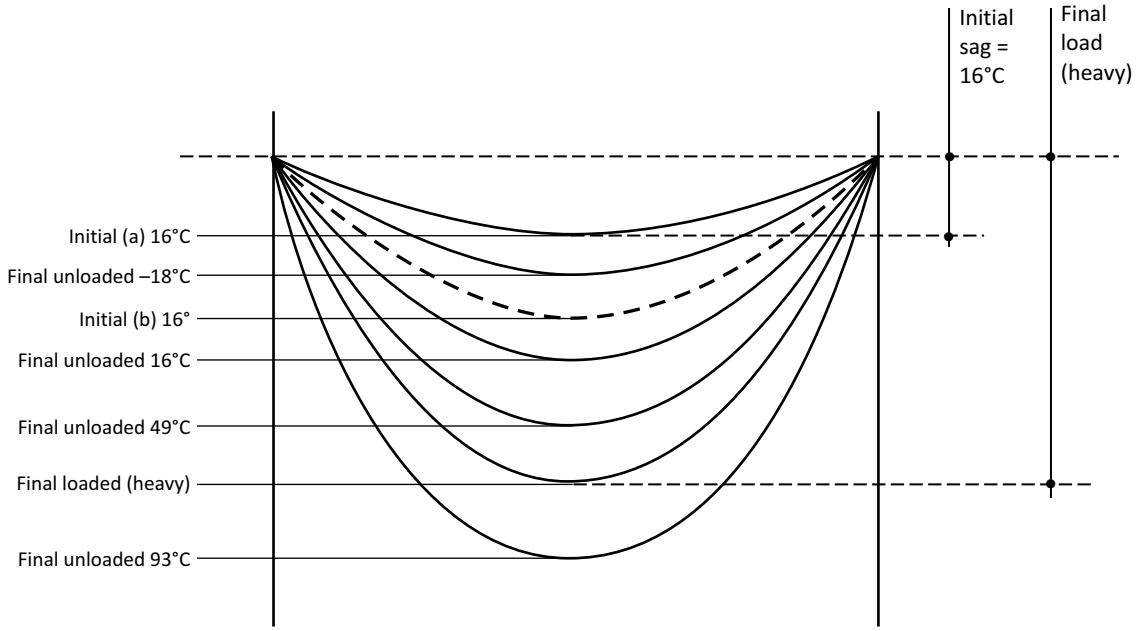
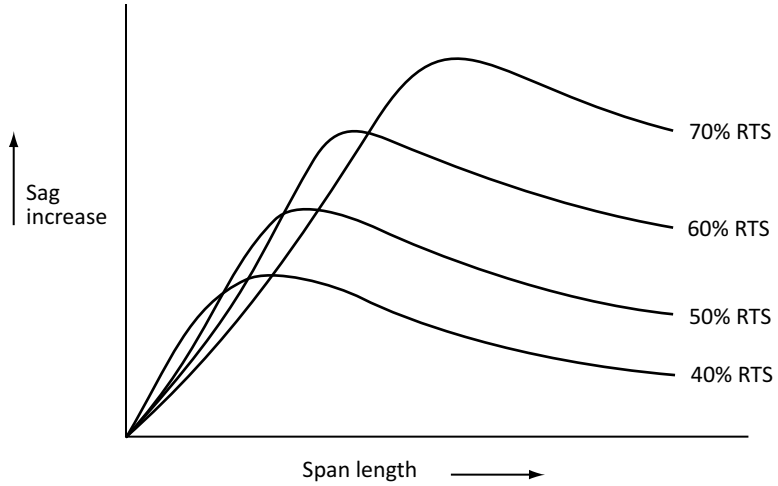


Figure B.4
Sag increases vs. span length
(See Clause B.3.)

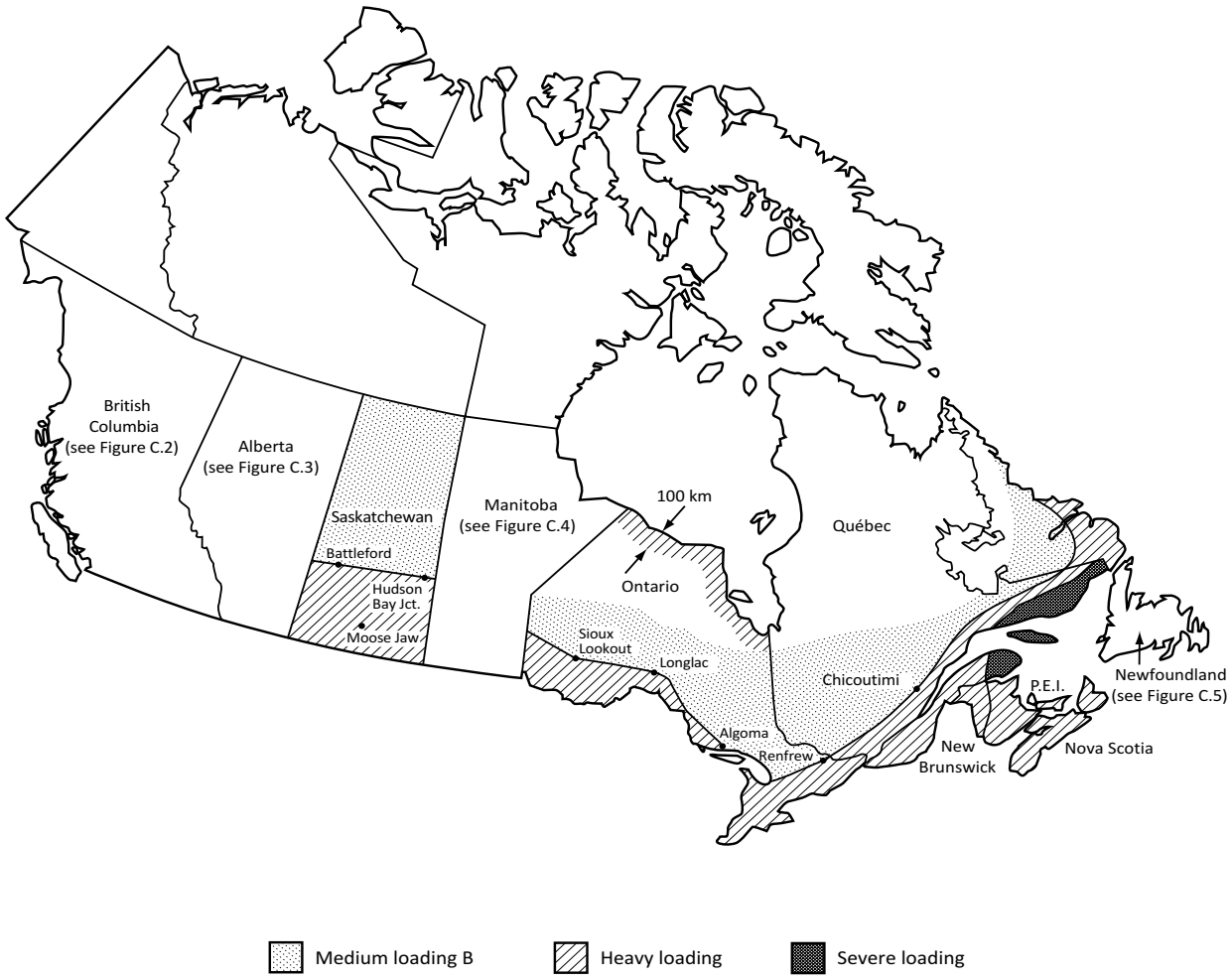


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Annex C (informative)
Loading maps

Note: *This Annex is not a mandatory part of this Standard.*

Figure C.1
Loading map of Canada
(See Clause 7.2 and Table 30.)



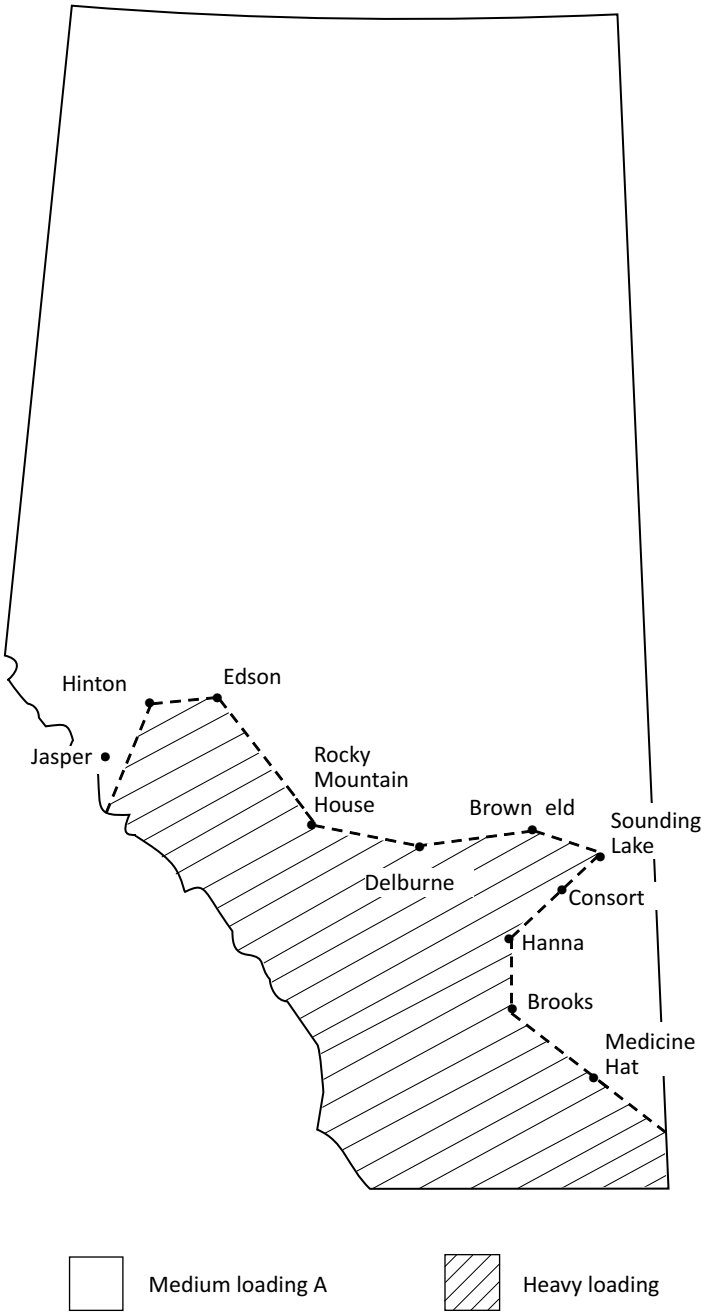
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Figure C.2
Loading map for British Columbia
(See Clause 7.2 and Table 30.)



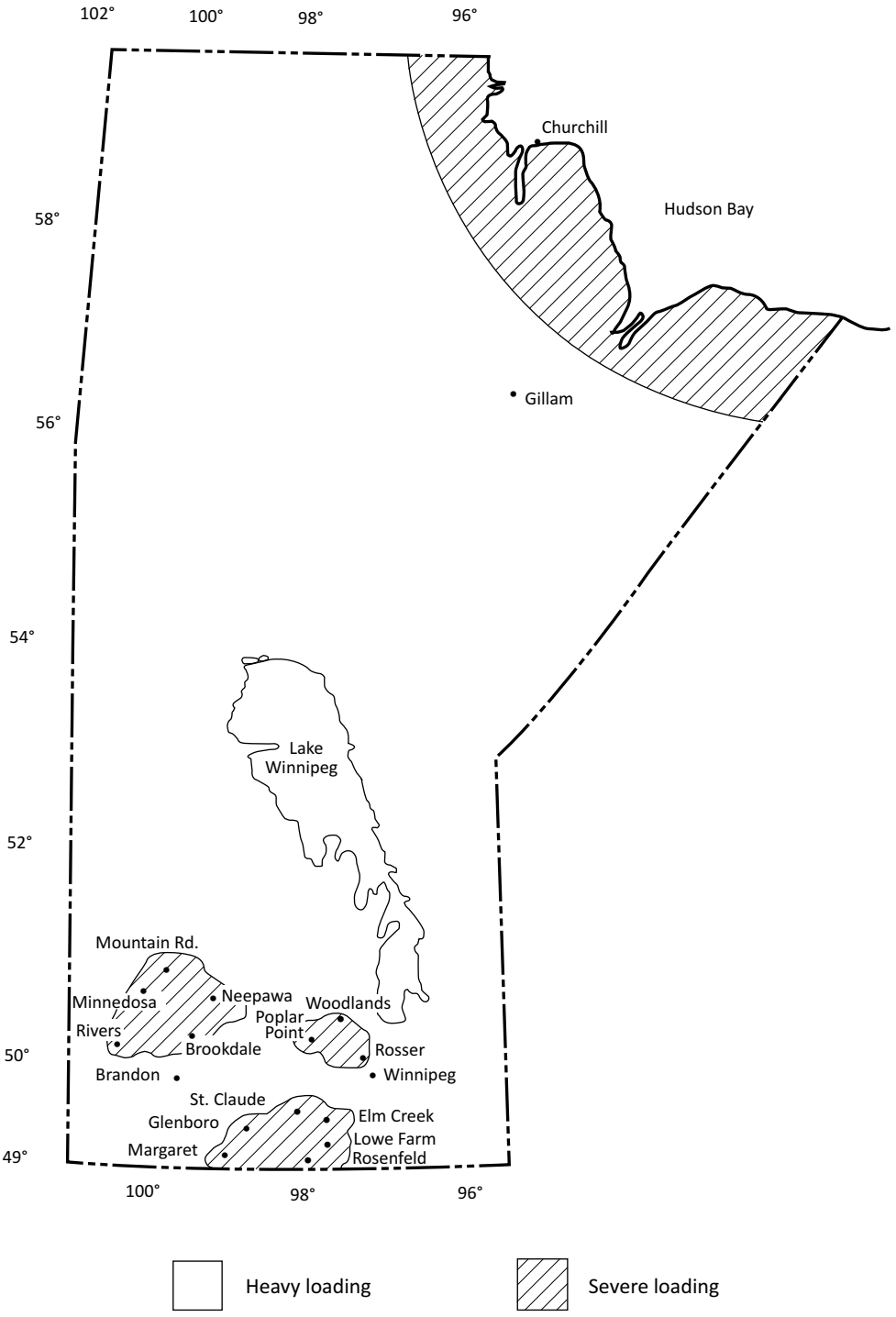
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Figure C.3
Loading map for Alberta
(See Clause 7.2 and Table 30.)



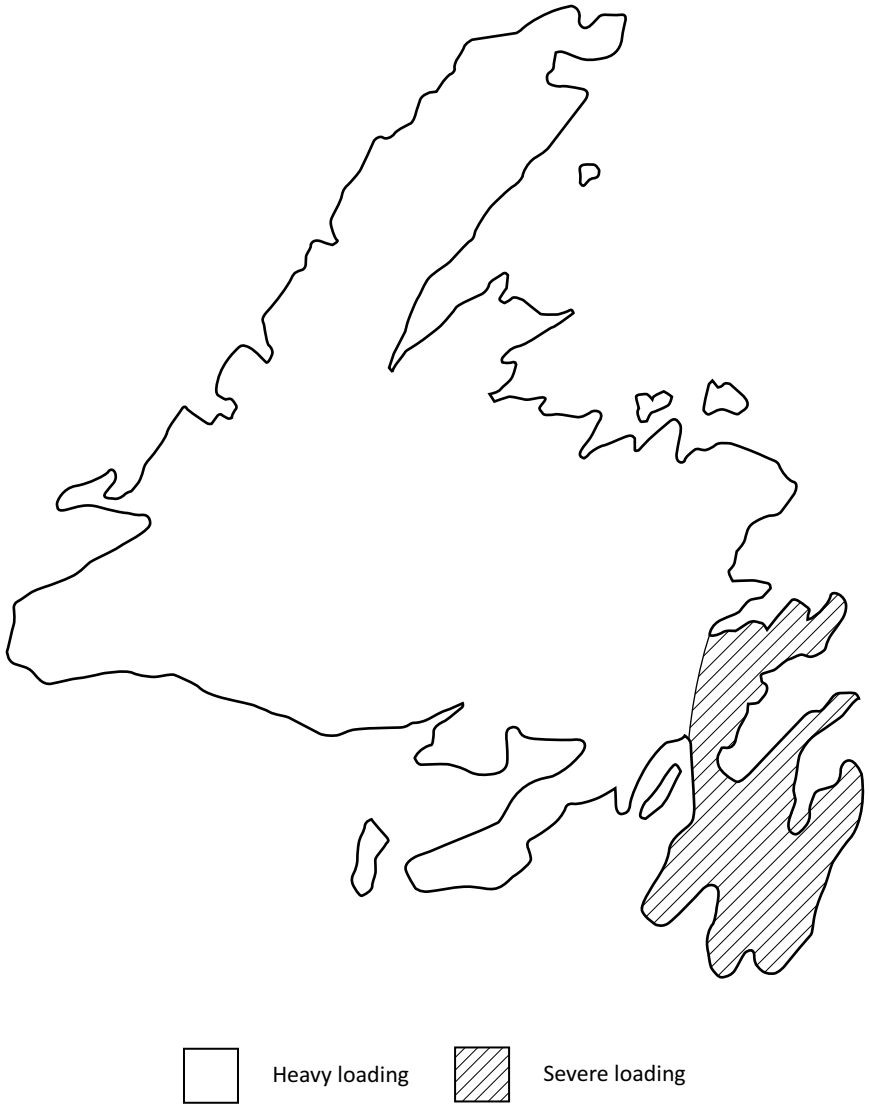
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Figure C.4
Loading map for Manitoba
(See Clause 7.2 and Table 30.)



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Figure C.5
Loading map for Newfoundland
(See Clause 7.2 and Table 30.)



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Annex D (informative)
Annual snow depth

Note: *This Annex is not a mandatory part of this Standard.*

D.1 Mean annual maximum snow depth on the ground

Clause 5.3.1.1(e) requires that, when determining the clearance of wires above ground, additional clearance equal to the mean annual maximum snow depth on the ground be provided where snow is not cleared. An estimate of the mean annual snow depth can be obtained using the following equation:

$$D = \frac{a \bullet S_s}{g \bullet p_s}$$

where

D = depth of snow on the ground, m

a = ratio of the mean annual maximum snow depth to the 50-year return period snow depth

= 0.55

S_s = 50-year return period deterministic design ground snow load as specified in the *National Building Code of Canada* (NBCC), Pa

g = acceleration due to gravity

= 9.81 m/s^2

p_s = mean annual snow pack density

= 205 kg/m^3, except as noted below:

Location	*ps*, kg/m^3
BC and the Yukon	260
Northern Canada (north of the tree line)	300

Values of S_s in the NBCC have been determined from an analysis of 50-year return period values of snow depth for more than 1600 Canadian locations, using Gumbel extreme value analysis.

The value of a reflects the average ratio (0.45) of mean annual snow depth to 50-year return period snow depth, plus one standard deviation (0.1). Because the value of this ratio varies from place to place, the value of a was increased from the mean value of 0.45 by one standard deviation to 0.55 to minimize underestimation of the annual maximum snow depth in the equation specified above.

Regionally representative values of S_s are available from provincial or municipal building codes. Recommendations of S_s values for locations not listed in building codes are available from Environment Canada.

Snow accumulation on the ground is sensitive to local characteristics such as surface roughness and vegetative cover. Designers should allow for increased snow depth in areas known to be prone to drifting or increased accumulation. Clearings in wooded areas, gullies, or depressions in otherwise flat farmland or grassland, or in the lee of natural or artificial snow breaks such as hedges, fences, or other structures, are also prone to increased snow depth.

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Because snow depths generally increase with elevation, especially in mountainous regions, and because the snow depth values obtained from the published values of S_s generally correspond to inhabited areas, designers should account for significantly increased snow depths at higher elevations. Mean annual maximum snow depths as high as 5 m can be expected in high-elevation locations in the mountains of western Canada, and as high as 2 m at higher elevations in eastern Canada.

Table D.1 provides estimates of the mean annual maximum snow depths for the locations listed in the NBCC. The values were calculated from the S_s values in the NBCC, using a value of a = 0.55 and locally appropriate values of D.

Table D.1
Mean annual maximum snow depth for selected Canadian locations
(See Clauses 5.3.1.1 and D.1.)

Location	Elevation, m	Mean annual maximum snow depth, m
British Columbia		
100 Mile House	1040	0.6
Abbotsford	10	0.4
Agassiz	15	0.5
Alberni	12	0.6
Ashcroft	305	0.4
Beatton River	840	0.7
Burnaby (Simon Fraser University)	110	0.6
Burns Lake	755	0.6
Cache Creek	455	0.4
Campbell River	20	0.7
Carmi	845	0.8
Castlegar	430	0.9
Chetwynd	605	0.5
Chilliwack	10	0.5
Cloverdale	10	0.5
Comox	15	0.6
Courtenay	10	0.6
Cranbrook	910	0.6
Crescent Valley	585	0.9
Crofton	5	0.4
Dawson Creek	665	0.5
Dog Creek	450	0.4
Duncan	10	0.4
Elko	1065	0.8
Fernie	1010	1.0

(Continued)

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Table D.1 (Concluded)

Location	Elevation, m	Mean annual maximum snow depth, m
Fort Nelson	465	0.5
Fort St. John	685	0.6
Glacier	1145	2.0
Golden	790	0.8
Grand Forks	565	0.6
Greenwood	745	0.8
Haney	10	0.5
Hope	40	0.6
Kamloops	355	0.4
Kaslo	545	0.6
Kelowna	350	0.4
Kimberley	1090	0.6
Kitimat Plant	15	1.2
Kitimat Townsite	130	1.4
Ladner	3	0.3
Langley	15	0.5
Lillooet	245	0.4
Lytton	325	0.6
Mackenzie	765	0.8
Masset	10	0.4
McBride	730	0.9
McLeod Lake	695	0.9
Merritt	570	0.4
Mission City	45	0.5
Montrose	615	0.9
Nakusp	445	0.9
Nanaimo	15	0.5
Nelson	600	0.9
New Westminster	10	0.5
North Vancouver	135	0.6
Ocean Falls	10	0.8
Osoyoos	285	0.2
Penticton	350	0.3
Port Alberni	15	0.6
Port Hardy	5	0.2

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Port McNeil	5	0.2
Powell River	10	0.4
Prince George	580	0.7
Prince Rupert	20	0.4
Princeton	685	0.6
Qualicum Beach	10	0.5
Quesnel	475	0.7
Revelstoke	440	1.2
Richmond	5	0.3
Salmon Arm	425	0.8
Sandspit	5	0.4
Sidney	10	0.2
Smithers	500	0.6
Smith River	660	0.6
Squamish	5	0.7
Stewart	10	1.7
Surrey (88 Ave. & 156 St.)	90	0.5
Taylor	515	0.5
Terrace	60	1.2
Tofino	10	0.2
Trail	440	0.9
Ucluelet	5	0.2
Vancouver	15	0.4
Vancouver (Granville & 41)	120	0.4
Vernon	405	0.5
Victoria	10	0.2
Victoria (Gonzales Hts.)	65	0.3
Victoria (Mt. Tolmie)	125	0.4
West Vancouver	45	0.5
Williams Lake	615	0.5
Youbou	200	0.8
Alberta		
Athabasca	515	0.4
Banff	1400	1.0
Barrhead	645	0.5

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Beaverlodge	730	0.7
Brooks	760	0.3
Calgary	1045	0.3
Campsie	660	0.5
Camrose	740	0.5
Cardston	1130	0.4
Claresholm	1030	0.4
Cold Lake	540	0.5
Coleman	1320	0.7
Coronation	790	0.6
Cowley	1175	0.4
Drumheller	685	0.3
Edmonton	645	0.5
Edson	920	0.6
Embarras Portage	220	0.5
Fairview	670	0.7
Fort MacLeod	945	0.3
Fort McMurray	255	0.4
Fort Saskatchewan	610	0.4
Fort Vermilion	270	0.6
Grande Prairie	650	0.6
Habay	335	0.7
Hardisty	615	0.5
High River	1040	0.4
Hinton	990	0.8
Jasper	1060	0.9
Keg River	420	0.7
Lac La Biche	560	0.4
Lacombe	855	0.6
Lethbridge	910	0.3
Manning	465	0.6
Medicine Hat	705	0.3
Peace River	330	0.6
Pincher Creek	1130	0.4
Ranfurly	670	0.5

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Red Deer	855	0.5
Rocky Mountain House	985	0.5
Slave Lake	590	0.5
Stettler	820	0.6
Stony Plain	710	0.5
Suffield	755	0.4
Taber	815	0.3
Turner Valley	1215	0.4
Valleyview	700	0.6
Vegreville	635	0.5
Vermilion	580	0.5
Wagner	585	0.5
Wainwright	675	0.5
Wetaskiwin	760	0.5
Whitecourt	690	0.5
Wimborne	975	0.4
Saskatchewan		
Assiniboia	740	0.4
Battrum	700	0.3
Biggar	645	0.6
Broadview	600	0.3
Dafoe	530	0.5
Dundurn	525	0.4
Estevan	565	0.4
Hudson Bay	370	0.5
Humbolt	565	0.6
Island Falls	305	0.6
Kamsack	455	0.6
Kindersley	685	0.4
Lloydminster	645	0.5
Maple Creek	765	0.3
Meadow Lake	480	0.5
Melfort	455	0.6
Melville	550	0.5
Moose Jaw	545	0.4

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Nipawin	365	0.5
North Battleford	545	0.5
Prince Albert	435	0.5
Qu'appelle	645	0.5
Regina	575	0.4
Rosetown	595	0.5
Saskatoon	500	0.5
Scott	645	0.5
Strasbourg	545	0.4
Swift Current	750	0.4
Uranium City	265	0.5
Weyburn	575	0.4
Yorkton	510	0.5
Manitoba		
Beausejour	245	0.5
Boissevain	510	0.6
Brandon	395	0.6
Churchill	10	0.8
Dauphin	295	0.5
Flin Flon	300	0.6
Gimli	220	0.5
Island Lake	240	0.7
Lac du Bonnet	260	0.5
Lynn Lake	350	0.7
Morden	300	0.6
Neepawa	365	0.6
Pine Falls	220	0.5
Portage La Prairie	260	0.6
Rivers	465	0.6
Sandilands	365	0.6
Selkirk	225	0.5
Split Lake	175	0.7
Steinbach	270	0.5
Swan River	335	0.5
The Pas	270	0.6

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Thompson	205	0.7
Virden	435	0.5
Winnipeg	235	0.5
Ontario		
Ailsa Craig	230	0.6
Ajax	95	0.3
Alexandria	80	0.7
Alliston	220	0.5
Almonte	120	0.7
Armstrong	340	0.7
Arnprior	85	0.7
Atikokan	400	0.7
Aurora	270	0.5
Bancroft	365	0.8
Barrie	245	0.7
Barriefield	100	0.6
Beaverton	240	0.6
Belleville	90	0.5
Belmont	260	0.5
Big Trout Lake	215	0.9
Borden (CFB)	225	0.6
Bowmanville	245	0.4
Bracebridge	310	0.8
Bradford	240	0.6
Brampton	215	0.4
Brantford	205	0.4
Brighton	95	0.4
Brockville	85	0.6
Brooklin	160	0.5
Burks Falls	305	0.7
Burlington	80	0.2
Caledonia	190	0.3
Cambridge	295	0.4
Campbellford	150	0.5
Cannington	255	0.6

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Carleton Place	135	0.7
Cavan	200	0.5
Centralia	260	0.6
Chapleau	425	1.1
Chatham	180	0.3
Chelmsford	270	0.7
Chesley	275	0.8
Clinton	280	0.7
Coboconk	270	0.7
Cobourg	90	0.3
Cochrane	245	0.8
Colborne	105	0.4
Collingwood	190	0.7
Cornwall	35	0.6
Corunna	185	0.3
Deep River	145	0.7
Deseronto	85	0.5
Dorchester	260	0.5
Dorion	200	0.8
Dresden	185	0.3
Dryden	370	0.7
Dunbarton	85	0.3
Dunnville	175	0.5
Durham	340	0.8
Dutton	225	0.4
Earlton	245	0.7
Edison	365	0.7
Elmvale	220	0.7
Embro	310	0.5
Englehart	205	0.7
Espanola	220	0.6
Etobicoke	160	0.3
Exeter	265	0.7
Fenelon Falls	260	0.6
Fergus	400	0.6

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Fonthill	230	0.6
Forest	215	0.5
Fort Erie	180	0.7
Fort Frances	340	0.6
Gananoque	80	0.6
Georgetown	255	0.4
Geraldton	345	0.8
Glencoe	215	0.4
Goderich	185	0.7
Gore Bay	205	0.7
Graham	495	0.7
Gravenhurst	255	0.7
Grimsby	85	0.2
Guelph	340	0.5
Guthrie	280	0.7
Hagersville	215	0.4
Haileybury	210	0.7
Haliburton	335	0.8
Hamilton	90	0.2
Hanover	270	0.7
Hastings	200	0.5
Hawkesbury	50	0.6
Hearst	245	0.8
Honey Harbour	180	0.7
Hornepayne	360	1.0
Huntsville	335	0.8
Ingersoll	280	0.5
Iroquois Falls	275	0.8
Jarvis	205	0.4
Jellicoe	330	0.7
Kapuskasing	245	0.8
Kemptville	90	0.6
Kenora	370	0.6
Killaloe	185	0.7
Kincardine	190	0.7

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Kingston	80	0.6
Kinmount	295	0.7
Kirkland Lake	325	0.8
Kitchener	335	0.5
Lakefield	240	0.6
Lansdowne House	240	0.8
Leamington	190	0.2
Lindsay	265	0.6
Lion's Head	185	0.7
Listowel	380	0.7
London	245	0.5
Lucan	300	0.6
Maitland	85	0.6
Markdale	425	0.9
Martin	485	0.7
Matheson	265	0.8
Mattawa	165	0.6
Midland	190	0.7
Milton	200	0.4
Milverton	370	0.7
Minden	270	0.7
Mississauga	160	0.3
Mitchell	335	0.7
Moosonee	10	0.6
Morrisburg	75	0.6
Mount Forest	420	0.7
Nakina	325	0.8
Napanee	90	0.5
Newcastle	115	0.4
New Liskeard	180	0.6
Newmarket	185	0.5
Niagara Falls	210	0.5
North Bay	210	0.6
North York	175	0.3
Norwood	225	0.6

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Oakville	90	0.2
Orangeville	430	0.6
Orillia	230	0.7
Oshawa	110	0.4
Ottawa	60	0.7
Owen Sound	215	0.8
Pagwa River	185	0.7
Paris	245	0.4
Parkhill	205	0.6
Parry Sound	215	0.8
Pembroke	125	0.7
Penetanguishene	220	0.8
Perth	130	0.6
Petawawa	135	0.7
Peterborough	200	0.5
Petrolia	195	0.4
Picton	95	0.5
Plattsville	300	0.5
Point Alexander	150	0.7
Porcupine	295	0.8
Port Burwell	195	0.3
Port Colborne	180	0.6
Port Credit	75	0.2
Port Dover	180	0.3
Port Elgin	205	0.8
Port Hope	100	0.3
Port Perry	270	0.7
Port Stanley	180	0.3
Prescott	90	0.6
Princeton	280	0.4
Raith	475	0.7
Red Lake	360	0.7
Renfrew	115	0.7
Ridgeway	190	0.7
Rockland	50	0.7

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Sarnia	190	0.3
Sault Ste. Marie	190	0.8
Scarborough	180	0.3
Schreiber	310	0.9
Seaforth	310	0.7
Simcoe	210	0.4
Sioux Lookout	375	0.7
Smiths Falls	130	0.6
Smithville	185	0.4
Smooth Rock Falls	235	0.7
Southhampton	180	0.7
South River	355	0.8
St. Catharines	105	0.3
Stirling	120	0.5
St. Mary's	310	0.6
Stratford	360	0.6
Strathroy	225	0.5
St. Thomas	225	0.4
Sturgeon Falls	205	0.6
Sudbury	275	0.7
Sundridge	340	0.8
Tavistock	340	0.6
Temagami	300	0.7
Thamesford	280	0.5
Thedford	205	0.6
Thunder Bay	210	0.8
Tillsonburg	215	0.4
Timmins	300	0.8
Toronto	105	0.2
Trenton	80	0.4
Trout Creek	330	0.7
Uxbridge	275	0.7
Vaughan (Woodbridge)	165	0.3
Vittoria	215	0.4
Walkerton	275	0.7

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Wallaceburg	180	0.2
Waterloo	330	0.5
Watford	240	0.5
Wawa	290	1.1
Welland	180	0.6
West Lorne	215	0.4
Whitby	85	0.3
White River	375	1.2
Wiarton	185	0.7
Windsor	185	0.2
Wingham	310	0.7
Woodstock	300	0.5
Wyoming	215	0.4
Québec		
Acton Vale	95	0.6
Alma	110	0.9
Amos	295	0.9
Asbestos	245	0.8
Aylmer	90	0.7
Bagotville	5	0.7
Baie-Comeau	60	1.2
Beaconsfield	25	0.6
Beauport	45	0.9
Bedford	55	0.6
Beloeil	25	0.7
Brome	210	0.7
Brossard	15	0.7
Buckingham	130	0.7
Campbell's Bay	115	0.7
Chambly	20	0.6
Chicoutimi	10	0.7
Coaticook	295	0.6
Contrecoeur	10	0.8
Cowansville	120	0.6
Deux-Montagnes	25	0.7

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Dolbeau	120	1.0
Dorval	25	0.7
Drummondville	85	0.7
Farnham	60	0.6
Fort-Coulonge	110	0.7
Gagnon	545	1.3
Gaspé	55	1.2
Gatineau	95	0.7
Gentilly	15	0.8
Gracefield	175	0.7
Granby	120	0.6
Harrington Harbour	30	1.3
Havre-Saint-Pierre	5	1.1
Hemmingford	75	0.7
Hull	65	0.7
Iberville	35	0.6
Inukjuak	5	0.8
Joliette	45	0.8
Jonquière	135	0.8
Kenogami	140	0.8
Kuujjuaq	25	1.3
Kuujjuarapik	20	1.2
Lachute	65	0.7
Lac-Mégantic	420	0.9
La Malbaie	25	0.8
L’Ancienne-Lorette	35	0.9
La Tuque	165	0.9
Laval	35	0.7
Lennoxville	155	0.5
Léry	30	0.6
Lévis	50	0.9
Loretteville	100	1.0
Louiseville	15	0.8
Magog	215	0.6

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Malartic	325	0.9
Maniwaki	180	0.7
Masson	50	0.7
Matane	5	1.0
Mont-Joli	90	1.1
Mont-Laurier	225	0.7
Montmagny	10	0.8
Montréal-Est	25	0.7
Montréal-Nord	20	0.7
Nitchequon	545	1.0
Noranda	305	0.9
Outremont	105	0.8
Percé	5	1.0
Pierrefonds	25	0.7
Pincourt	25	0.6
Plessisville	145	0.8
Port-Cartier	20	1.1
Québec	120	1.0
Richmond	150	0.6
Rimouski	30	1.0
Rivière-du-Loup	55	0.9
Roberval	100	1.0
Rock Island	160	0.5
Rosemère	25	0.7
Rouyn	300	0.8
Salaberry-de-Valleyfield	50	0.6
Schefferville	550	1.2
Senneterre	310	0.9
Sept-Îles	5	1.1
Shawinigan	60	0.8
Shawville	170	0.8
Sherbrooke	185	0.6
Sillery	10	0.8
Sorel	10	0.8
Ste-Agathe-des-Monts	360	0.9

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Ste-Anne-de-Bellevue	35	0.6
Ste-Foy	115	1.0
St-Félicien	105	1.0
St-Georges-de-Cocouna	35	0.9
St-Hubert	25	0.7
St-Hubert-de-Témiscouata	310	1.2
St-Hyacinthe	35	0.6
St-Jean	35	0.6
St-Jérôme	95	0.7
St-Jovite	230	0.8
St-Lambert	15	0.7
St-Laurent	45	0.7
St-Nicolas	65	1.0
Sutton	185	0.7
Tadoussac	65	0.9
Témiscaming	240	0.7
Thetford Mines	330	0.9
Thurso	50	0.7
Trois-Rivières	25	0.8
Val-d'Or	310	0.9
Varennes	15	0.7
Verchères	15	0.7
Verdun	20	0.7
Victoriaville	125	0.7
Ville-Marie	200	0.6
Waterloo	330	0.7
Windsor	185	0.6
New Brunswick		
Alma	5	0.6
Bathurst	10	1.0
Campbellton	30	1.0
Chatham	5	0.9
Edmundston	160	0.9
Fredericton	15	0.8
Gagetown	20	0.7

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Grand Falls	115	0.7
Moncton	20	0.8
Oromocto	20	0.8
Sackville	15	0.7
Saint John	5	0.6
Shippegan	5	0.9
St. Stephen	20	0.7
Woodstock	60	0.8
Nova Scotia		
Amherst	25	0.7
Antigonish	10	0.6
Bridgewater	10	0.5
Canso	5	0.4
Dartmouth	10	0.4
Debert	45	0.6
Digby	35	0.6
Greenwood (CFB)	105	0.7
Halifax	55	0.5
Kentville	25	0.7
Liverpool	20	0.4
Lockeport	5	0.4
Louisbourg	5	0.6
Lunenburg	25	0.5
New Glasgow	30	0.6
North Sydney	20	0.7
Pictou	25	0.6
Port Hawkesbury	40	0.6
Springhill	185	0.8
Stewiacke	25	0.5
Sydney	5	0.6
Tatamagouche	25	0.6
Truro	25	0.5
Wolfville	35	0.7
Yarmouth	10	0.5

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Table D.1 (Continued)

Location	Elevation, m	Mean annual maximum snow depth, m
Prince Edward Island		
Charlottetown	5	0.7
Souris	5	0.7
Summerside	10	0.8
Tignish	10	0.9
Newfoundland and Labrador		
Argentia	15	0.7
Bonavista	15	0.7
Buchans	255	1.3
Cape Harrison	5	1.7
Cape Race	5	0.6
Channel-Port aux Basques	5	0.8
Corner Brook	35	1.0
Gander	125	1.0
Goose Bay	40	1.4
Grand Bank	5	0.7
Grand Falls	60	0.9
Labrador City	550	1.2
St. Anthony	10	1.4
Stephenville	25	1.0
St. John’s	65	0.8
Twin Falls	425	1.3
Wabana	75	0.8
Yukon Territory		
Aishihik	920	0.4
Dawson City	330	0.6
Destruction Bay	815	0.4
Snag	595	0.5
Teslin	690	0.6
Watson Lake	685	0.7
Whitehorse	655	0.4
Northwest Territories		
Aklavik	5	0.6
Alert	5	0.3
Eureka	5	0.3

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Table D.1 (Concluded)

Location	Elevation, m	Mean annual maximum snow depth, m
Fort Good Hope	100	0.8
Fort Providence	150	0.7
Fort Resolution	160	0.6
Fort Simpson	120	0.6
Fort Smith	205	0.6
Hay River	45	0.7
Holman	10	0.4
Inuvik	45	0.6
Isachsen	10	0.3
Mould Bay	5	0.3
Norman Wells	65	0.7
Nottingham Island	30	0.9
Port Radium	195	0.8
Rae-Edzo	160	0.6
Resolution Island	5	1.0
Tungsten	1340	0.8
Yellowknife	160	0.6
Nunavut Territory		
Arctic Bay	15	0.4
Arviat (formerly Eskimo Point)	5	0.6
Baker Lake	5	0.6
Cambridge Bay	15	0.3
Chesterfield Inlet	10	0.6
Clyde River	5	0.7
Coral Harbour	15	0.7
Iqaluit	45	0.6
Kugluktuk (formerly Coppermine)	10	0.5
Rankin Inlet	10	0.6
Resolute	25	0.3

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Annex E (informative)
Bibliography

Notes:
1) *This Annex is not a mandatory part of this Standard.*
2) *The following publications relate to overhead lines. They are not specifically referenced in this Standard and are listed in this Annex for information only.*

CSA Group

C49.2-10 (R2014)
Compact round aluminum conductors steel reinforced (ACSR)

C49.3-10 (R2014)
Aluminum 1350 wire, annealed and intermediate tempers, for electrical purposes

C49.5-10
Compact round aluminum stranded conductors (compact round ASC)

CAN/CSA-C60104-03 (R2011)
Aluminum-magnesium-silicon alloy wire for overhead line conductors

CAN/CSA-C60888-03 (R2011)
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Hard-drawn aluminum wire for overhead line conductors

G40.20-13/G40.21-13
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O86-14
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S16-14
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CAN/CSA-S136-12
North American specification for the design of cold-formed steel structural members

ASCE (American Society of Civil Engineers)

10-97
Design of Latticed Steel Transmission Structures

ASCE/SEI (American Society of Civil Engineers/Structural Engineering Institute)

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IEEE (Institute of Electrical and Electronics Engineers)

751-91

IEEE Trial-Use Design Guide for Wood Transmission Structures

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